

|                                                                                   |                                                          |         |               |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|---------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |               |
|                                                                                   | V23                                                      | Rev 3.3 | Page 1 of 146 |

---

# QSMG\_EEA\_4001\_UDS 诊断服务

Version No: 3.3

Date: 2023/11/30

# Contents

|                                                                                          |    |
|------------------------------------------------------------------------------------------|----|
| Revision History .....                                                                   | 7  |
| 1 Introduction .....                                                                     | 9  |
| 1.1 Purpose/ Scope .....                                                                 | 9  |
| 1.2 References .....                                                                     | 9  |
| 1.2.1 ISO .....                                                                          | 9  |
| 1.2.2 AutoSAR .....                                                                      | 9  |
| 1.2.3 Chinese Standard .....                                                             | 10 |
| 1.2.4 SmartMi Global Standards .....                                                     | 10 |
| 1.2.5 Other National Standards .....                                                     | 10 |
| 1.2.6 Industry Documents .....                                                           | 10 |
| 1.3 Symbols and abbreviated terms .....                                                  | 10 |
| 2 Diagnostic OSI model .....                                                             | 12 |
| 3 Application protocol .....                                                             | 13 |
| 3.1 Diagnostic Addressing .....                                                          | 13 |
| 3.1.1 Physical and Functional .....                                                      | 13 |
| 3.1.2 Diagnostic clients address allocation .....                                        | 13 |
| 3.2 Server (ECU) response implementation rules .....                                     | 14 |
| 3.2.1 Sub Function .....                                                                 | 14 |
| 3.2.2 General server response behaviour .....                                            | 14 |
| 3.2.3 Response Behaviour with Sub-function .....                                         | 15 |
| 3.2.4 Request message without sub-function parameter and server response behaviour ..... | 19 |
| 3.2.5 Multiple concurrent request messages with physical and functional addressing ..... | 19 |
| 3.3 Diagnostics and Node Management .....                                                | 22 |
| 4 Diagnostic Services .....                                                              | 22 |
| 4.1 Required Diagnostic Services Overview .....                                          | 22 |
| 4.1.1 Support Services .....                                                             | 22 |
| 4.1.2 Services allowed .....                                                             | 23 |
| 4.2 Diagnostic Session Control (0x10) .....                                              | 26 |
| 4.2.1 Request message .....                                                              | 26 |
| 4.2.2 Positive response message .....                                                    | 27 |
| 4.2.3 Negative response message & Supported negative response codes (NRC_) .....         | 27 |
| 4.3 ECUReset (11 hex) service .....                                                      | 29 |
| 4.3.1 Request message .....                                                              | 29 |
| 4.3.2 Positive response message .....                                                    | 29 |
| 4.3.3 Negative response message & upported negative response codes (NRC_) .....          | 30 |
| 4.4 SecurityAccess (27 hex) service .....                                                | 31 |
| 4.4.1 Service description .....                                                          | 31 |
| 4.4.2 Security Access State Chart .....                                                  | 32 |
| 4.4.3 Request message .....                                                              | 35 |
| 4.4.4 Positive response message .....                                                    | 36 |
| 4.4.5 Negative response message & Supported negative response codes (NRC_) .....         | 37 |
| 4.5 CommunicationControl (28 hex) service .....                                          | 38 |
| 4.5.1 Request message .....                                                              | 38 |
| 4.5.2 Positive response message .....                                                    | 40 |
| 4.5.3 Negative response message & upported negative response codes (NRC_) .....          | 40 |
| 4.6 TesterPresent (3E hex) service .....                                                 | 41 |
| 4.6.1 Request message .....                                                              | 42 |

|                                                                                   |    |
|-----------------------------------------------------------------------------------|----|
| 4.6.2 Positive response message .....                                             | 42 |
| 4.6.3 Negative response message & Supported negative response codes (NRC_) .....  | 42 |
| 4.7 ReadDataByIdentifier (22 hex) service .....                                   | 43 |
| 4.7.1 Request message .....                                                       | 44 |
| 4.7.2 Positive response message .....                                             | 44 |
| 4.7.3 Negative response message & Supported negative response codes (NRC_) .....  | 45 |
| 4.8 WriteDataByIdentifier (2E hex) service .....                                  | 46 |
| 4.8.1 Request message .....                                                       | 46 |
| 4.8.2 Positive response message .....                                             | 47 |
| 4.8.3 Negative response message & Supported negative response codes (NRC_) .....  | 47 |
| 4.9 ControlDTCSetting (85 hex) service .....                                      | 48 |
| 4.9.1 Request message .....                                                       | 49 |
| 4.9.2 Positive response message .....                                             | 49 |
| 4.9.3 Negative response message & Supported negative response codes (NRC_) .....  | 49 |
| 4.10 ClearDiagnosticInformation (14 hex) service .....                            | 51 |
| 4.10.1 Request message .....                                                      | 51 |
| 4.10.2 Positive response message .....                                            | 51 |
| 4.10.3 Negative response message & Supported negative response codes (NRC_) ..... | 51 |
| 4.11 ReadDTCInformation (19 hex) service .....                                    | 53 |
| 4.11.1 Request message .....                                                      | 53 |
| 4.11.2 Positive response message .....                                            | 54 |
| 4.11.3 Negative response message & Supported negative response codes (NRC_) ..... | 58 |
| 4.12 InputOutputControlByIdentifier (2F hex) service .....                        | 59 |
| 4.12.1 Request message .....                                                      | 60 |
| 4.12.2 Positive response message .....                                            | 62 |
| 4.12.3 Negative response message & Supported negative response codes (NRC_) ..... | 62 |
| 4.13 RoutineControl (31 hex) service .....                                        | 64 |
| 4.13.1 Request message .....                                                      | 64 |
| 4.13.2 Positive response message .....                                            | 65 |
| 4.13.3 Negative response message & Supported negative response codes (NRC_) ..... | 66 |
| 4.14 RequestDownload (34 hex) service .....                                       | 67 |
| 4.14.1 Request message .....                                                      | 67 |
| 4.14.2 Positive response message .....                                            | 68 |
| 4.14.3 Negative response message & Supported negative response codes (NRC_) ..... | 69 |
| 4.15 RequestUpload (0x35) service .....                                           | 70 |
| 4.15.1 Request message .....                                                      | 70 |
| 4.15.2 Positive response message .....                                            | 71 |
| 4.15.3 Negative response message & Supported negative response codes (NRC_) ..... | 72 |
| 4.16 TransferData (36 hex) service .....                                          | 74 |
| 4.16.1 Request message .....                                                      | 74 |
| 4.16.2 Positive response message .....                                            | 75 |
| 4.16.3 Negative response message & Supported negative response codes (NRC_) ..... | 75 |
| 4.17 RequestTransferExit (37 hex) service .....                                   | 77 |
| 4.17.1 Request message .....                                                      | 77 |
| 4.17.2 Positive response message .....                                            | 77 |
| 4.17.3 Negative response message & Supported negative response codes (NRC_) ..... | 77 |
| 4.18 RequestFileTransfer (38 hex) service .....                                   | 78 |
| 4.18.1 Request message .....                                                      | 79 |
| 4.18.2 Positive response message .....                                            | 81 |
| 4.18.3 Negative response message & Supported negative response codes (NRC_) ..... | 82 |

|                                                                                   |                                                          |         |               |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|---------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |               |
|                                                                                   | V23                                                      | Rev 3.3 | Page 4 of 146 |

|        |                                                                               |     |
|--------|-------------------------------------------------------------------------------|-----|
| 5      | Diagnostic Error Storage .....                                                | 83  |
| 5.1    | Time Message .....                                                            | 83  |
| 5.2    | Diagnostic Fault Detection and DTC Clear/Setting Requirement .....            | 84  |
| 5.2.1  | Diagnostic Fault Detection .....                                              | 84  |
| 5.2.2  | Operation Cycle .....                                                         | 84  |
| 5.2.3  | Content of Error Storage .....                                                | 85  |
| 5.2.4  | DTC Primary Memory .....                                                      | 92  |
| 5.2.5  | DTC Active Message .....                                                      | 93  |
| 5.2.6  | DTC Active Message Transmit Requirement .....                                 | 94  |
| 5.2.7  | Clear DTC .....                                                               | 100 |
| 6      | The Vehicle Learn Processing .....                                            | 100 |
| 6.1    | Learn Processing Category .....                                               | 100 |
| 6.1.1  | Automatic Learns .....                                                        | 101 |
| 6.1.2  | Manual Learns .....                                                           | 101 |
| 6.2    | Learn Processing Requirements .....                                           | 101 |
| 6.2.1  | Learn Status .....                                                            | 101 |
| 6.3    | Learn with OTA .....                                                          | 102 |
| 7      | Appendix A-ECU Address .....                                                  | 102 |
| 8      | Appendix B-Corporate Standard DIDs .....                                      | 104 |
| 8.1    | Data Identifier (DID) Ranges .....                                            | 104 |
| 8.2    | Read and Write DIDs .....                                                     | 104 |
| 8.2.1  | DID 0xF085: Power Moding State .....                                          | 106 |
| 8.2.2  | DID 0xF087: Vehicle Mode Status .....                                         | 106 |
| 8.2.3  | DID 0xF0A9: XCP CANID Persistence flag .....                                  | 107 |
| 8.2.4  | DID 0xF0B3: Supplier component Traceability Characters .....                  | 107 |
| 8.2.5  | DID 0xF0B4: Manufacturing Traceability Characters .....                       | 108 |
| 8.2.6  | DID 0xD004: Vehicle Odometer .....                                            | 108 |
| 8.2.7  | DID 0xF0F0: Programmed State .....                                            | 109 |
| 8.2.8  | DID 0xF0F4: Safe Bypass .....                                                 | 109 |
| 8.2.9  | DID 0xF0FE: Total Vehicle Operation Time .....                                | 110 |
| 8.2.10 | DID 0xF186: Active Diagnostic Session Data Identifier .....                   | 110 |
| 8.2.11 | DID 0xF187: VehicleManufacturerSparePart Number .....                         | 111 |
| 8.2.12 | DID 0xF189: Vehicle Manufacturer ECU SW Part Number and Version Number .....  | 112 |
| 8.2.13 | DID 0xF1C1: Vehicle Manufacturer Boot SW Part Number and Version Number ..... | 113 |
| 8.2.14 | DID 0xF1C2: Vehicle Manufacturer APP SW Part Number and Version Number .....  | 114 |
| 8.2.15 | DID 0xF1C3: Vehicle Manufacturer Cal SW Part Number and Version Number .....  | 115 |
| 8.2.16 | DID 0xF18A: System Supplier Identifier .....                                  | 115 |
| 8.2.17 | DID 0xF18B: ECU Manufacturing Date .....                                      | 116 |
| 8.2.18 | DID 0xF18C: ECU Serial Number .....                                           | 116 |
| 8.2.19 | DID 0xF190: Vehicle Identification Number .....                               | 116 |
| 8.2.20 | DID 0xF193: System Supplier ECU Hardware Version Number .....                 | 117 |
| 8.2.21 | DID 0xF195: System Supplier ECU Software Version Number .....                 | 117 |
| 8.2.22 | DID 0xF197: ECU Name .....                                                    | 117 |
| 8.2.23 | DID 0xF198: Repair Shop Code Or Tester Serial Number .....                    | 117 |
| 8.2.24 | DID 0xF199: Programming Date .....                                            | 118 |
| 8.2.25 | DID 0xF194: System Supplier ECU Software Part Number .....                    | 118 |
| 8.2.26 | DID 0xF19E: ECU Diagnostic Database Version .....                             | 118 |
| 8.2.27 | DID 0xF1A2: ECU Matrix Version of the ECU product .....                       | 119 |
| 8.2.28 | DID 0xF1B0: ECU Address .....                                                 | 119 |
| 8.2.29 | DID 0xF1B1: ECU PRID .....                                                    | 120 |

|                                                                                   |                                                          |         |               |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|---------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |               |
|                                                                                   | V23                                                      | Rev 3.3 | Page 5 of 146 |

|        |                                                                                                                      |     |
|--------|----------------------------------------------------------------------------------------------------------------------|-----|
| 8.2.30 | DID 0xF1B2: Log File Information .....                                                                               | 120 |
| 8.2.31 | 0xF1E5: Link Status .....                                                                                            | 120 |
| 8.2.32 | 0xF1E6: 100BASE-T1 SQI threshold .....                                                                               | 120 |
| 8.2.33 | 0xF1E7: 1000BASE-T1 SQI threshold .....                                                                              | 121 |
| 8.2.34 | 0xF1E8: SQI .....                                                                                                    | 121 |
| 8.2.35 | 0xF1E9: Polarity detection .....                                                                                     | 121 |
| 8.2.36 | 0xF1EB: Switch counter .....                                                                                         | 121 |
| 8.2.37 | 0xF1EC: MAC address .....                                                                                            | 121 |
| 8.2.38 | 0xF1D3: Signal lines short/open circuits Number .....                                                                | 121 |
| 8.2.39 | 0xF1D5: Signal lines current status .....                                                                            | 121 |
| 8.2.40 | 0xF1D7: MASTER/SLAVE status .....                                                                                    | 121 |
| 8.2.41 | 0xF1D8: Test mode status .....                                                                                       | 121 |
| 8.2.42 | 0xF1D9: Switch ARL table .....                                                                                       | 122 |
| 8.2.43 | 0xF1DA: Ethernet Package Error Rate .....                                                                            | 122 |
| 8.2.44 | 0xF1DB: IP Address .....                                                                                             | 122 |
| 8.2.45 | 0xF100: Vehicle Configuration .....                                                                                  | 122 |
| 8.2.46 | 0xF101: ECU Configuration .....                                                                                      | 122 |
| 8.3    | Routine Identification .....                                                                                         | 122 |
| 8.3.1  | RID 0x0202: Check Memory Routine .....                                                                               | 123 |
| 8.3.2  | RID 0x0203: Check Programming Precondition .....                                                                     | 124 |
| 8.3.3  | RID 0xFF00: Erase Memory .....                                                                                       | 124 |
| 8.3.4  | RID 0xFF01: Check Programming Dependencies .....                                                                     | 125 |
| 8.3.5  | RID 0x0204: Disable Programming Prohibition Rollback function .....                                                  | 126 |
| 8.3.6  | RID 0x0205: Upgrade Package SHA256 Check .....                                                                       | 127 |
| 8.3.7  | RID 0x0206: Check Programming Integrity-SHA256 .....                                                                 | 127 |
| 8.3.8  | RID 0x0207: Upgrade Package Installation .....                                                                       | 127 |
| 8.3.9  | RID 0x0208: Program Rollback .....                                                                                   | 127 |
| 8.3.10 | RID 0x0400: SecurityKey Generation .....                                                                             | 127 |
| 8.3.11 | 0xA1A1: Test Mode Active .....                                                                                       | 128 |
| 8.3.12 | 0xA1A2: Cable Diagnostic Active .....                                                                                | 128 |
| 8.3.13 | 0xA1A3: Reset Switch Counter .....                                                                                   | 128 |
| 8.3.14 | 0xA1AF: Ethernet Message Mirror .....                                                                                | 128 |
| 9      | Appendix C-Failure Type .....                                                                                        | 129 |
| 9.1    | Definition and Appropriate Uses Of DTC Failure Type Value \$00 .....                                                 | 129 |
| 9.2    | DTC Failure Type Definition by Category .....                                                                        | 130 |
| 9.2.1  | DTC Failure Sub Type (Symptom) Definition of General Failure Information .....                                       | 130 |
| 9.2.2  | DTC Failure Sub Type (Symptom) Definition of General Electrical Failures .....                                       | 131 |
| 9.2.3  | DTC Failure Sub Type (Symptom) Definition of General Signal Failures .....                                           | 132 |
| 9.2.4  | DTC Failure Sub Type (Symptom) Definition of FM (Frequency Modulation) / PWM (Pulse Width Modulation) Failures ..... | 133 |
| 9.2.5  | DTC Failure Sub Type (Symptom) Definition of System Internal Failures .....                                          | 134 |
| 9.2.6  | DTC Failure Sub Type (Symptom) Definition of System Programming Failures .....                                       | 135 |
| 9.2.7  | DTC Failure Sub Type (Symptom) Definition of Algorithm Based Failures .....                                          | 136 |
| 9.2.8  | DTC Failure Sub Type (Symptom) Definition of Mechanical Failures .....                                               | 137 |
| 9.2.9  | DTC Failure Sub Type (Symptom) Definition of Bus Signal Failures .....                                               | 138 |
| 10     | Appendix D-Corporate Standard DTC .....                                                                              | 138 |
| 10.1   | ECU Communication CAN Bus Off .....                                                                                  | 138 |
| 10.2   | Lost Communication with CAN ECU .....                                                                                | 139 |
| 10.3   | Invalid Data Received From CAN ECU .....                                                                             | 140 |
| 10.4   | No Communication on LIN Bus .....                                                                                    | 141 |

|       |                                                                  |     |
|-------|------------------------------------------------------------------|-----|
| 10.5  | Lost Communication with LIN Slave ECU .....                      | 141 |
| 10.6  | Loss of Link on Ethernet Port X .....                            | 142 |
| 10.7  | Invalid Data Received from Ethernet ECU (Checksum Failure) ..... | 143 |
| 10.8  | Lost Communication with Ethernet ECU .....                       | 143 |
| 10.9  | ECU Power Under Voltage DTC .....                                | 144 |
| 10.10 | ECU Power Over Voltage DTC .....                                 | 144 |
| 10.11 | Some ECU Hardware DTCs .....                                     | 145 |
| 10.12 | Some ECU Software DTCs .....                                     | 146 |

## Revision History

| Revision | Date       | Author | Remark                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0      | 2022/02/22 | ZH     | Initial version.                                                                                                                                                                                                                                                                                                                                                                                          |
| 2.0      | 2022/11/30 | ZH     | Add the Corporate DIDs and DTCs                                                                                                                                                                                                                                                                                                                                                                           |
|          |            | ZH     | Add ISP ECU address<br>Delete the DoIP logical Address Tester F1                                                                                                                                                                                                                                                                                                                                          |
| 3.0      | 2023/03/17 | CLF    | Fix some formatting errors<br>Add the description of 0x28 service 0x01 subfunction<br>Add 38 service description<br>Add description of how to use controlmask in 2F service<br>Add DTC extended data 04-DTC first occurrence time<br>Correct some DID Description errors                                                                                                                                  |
|          | 2023/03/31 | CLF    | Add DID F1B1 (PRID)                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.1      | 2023/08/10 | CLF    | Add DID F1B2 (Log File Information)<br>Add RID FF03 (Upgrade Package SHA256 Check)<br>Add RID FF04 (Check Programming Integrity-SHA256)<br>Add RID FF05 (Upgrade Package Installation)<br>Add RID FF06 (Program Rollback)<br>Add reference information<br>Change DID F198/F199 writable in programming session                                                                                            |
| 3.2      | 2023/10/23 | CLF    | Update DID support list<br>Update RID support list<br>Add RID 0400 (SecurityKey Generation)<br>Change RID FF02 to 0204<br>Change RID FF03 to 0205<br>Change RID FF04 to 0206<br>Change RID FF05 to 0207<br>Change RID FF06 to 0208<br>Update DTC range of “Loss of Link on Ethernet Port X”<br>Delete descriptions of DTC in SAE_J2012-DA_DTC Format<br>Update DTC range of “No Communication on LIN Bus” |

|     |            |     |                                                                                                                                          |
|-----|------------|-----|------------------------------------------------------------------------------------------------------------------------------------------|
|     |            |     | Update DTC range of “Lost Communication with Ethernet ECU”<br>Change parameter “DTCFormatIdentifier” in 19 service to “ISO 14229 Format” |
| 3.3 | 2023/11/30 | CLF | The year of publication of the referenced standard is removed<br>suppressPosRspMsgIndicationBit is not supported for service 0x27        |

# 1 Introduction

## 1.1 Purpose/ Scope

This document is intended for project managers, design developers, testers, quality assurance personnel, and document writers. This document provides complete, correct, feasible and necessary technical specifications for network management, provides guidance for project management, provides a basis for design and development, and provides a benchmark for testing.

## 1.2 References

### 1.2.1 ISO

| No. | Document                                                                                                                                                        | Description |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1   | ISO 14229-1 Road vehicles — Unified diagnostic services (UDS) Part 1: Specification and requirements                                                            | ISO 14229-1 |
| 2   | ISO 14229-2 Road vehicles — Unified diagnostic services (UDS) Part 2: Session layer services                                                                    | ISO 14229-2 |
| 3   | ISO 14229-3 Road vehicles — Unified diagnostic services (UDS) Part 3: Unified diagnostic services on CAN implementation (UDSonCAN)                              | ISO 14229-3 |
| 4   | ISO 15765 Road vehicles — Diagnostic communication over Controller Area Network (DoCAN)                                                                         | ISO 15765   |
| 5   | ISO 15031-6 Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics —Part 6: Diagnostic trouble code definitions | ISO 15031-6 |

### 1.2.2 AutoSAR

| No. | Document                                   | Description |
|-----|--------------------------------------------|-------------|
| 1   | AUTOSAR_SWS_DiagnosticCommunicationManager |             |
| 2   | AUTOSAR_SWS_DiagnosticEventManager         |             |
| 3   | AUTOSAR_SWS_CANTransceiverDriver           |             |
| 4   | AUTOSAR_SWS_CANTransportLayer              |             |
| 5   | AUTOSAR_SWS_COM                            |             |
| 6   | AUTOSAR_SWS_COMManager                     |             |
| 7   | AUTOSAR_SWS_NetworkManagementInterface     |             |
| 8   | AUTOSAR_SWS_PDURouter                      |             |

### 1.2.3 Chinese Standard

N/A

### 1.2.4 SmartMi Global Standards

| No. | Document                   | Description |
|-----|----------------------------|-------------|
| 1   | SMG-EEA-1002-网络管理需求规范      |             |
| 2   | SMG-EEA-0010-软硬件版本规则       |             |
| 3   | SMG-EEA-1001-CAN (FD) 需求规范 |             |
| 4   | SMG-EEA-3001-Ethernet需求规范  |             |
| 5   | SMG-EEA-4002-UDSonCAN需求规范  |             |
| 6   | SMG-EEA-4003-UDSonLIN需求规范  |             |
| 7   | SMG-EEA-4004-DOIP需求规范      |             |
| 8   | SMG-EEA-4005-远程诊断需求规范      |             |
| 9   | SMG-EEA-4021-安全诊断需求规范      |             |
| 10  | SMG-EEA-6001-OTA需求规范       |             |
| 11  | QSMG-EEA-4012-文件系统刷写规范     |             |

### 1.2.5 Other National Standards

N/A

### 1.2.6 Industry Documents

N/A

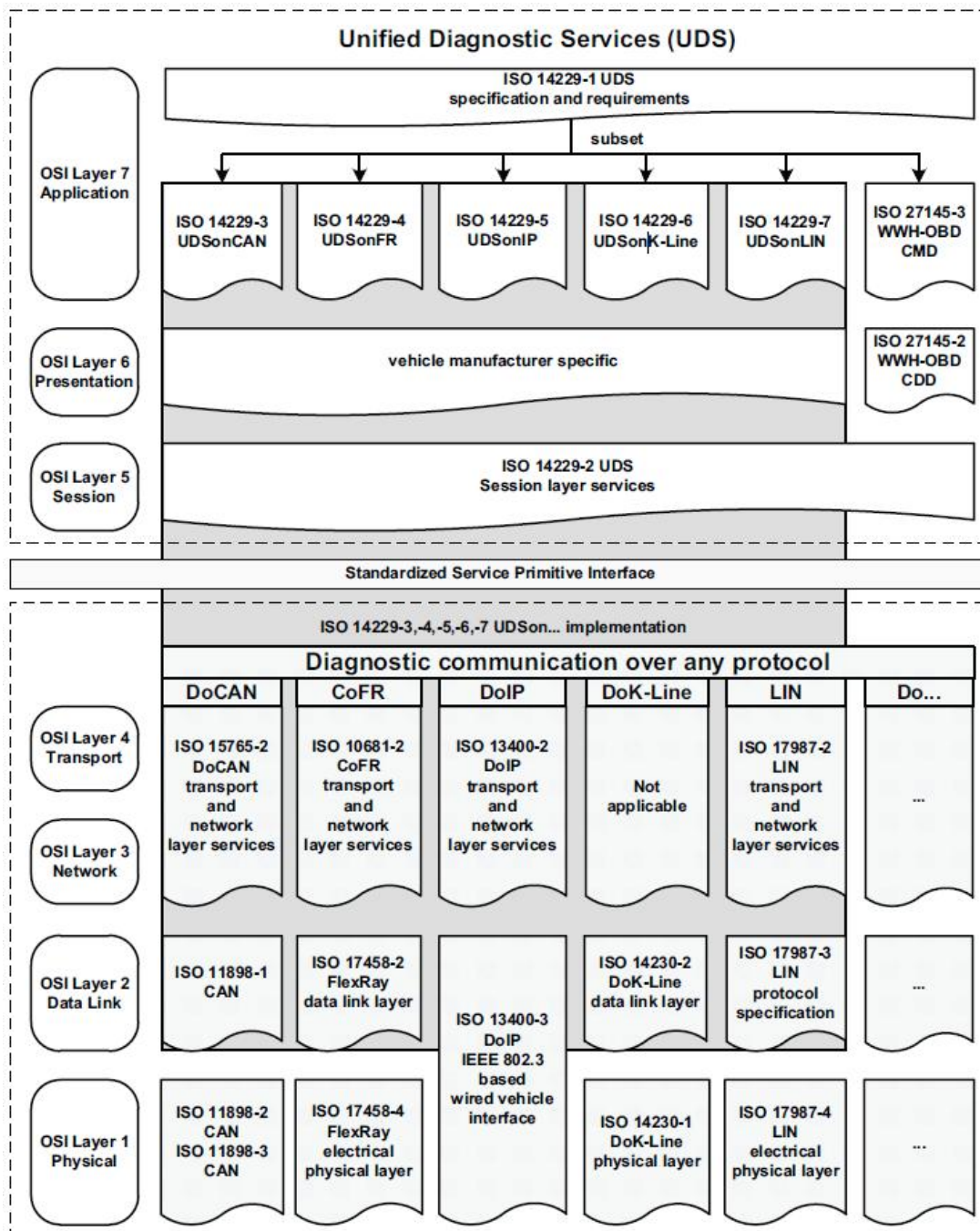
## 1.3 Symbols and abbreviated terms

| Abbreviated | Description                         |
|-------------|-------------------------------------|
| ACK         | Acknowledge                         |
| AUTOSAR     | Automotive Open System Architecture |
| BS          | BlockSize                           |
| CAN         | Controller Area Network             |
| CF          | Consecutive Frame                   |
| DL          | Data Length                         |
| DRE         | Design Release Engineer             |

|                                |                                                                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| ECU                            | Electronic Control Unit                                                                                                                          |
| FC                             | Flow Control                                                                                                                                     |
| FF                             | First Frame                                                                                                                                      |
| FF_DL                          | First Frame Data Length                                                                                                                          |
| FS                             | Flow Status                                                                                                                                      |
| N_USData                       | Network layer unacknowledged segmented data transfer service name                                                                                |
| PCI                            | Protocol Control Information                                                                                                                     |
| SF                             | Single Frame                                                                                                                                     |
| SN                             | Sequence Number                                                                                                                                  |
| SSTS                           | Subsystem Technical Specification                                                                                                                |
| STmin                          | Separation Time minimum                                                                                                                          |
| UDS                            | Unified Diagnostic Services                                                                                                                      |
| suppressPosRspMsgIndicationBit | TRUE = server shall NOT send a positive response message<br>FALSE = server shall send a positive or negative response message                    |
| USDT                           | Unacknowledged Segmented Data Transfer – used to signify messages which may be segmented into multiple frames due to data length of the message. |
| UUDT                           | Unacknowledged Unsegmented Data Transfer – used to signify messages which are single frame ONLY.                                                 |

## 2 Diagnostic OSI model

The following figure depicts the implementation of UDS reference according to OSI model.



## 3 Application protocol

The Application Layer Protocol shall follow the requirements stated in ISO14229-1. In addition to, the following implementation specific requirements shall be considered.

### 3.1 Diagnostic Addressing

#### 3.1.1 Physical and Functional

The diagnostic message Id in a request message also determines the type of addressing used for that message, either physically addressed (targeted to a single server) or functionally addressed (targeted to one or multiple servers).

- Physical addressing (1 to 1 communication) shall be supported for all types of network layer messages.
- Functional addressing (1 to n communication) shall only be supported for SingleFrame communication.

All diagnostic response messages will be physically addressed.

For a physically addressed request message the ‘suppressPosRspMsgIndicationBit’ shall always be set to ‘FALSE’ (response required).

For a functionally addressed request message the ‘suppressPosRspMsgIndicationBit’ shall follow the individual service description.

#### 3.1.2 Diagnostic clients address allocation

Each type of client address consists of one byte.

The diagnostic address consists of the client address and the ECU source address. Refer to < SMG-EEA-4002-UDSonCAN 需求规范> and < SMG-EEA-4004-DOIP 需求规范> .

The client addresses are used according to the following table.

| Client ID                                                                                                                      | Use case                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xF1                                                                                                                           | This client ID is used by CAN testers(UDSonCAN) that send diagnostic messages through the DLC interface. For example, factory or aftermarket equipment. |
| 0xF2                                                                                                                           | This client ID is used by DoIP testers(DoIP) that send diagnostic messages through the DLC interface. For example, factory or aftermarket equipment.    |
| 0xF4                                                                                                                           | On board testers. Using for OTA. The OTA master(DoIP) send diagnostic messages shall use this client ID                                                 |
| 0xF5                                                                                                                           | On board testers.                                                                                                                                       |
| 0xF6                                                                                                                           | On board testers. Using for RemoteDiagnosticSystem(RDS). The RDS master(DoIP) send diagnostic messages shall use this client ID                         |
| Note1: All ECUs shall be capable of establishing diagnostic communication with clients having diagnostic addresses 0xF1, 0xF2, |                                                                                                                                                         |

0xF4, 0xF5, 0xF6.

Note2: All ECUs shall support the all-nodes functional diagnostic address, 0xFE.

Each ECU has an ECU address that can be used to construct diagnostic addresses, NM addresses, and DoIP logical addresses. Refer to Appendix A for more details.

## 3.2 Server (ECU) response implementation rules

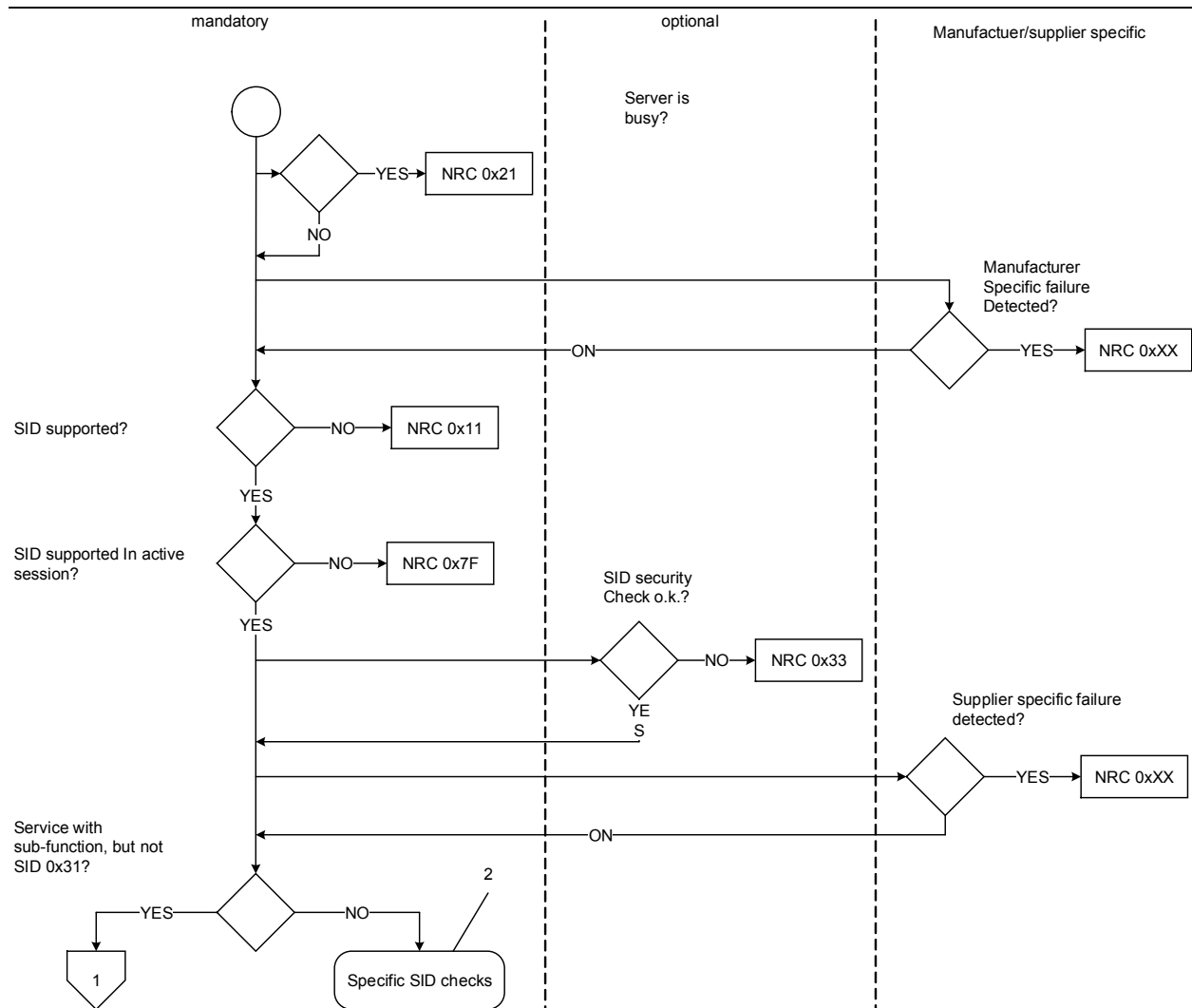
### 3.2.1 Sub Function

For the services which support sub function, the sub function byte is divided into 2 part as definition in below table.

| Bit position | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7            | <p><b>suppressPosRspMsgIndicationBit</b></p> <p>This bit indicates if a positive response message shall be suppressed by the server.</p> <p>‘0’ = FALSE, do not suppress a positive response message (a positive response message is required).</p> <p>‘1’ = TRUE, suppress response message (a positive response message shall not be sent; the server being addressed shall not send a positive response message).</p> <p>Independent of the suppressPosRspMsgIndicationBit, negative response messages are sent by the server(s) according to the restrictions specified in 7.5 of Reference[5].</p> <p>Even if a positive response is not required (i.e., SPRMIB = true), the execution of the service must be completely passed to keep the implementation consistent regardless of SPRMIB value.</p> <p>SuppressPosRspMsgIndicationBit values of both ‘0’ and ‘1’ shall be supported for all sub-function values (i.e., bits 6-0 of the sub-function structure) supported by the server for any given service.</p> |
| 6-0          | <p><b>sub-function parameter value</b></p> <p>The bits 0-6 of the sub-function parameter contain the sub-function parameter value of the service (0x00 – 0x7F).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### 3.2.2 General server response behaviour

The following Figure depicts the general server response behaviour.

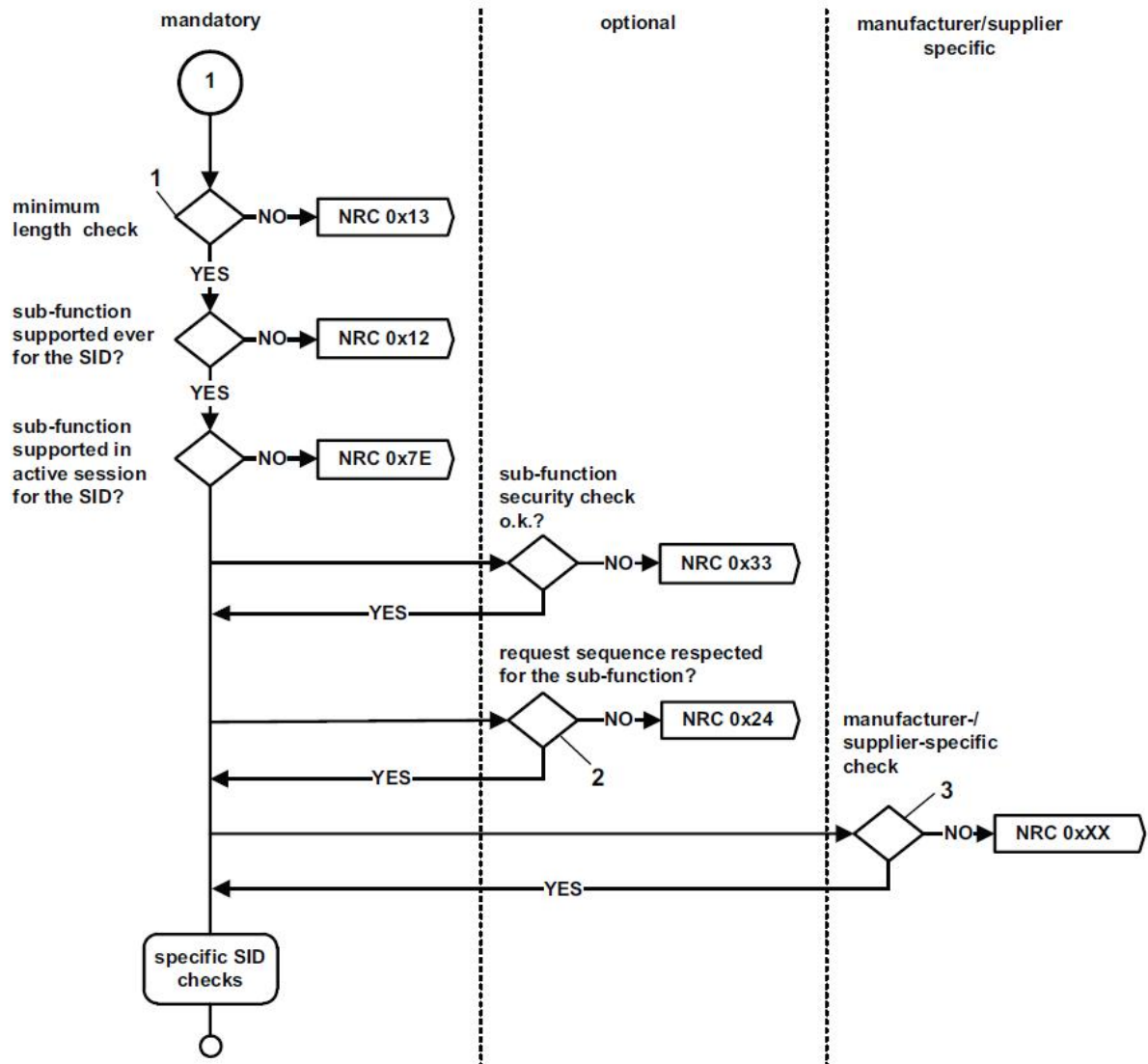


Note1: Diagnostic request can not be accepted because another diagnostic task is already requested and in progress by a different client.

Note2 : refer to the response behaviour (supported negative response codes) of each service.

### 3.2.3 Response Behaviour with Sub-function

The following figure depicts the general server response behavior for request message with sub-function parameter.



Note1: at least 2 (SID+SubFunction Parameter).

Note2: if sub-function is subject to sequence check, e.g. LinkControl or SecurityAccess.

Note3: refer to the response behaviour (supported negative response codes) of each service.

### 3.2.3.1 communication schemes with physical addressing

| Server case # | Client request message |                                                  | Server capability |                       |                                               | Server response |                      | Comments to server response                                                                              |
|---------------|------------------------|--------------------------------------------------|-------------------|-----------------------|-----------------------------------------------|-----------------|----------------------|----------------------------------------------------------------------------------------------------------|
|               | Addressing scheme      | sub-function (suppress-PosRspMsg-Indication-Bit) | SI supported      | SubFunction supported | Data parameter supported (only if applicable) | Message         | NRC                  |                                                                                                          |
| a)            | physical               | FALSE (bit = 0)                                  | YES               | YES                   | At least 1                                    | PosRsp          | ---                  | Server sends positive response                                                                           |
| b)            |                        |                                                  |                   |                       | At least 1                                    | NegRsp          | NRC= 0xXX            | Server sends negative response because error occurred reading the data-parameters of the request message |
| c)            |                        |                                                  |                   |                       | None                                          |                 | NRC= ROOR            | Negative response with NRC 0x31                                                                          |
| d)            |                        |                                                  | NO                | --                    | --                                            |                 | NRC= SNS or SNSIAS   | Negative response with NRC 0x11 or NRC 0x7F                                                              |
| e)            |                        |                                                  | YES               | NO                    | --                                            |                 | NRC= SFNS or SFNSIAS | Negative response with NRC 0x12 or NRC 0x7E                                                              |
| f)            |                        | TRUE (bit = 1)                                   | YES               | YES                   | At least 1                                    | NoRsp           | ---                  | Server does NOT send a response                                                                          |
| g)            |                        |                                                  |                   |                       | At least 1                                    | NegRsp          | NRC= 0xXX            | Server sends negative response because error occurred reading the data-parameters of the request message |
| h)            |                        |                                                  |                   |                       | None                                          |                 | NRC= ROOR            | Negative response with NRC 0x31                                                                          |
| i)            |                        |                                                  | NO                | --                    | --                                            |                 | NRC= SNS             | Negative response with NRC 0x11                                                                          |
| j)            |                        |                                                  | YES               | NO                    | --                                            |                 | NRC= SFNS            | Negative response with NRC 0x12                                                                          |

Refer to ISO 14229-1 for detail.

### 3.2.3.2 communication schemes with functional addressing

| Server case # | Client request message |                                                | Server capability |                        |                                               | Server response |                                 | Comments to server response                                                                              |
|---------------|------------------------|------------------------------------------------|-------------------|------------------------|-----------------------------------------------|-----------------|---------------------------------|----------------------------------------------------------------------------------------------------------|
|               | Addressing scheme      | sub-function (suppressPosRspMsg-IndicationBit) | SI supported      | Sub-Function supported | Data parameter supported (only if applicable) | Message         | NRC                             |                                                                                                          |
| a)            | functional             | FALSE (bit = 0)                                | YES               | YES                    | At least 1                                    | PosRsp          | ---                             | Server sends positive response                                                                           |
| b)            |                        |                                                |                   |                        | At least 1                                    | NegRsp          | NRC= 0xXX                       | Server sends negative response because error occurred reading the data-parameters of the request message |
| c)            |                        |                                                |                   |                        |                                               |                 |                                 |                                                                                                          |
| d)            |                        |                                                | NO                | --                     | --                                            | ---             | Server does NOT send a response |                                                                                                          |
| e)            |                        |                                                | YES               | NO                     | --                                            | ---             | Server does NOT send a response |                                                                                                          |
| f)            |                        | TRUE (bit = 1)                                 | YES               | YES                    | At least 1                                    | NoRsp           | ---                             | Server does NOT send a response                                                                          |
| g)            |                        |                                                |                   |                        | At least 1                                    | NegRsp          | NRC= 0xXX                       | Server sends negative response because error occurred reading the data-parameters of the request message |
| h)            |                        |                                                |                   |                        |                                               |                 |                                 |                                                                                                          |
| i)            |                        |                                                | NO                | --                     | --                                            | NoRsp           | ---                             | Server does NOT send a response                                                                          |
| j)            |                        |                                                | YES               | NO                     | --                                            |                 | ---                             | Server does NOT send a response                                                                          |

Refer to ISO 14229-1 for detail.

### 3.2.4 Request message without sub-function parameter and server response behaviour

#### 3.2.4.1 communication schemes with physical addressing

| Server case # | Client request message | Server capability |                     | Server response |                   | Comments to server response                                                                      |
|---------------|------------------------|-------------------|---------------------|-----------------|-------------------|--------------------------------------------------------------------------------------------------|
|               | Addressing scheme      | SI supported      | Parameter supported | Message         | NRC               |                                                                                                  |
| a)            | physical               | YES               | ALL                 | PosRsp          | ---               | Server sends positive response                                                                   |
| b)            |                        |                   | At least 1          |                 | ---               | Server sends positive response                                                                   |
| c)            |                        |                   | At least 1          | NegRsp          | NRC=0xXX<br>X     | Server sends negative response because error occurred reading data-parameters of request message |
| d)            |                        |                   | NONE                |                 | NRC=ROOR          | Negative response with NRC 0x31                                                                  |
| e)            |                        | NO                | ---                 |                 | NRC=SNS or SNSIAS | Negative response with NRC 0x11 or NRC 0x7F                                                      |

Refer to ISO 14229-1 for detail.

#### 3.2.4.2 communication schemes with functional addressing

| Server case # | Client request message | Server capability |                     | Server response |          | Comments to server response                                                                      |
|---------------|------------------------|-------------------|---------------------|-----------------|----------|--------------------------------------------------------------------------------------------------|
|               | Addressing scheme      | SI supported      | Parameter supported | Message         | NRC      |                                                                                                  |
| a)            | functional             | YES               | ALL                 | PosRsp          | ---      | Server sends positive response                                                                   |
| b)            |                        |                   | At least 1          |                 | ---      | Server sends positive response                                                                   |
| c)            |                        |                   | At least 1          | NegRsp          | NRC=0xXX | Server sends negative response because error occurred reading data-parameters of request message |
| d)            |                        |                   | NONE                | NoRsp           | ---      | Server does NOT send a response                                                                  |
| e)            |                        | NO                | ---                 |                 | ---      | Server does NOT send a response                                                                  |

Refer to ISO 14229-1 for detail.

### 3.2.5 Multiple concurrent request messages with physical and functional addressing

A common server implementation has only one diagnostic protocol instance available in the server. One

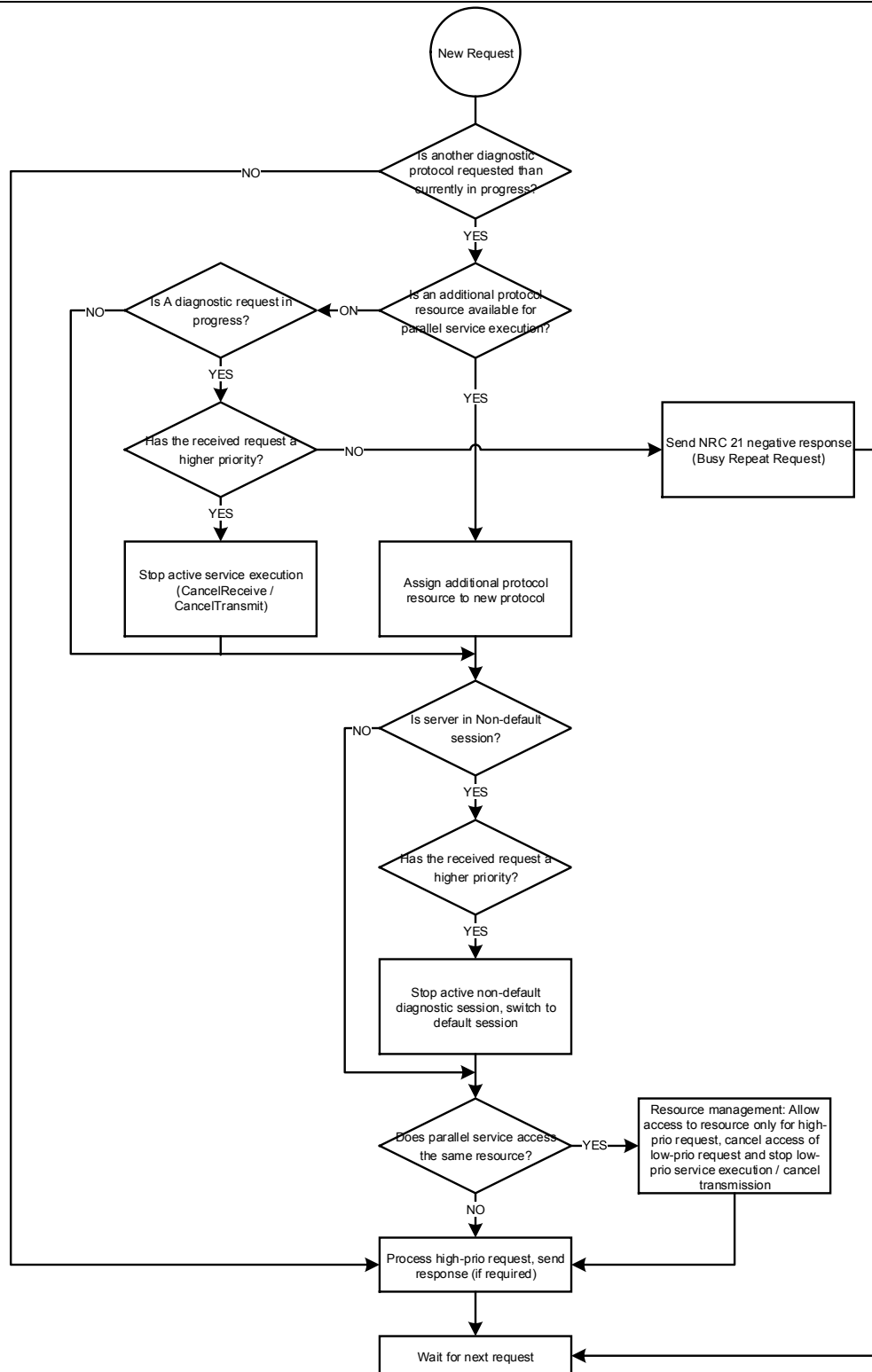
|                                                                                   |                                                          |         |                |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|----------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |                |
|                                                                                   | V23                                                      | Rev 3.3 | Page 20 of 146 |

diagnostic protocol instance can only handle one request at a time. The rule is that any received message (regardless of addressing mode physical or functional) occupies this resource until the request message is processed (with final response sent or application call without response).

There are only two exceptions which have to be treated separately:

- The keep-alive logic is used by a client to keep a previously enabled session active in one or multiple servers. Keep-Alive-Logic is defined as the functionally addressed valid TesterPresent message with SPRMIB=true(3E 80) and has to be processed by bypass logic. It is up to the server to make sure that this specific message cannot “block” the server’s application layer and that an immediately following addressed message can be processed.
- If a server supports one or more legislated diagnostic requests(e.g.,OBD diagnostic) and one of these requests is received while a non-legislated service (e.g., enhanced diagnostics) is active, then the active service shall be aborted, the default session shall be started and the legislated diagnostic service shall be processed. This requirement does not apply if the programming session is active.

If a server implements multiple client handling on server level the implementation shall adhere to the flow chart depicted in following figure.



Legistated Service Request from OBD client shall have highest priority.

An ECU shall process a legislated service request only when ECU is in “non-OFF” power mode.

The Client priority show in following table.

|                  |                                                                                         |    |    |    |    |
|------------------|-----------------------------------------------------------------------------------------|----|----|----|----|
| Priority         | High  |    |    |    |    |
| Client addresses | F1                                                                                      | F2 | F4 | F6 | F5 |

Note: Other Client addresses shall not interrupt the OTA Client, When the OTA is ongoing (Entered the OTA mode). The gateway and OTA master should have the ability to handle this case.

### 3.3 Diagnostics and Node Management.

The following sections describe the interaction between diagnostic services and Node Management.

In the default session, every diagnostic request message including functional request message(except . 02 3E 80) shall keep nodes awake for diagnostic Communication.

The Non default session state shall keep nodes awake at all time for diagnostic Communication.

## 4 Diagnostic Services

The ISO 14229-1 shall be replayed.

### 4.1 Required Diagnostic Services Overview

#### 4.1.1 Support Services

All supported services can be found in following table.

| UDS Diagnostic Services ID | sub-function | suppressPosRsp<br>MsgIndicationBit =<br>TRUE (No response)<br>supported | Request Message  |                    | Application Software | Bootloader Software |
|----------------------------|--------------|-------------------------------------------------------------------------|------------------|--------------------|----------------------|---------------------|
|                            |              |                                                                         | Physical Request | Functional Request |                      |                     |
| 0x10                       | 01           | Y                                                                       | Y                | Y                  | M                    | M                   |
|                            | 02           | Y                                                                       | Y                | N                  | M                    | M                   |
|                            | 03           | Y                                                                       | Y                | Y                  | M                    | M                   |
|                            | 60           | Y                                                                       | Y                | Y                  | U                    | N/A                 |
| 0x11                       | 01           | Y                                                                       | Y                | Y                  | M                    | M                   |
| 0x27                       | 01           | N                                                                       | Y                | Y                  | M                    | N/A                 |
|                            | 02           | N                                                                       | Y                | N                  | M                    | N/A                 |
|                            | 03           | N                                                                       | Y                | Y                  | M                    | N/A                 |
|                            | 04           | N                                                                       | Y                | N                  | M                    | N/A                 |
|                            | 05           | N                                                                       | Y                | Y                  | M                    | M                   |
|                            | 06           | N                                                                       | Y                | N                  | M                    | M                   |
|                            | 09           | N                                                                       | Y                | Y                  | M                    | M                   |
|                            | 0A           | N                                                                       | Y                | N                  | M                    | M                   |

|      |    |   |   |   |     |     |
|------|----|---|---|---|-----|-----|
|      | 61 | N | Y | Y | U   | N/A |
|      | 62 | N | Y | N | U   | N/A |
| 0x28 | 00 | Y | Y | Y | M   | M   |
|      | 01 | Y | Y | Y | U   | U   |
|      | 03 | Y | Y | Y | M   | M   |
| 0x3E | 00 | Y | Y | Y | M   | M   |
| 0x22 | -- | N | Y | Y | M   | M   |
| 0x2E | -- | N | Y | Y | M   | N/A |
| 0x85 | 01 | Y | Y | Y | M   | M   |
|      | 02 | Y | Y | Y | M   | M   |
| 0x14 | -- | N | Y | Y | M   | N/A |
| 0x19 | 01 | Y | Y | Y | M   | N/A |
|      | 02 | Y | Y | Y | M   | N/A |
|      | 03 | Y | Y | Y | M   | N/A |
|      | 04 | Y | Y | Y | M   | N/A |
|      | 06 | Y | Y | Y | M   | N/A |
|      | 0A | Y | Y | Y | M   | N/A |
| 0x2F | -- | N | Y | N | C1  | N/A |
| 0x31 | 01 | Y | Y | Y | M   | M   |
|      | 02 | Y | Y | Y | U   | U   |
|      | 03 | Y | Y | Y | U   | U   |
| 0x34 | -- | N | Y | N | N/A | M   |
| 0x35 | -- | N | Y | N | U   | U   |
| 0x36 | -- | N | Y | N | N/A | M   |
| 0x37 | -- | N | Y | N | N/A | M   |
| 0x38 | -- | N | Y | N | U   | U   |

C1 = this service shall be supported if the ECUs need to support relatively simple (e.g., static) input substitution/output control.

- **M** mandatory, shall always be present.
- **U** user optional; the parameter shall or shall not be supplied.
- **C** conditional; the presence of the parameter depends upon other parameters within the service

#### 4.1.2 Services allowed

All supported services in each session and Security Access for Diagnostic can be found in following table.

| UDS Diagnostic Services ID | sub-function | Request Message |                             | Security Access requirement |
|----------------------------|--------------|-----------------|-----------------------------|-----------------------------|
|                            |              | Default Session | Extended Diagnostic Session |                             |
| 0x10                       | 01           | Y               | Y                           | N                           |
|                            | 02           | N               | Y                           | Y                           |
|                            | 03           | Y               | Y                           | N                           |
|                            | 60           | Y               | Y                           | N                           |

|      |    |                   |   |                   |
|------|----|-------------------|---|-------------------|
| 0x11 | 01 | N                 | Y | Y                 |
| 0x27 | 01 | N                 | Y | N                 |
|      | 02 | N                 | Y | N                 |
|      | 03 | N                 | Y | N                 |
|      | 04 | N                 | Y | N                 |
|      | 05 | N                 | Y | N                 |
|      | 06 | N                 | Y | N                 |
|      | 09 | N                 | Y | N                 |
|      | 0A | N                 | Y | N                 |
|      | 61 | N                 | Y | N                 |
|      | 62 | N                 | Y | N                 |
| 0x28 | 00 | N                 | Y | N                 |
|      | 01 | N                 | Y | N                 |
|      | 03 | N                 | Y | N                 |
| 0x3E | 00 | Y                 | Y | N                 |
| 0x22 | -- | Note <sup>1</sup> | Y | Note <sup>1</sup> |
| 0x2E | -- | Note <sup>1</sup> | Y | Note <sup>1</sup> |
| 0x85 | 01 | N                 | Y | N                 |
|      | 02 | N                 | Y | N                 |
| 0x14 | -- | Y                 | Y | N                 |
| 0x19 | 01 | Y                 | Y | N                 |
|      | 02 | Y                 | Y | N                 |
|      | 03 | Y                 | Y | N                 |
|      | 04 | Y                 | Y | N                 |
|      | 06 | Y                 | Y | N                 |
|      | 0A | Y                 | Y | N                 |
| 0x2F | -- | N                 | Y | Note <sup>2</sup> |
| 0x31 | 01 | Note <sup>3</sup> | Y | Note <sup>3</sup> |
|      | 02 | Note <sup>3</sup> | Y | Note <sup>3</sup> |
|      | 03 | Note <sup>3</sup> | Y | Note <sup>3</sup> |
| 0x34 | -- | N                 | N | Y                 |
| 0x35 | -- | N                 | Y | Y                 |
| 0x36 | -- | N                 | N | Y                 |
| 0x37 | -- | N                 | N | Y                 |
| 0x38 | -- | N                 | N | Y                 |

Note<sup>1</sup>: It is depended on the DIDs. If a DID is secured, the security access and non-default diagnostic session shall be used.

Note<sup>2</sup>: It is depended on the IO Control IDs. If a IOControlID is secured, the security access shall be used.

Note<sup>3</sup>: Secured routines require a SecurityAccess service and therefore a non-default diagnostic session. A routine that requires to be stopped actively by the client also requires a non-default session. Non-secured routines should support default and non-default diagnostic session.

All supported services in each session and Security Access for BootLoader can be found in following table.

| UDS Diagnostic Services ID | sub-function | Request Message      |                             |                      | Security Access requirement |
|----------------------------|--------------|----------------------|-----------------------------|----------------------|-----------------------------|
|                            |              | Default Session      | Extended Diagnostic Session | Programming Session  |                             |
| 0x10                       | 01           | Y                    | Y                           | Y                    | N                           |
|                            | 02           | N                    | Y                           | Y                    | Y                           |
|                            | 03           | Y                    | Y                           | N                    | N                           |
|                            | 60           | N                    | N                           | N                    | N                           |
| 0x11                       | 01           | N                    | Y                           | Y                    | Y                           |
| 0x27                       | 01           | N                    | N                           | N                    | N                           |
|                            | 02           | N                    | N                           | N                    | N                           |
|                            | 03           | N                    | N                           | N                    | N                           |
|                            | 04           | N                    | N                           | N                    | N                           |
|                            | 05           | N                    | N                           | Y                    | N                           |
|                            | 06           | N                    | N                           | Y                    | N                           |
|                            | 09           | N                    | N                           | Y                    | N                           |
|                            | 0A           | N                    | N                           | Y                    | N                           |
|                            | 61           | N                    | N                           | N                    | N                           |
|                            | 62           | N                    | N                           | N                    | N                           |
| 0x28                       | 00           | N                    | Y                           | N                    | N                           |
|                            | 01           | N                    | Y                           | N                    | N                           |
|                            | 03           | N                    | Y                           | N                    | N                           |
| 0x3E                       | 00           | Y                    | Y                           | Y                    | N                           |
| 0x22                       | --           | Y /Note <sup>1</sup> | Y                           | Y /Note <sup>1</sup> | Note <sup>1</sup>           |
| 0x2E                       | --           | N                    | Y                           | N                    | Note <sup>1</sup>           |
| 0x85                       | 01           | N                    | Y                           | N                    | N                           |
|                            | 02           | N                    | Y                           | N                    | N                           |
| 0x14                       | --           | N                    | N                           | N                    | N                           |
| 0x19                       | 01           | N                    | N                           | N                    | N                           |
|                            | 02           | N                    | N                           | N                    | N                           |
|                            | 03           | N                    | N                           | N                    | N                           |
|                            | 04           | N                    | N                           | N                    | N                           |
|                            | 06           | N                    | N                           | N                    | N                           |
|                            | 0A           | N                    | N                           | N                    | N                           |
| 0x2F                       | --           | N                    | N                           | N                    | Note <sup>2</sup>           |
| 0x31                       | 01           | Y /Note <sup>3</sup> | Y                           | Y                    | Note <sup>3</sup>           |
|                            | 02           | Y /Note <sup>3</sup> | Y                           | Y                    | Note <sup>3</sup>           |
|                            | 03           | Y /Note <sup>3</sup> | Y                           | Y                    | Note <sup>3</sup>           |
| 0x34                       | --           | N                    | N                           | Y                    | Y                           |
| 0x35                       | --           | N                    | N                           | Y                    | Y                           |
| 0x36                       | --           | N                    | N                           | Y                    | Y                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |   |   |   |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|---|---|---|
| 0x37                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -- | N | N | Y | Y |
| 0x38                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -- | N | N | Y | Y |
| Note <sup>1</sup> : It is depended on the DIDs. If a DID is secured, the security access and non-default diagnostic session shall be used.<br>Note <sup>2</sup> : It is depended on the IO Control IDs. If a IOControlID is secured, the security access shall be used.<br>Note <sup>3</sup> : Secured routines require a SecurityAccess service and therefore a non-default diagnostic session. A routine that requires to be stopped actively by the client also requires a non-default session. Non-secured routines should support default and non-default diagnostic session. |    |   |   |   |   |

## 4.2 Diagnostic Session Control (0x10)

The DiagnosticSessionControl service is used to enable different diagnostic sessions in the server(s).

A diagnostic session enables a specific set of diagnostic services and/or functionality in the server(s). This service provides the capability that the server(s) can report data link layer specific parameter values valid for the enabled diagnostic session (e.g. timing parameter values). The user of this International Standard shall define the exact set of services and/or functionality enabled in each diagnostic session.

There shall always be exactly one diagnostic session active in a server. A server shall always start the default diagnostic session when powered up. If no other diagnostic session is started, then the default diagnostic session shall be running as long as the server is powered.

Diagnostic apparatus switch to different diagnostic session mode by using this service. For execute diagnostic services, relevant session mode should be started firstly.

Default diagnostic Session should not description move to programming diagnostic session via the DiagnosticSessionControl (10 hex) service.

Programming diagnostic session should not description move to extended diagnostic session via the DiagnosticSessionControl (10 hex) service.

### 4.2.1 Request message

#### 4.2.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                              | Cvt | Hex value | Mnemonic |
|-------------|---------------------------------------------|-----|-----------|----------|
| #1          | DiagnosticSessionControl Request Service Id | M   | 10        | DSC      |
| #2          | sub-function = [ diagnosticSessionType ]    | M   | 00-FF     | LEV_DS_  |

#### 4.2.1.2 Request message sub-function parameter \$Level (LEV\_) definition

Table below specifies the sub-function parameter values defined for this service [suppressPosRspMsgIndicationBit (bit 7) not shown].

Request message sub-function parameter definition

| Hex | Description | Mnemonic |
|-----|-------------|----------|
|-----|-------------|----------|

| (bit 6-0) |                           |        |
|-----------|---------------------------|--------|
| 01        | defaultSession            | DS     |
| 02        | programmingSession        | PRGS   |
| 03        | extendedDiagnosticSession | EXTDS  |
| 60        | SupplierDiagnosticSession | SDS    |
| Others    | Reserved                  | RESRVD |

Before entering the programmingSession, the 0x27 service of Level 05 or Level 09 shall be performed.

## 4.2.2 Positive response message

### 4.2.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                               | Cvt | Hex value | Mnemonic |
|-------------|----------------------------------------------|-----|-----------|----------|
| #1          | DiagnosticSessionControl Response Service Id | M   | 50        | DSCPR    |
| #2          | diagnosticSessionType                        | M   | 00-7F     | DS_      |
| #3          | sessionParameterRecord[]#1 = [               | M   | 00-FF     | SPREC_   |
| #4          | P2 Server_max (high byte)                    | M   | 00-FF     | P2SMH    |
| #5          | P2 Server_max (low byte)                     | M   | 00-FF     | P2SML    |
| #6          | P2* Server_max (high byte)                   | M   | 00-FF     | P2ESMH   |
| #6          | P2* Server_max (low byte) ]                  | M   | 00-FF     | P2ESML   |

### 4.2.2.2 Positive response message data parameter definition

Response message data parameter definition

| Definition                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------|
| <b>diagnosticSessionType</b><br>This parameter is an echo of bits 6-0 of the sub-function parameter from the request message. |

SessionParameterRecord content definition

| Parameter      | Description                                                                                 | # Of bytes | Resolution | minimum value | maximum value |
|----------------|---------------------------------------------------------------------------------------------|------------|------------|---------------|---------------|
| P2 Server_max  | Default P2 Server_max timing supported by the server for the activated diagnostic session.  | 2          | 1 ms       | 0 ms          | 65 535 ms     |
| P2* Server_max | Enhanced(NRC0x78)P2Server_max supported by the server for the activated diagnostic session. | 2          | 10 ms      | 0 ms          | 655 350 ms    |

## 4.2.3 Negative response message & Supported negative response codes (NRC\_)

Negative response message definition

| A_Data byte | Parameter name | Cvt | Hex value | Mnemonic |
|-------------|----------------|-----|-----------|----------|
|-------------|----------------|-----|-----------|----------|

|    |                                             |   |       |            |
|----|---------------------------------------------|---|-------|------------|
| #1 | Negative Response Service Id                | M | 7F    | SIDRSIDNRQ |
| #2 | DiagnosticSessionControl Request Service Id | M | 10    | DSC        |
| #3 | responseCode                                | M | 00-7F | NRC_       |

Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cvt | Mnemonic |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | GR       |
| 12  | <b>subFunctionNotSupported</b><br>Send if the sub-function parameter in the request message is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | SFNS     |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the “identical request message” or “another request message”. The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.<br>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services. | M   | BRR      |
| 22  | <b>conditionsNotCorrect</b><br>This code shall be returned if the criteria for the request DiagnosticSessionControl are not met.<br>If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTS, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | U   | CNC      |
| 33  | <b>securityAccessDenied</b><br>This NRC shall be sent if the requested programmingSession is secured and the server is not in an unlocked state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | M   | SAD      |
| 78  | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M   | RCRRP    |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |         |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|
|    | This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.                                                                                                                                                                                                                                                                                                                                                               |   |         |
| 7E | <b>subFunctionNotSupportedInActiveSession</b><br>This response code indicates that the requested action will not be taken because the server does not support the requested sub-function in the session currently active. Within the programmingSession, negative response code SFNS (subFunctionNotSupported) may optionally be reported instead of negative response code SNFSIAS (subFunctionNotSupportedInActiveSession). This response code shall only be used when the requested sub-function is known to be supported in another session, otherwise response code SFNS (subFunctionNotSupported) shall be used. | M | SFNSIAS |

### 4.3 ECUReset (11 hex) service

The ECUReset service is used by the client to request a server reset.

This service requests the server to effectively perform a server reset based on the content of the resetType parameter value embedded in the ECUReset request message. The ECUReset positive response message (if required) shall be sent before the reset is executed in the server(s). After a successful server reset the server shall activate the defaultSession.

#### 4.3.1 Request message

##### 4.3.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name               | Cvt | Hex value | Mnemonic   |
|-------------|------------------------------|-----|-----------|------------|
| #1          | ECUReset Request Service Id  | M   | 11        | ER         |
| #2          | sub-function = [ ResetType ] | M   | 00-FF     | LEV_<br>RT |

##### 4.3.1.2 Request message sub-function parameter \$Level (LEV\_) definition

Table below specifies the sub-function parameter values defined for this service [suppressPosRspMsgIndicationBit (bit 7) not shown].

Request message sub-function parameter definition

| Hex (bit 6-0) | Description      | Mnemonic |
|---------------|------------------|----------|
| 01            | <b>hardReset</b> | HR       |

#### 4.3.2 Positive response message

##### 4.3.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name               | Cvt | Hex value | Mnemonic |
|-------------|------------------------------|-----|-----------|----------|
| #1          | ECUReset Response Service Id | M   | 51        | ERPR     |
| #2          | sub-function = [ resetType ] | M   | 00-7F     | LEV_RT   |

### 4.3.3 Negative response message & upported negative response codes (NRC\_)

Negative response message definition

| A_Data byte | Parameter name               | Cvt | Hex value | Mnemonic   |
|-------------|------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id | M   | 7F        | SIDRSIDNRQ |
| #2          | ECUReset Request Service Id  | M   | 11        | ER         |
| #3          | responseCode                 | M   | 00-7F     | NRC_       |

Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cvt | Mnemonic |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | GR       |
| 12  | <b>subFunctionNotSupported</b><br>Send if the sub-function parameter in the request message is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | SFNS     |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the “identical request message” or “another request message”. The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.<br>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services. | M   | BRR      |
| 22  | <b>conditionsNotCorrect</b><br>This NRC shall be returned if the criteria for the ECUReset request is not met. If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTs, or supplemental diagnostic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | U   | CNC      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |         |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|
|    | specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |         |
| 33 | <b>securityAccessDenied</b><br>This NRC shall be sent if the requested reset is secured and the server is not in an unlocked state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | M | SAD     |
| 78 | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row. | M | RCRRP   |
| 7E | <b>subFunctionNotSupportedInActiveSession</b><br>This response code indicates that the requested action will not be taken because the server does not support the requested sub-function in the session currently active. Within the programmingSession, negative response code SFNS (subFunctionNotSupported) may optionally be reported instead of negative response code SNFSIAS (subFunctionNotSupportedInActiveSession). This response code shall only be used when the requested sub-function is known to be supported in another session, otherwise response code SFNS (subFunctionNotSupported) shall be used.                                                                                                                               | U | SFNSIAS |
| 7F | <b>serviceNotSupportedInActiveSession</b><br>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.                                                                                                                                                                                                                                                                                                                                                      | M | SNSIAS  |

## 4.4 SecurityAccess (27 hex) service

### 4.4.1 Service description

The purpose of this service is to provide a means to access data and/or diagnostic services which have restricted access for security, emissions, or safety reasons. Diagnostic modes for downloading routines or data into a node and reading specific memory locations from a node are situations where security access may be required. Improper routines or data downloaded into a node could potentially damage the electronics or other vehicle components or risk the vehicle's compliance to emission, safety, or security standards.

If a device supports security, but is already unlocked when a SecurityAccess 'requestSeed' message is received, that node shall respond with a SecurityAccess 'requestSeed' positive response message service with zero seed. If a device supports security and it's locked when a SecurityAccess 'requestSeed' message is received, that node shall respond with a SecurityAccess 'requestSeed' positive response message service with non-zero seed.

A tester shall use this method to determine if a node is locked by checking for a non-zero seed.

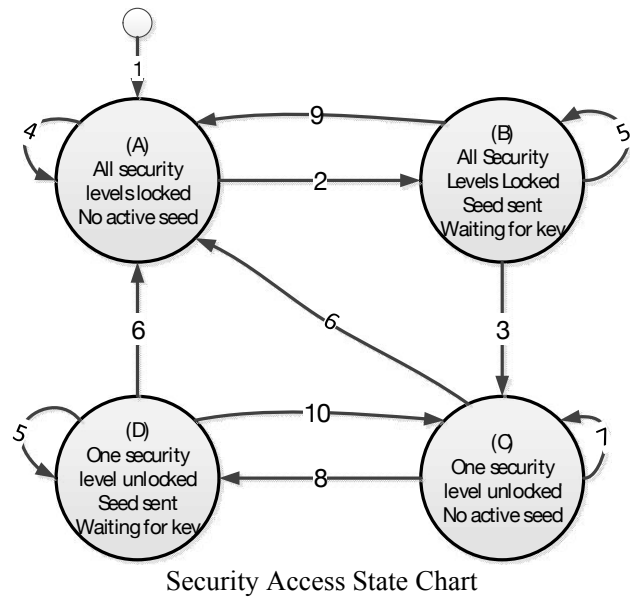
After 3 unsuccessful attempts(invalid Key) of Security Access procedures the ECU shall insert a 10 seconds delay timer before allowing further unlocking attempts. A power loss or key cycle must not reduce or stop the remaining timer.The default 10s delay time shall not be invoked during power on/start up if there is no delay time remaining during previous 3 unsuccessful attempts.

Store attempt counter in non volatile memory.

The security algorithms is refer to < SMG-EEA-4021-安全诊断需求规范>.

4.4.2 Security Access State Chart

The following figure, from Annex I of ISO 14229-1, shows the ECU security access behavior and how it is affected by diagnostic session transitions.



- Key
- (A) All security levels locked. No active seed
  - (B) All Security Levels Locked. Seed sent. Waiting for key.
  - (C) One security level unlocked. No active seed.
  - (D) One security level unlocked. Seed sent. Waiting for key.

| No. | Operation | Condition                                                                                                    | Action |
|-----|-----------|--------------------------------------------------------------------------------------------------------------|--------|
| 1   |           | Start / restart of ECU Application<br>(e.g.,ECU reset, power cycle, key cycle, sleep wake transition, etc.). |        |

| No. | Operation |     | Condition                                                                                                                 | Action                                                                                                                                                                                                                                |
|-----|-----------|-----|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   |           | AND | SecurityAccess requestSeed received.                                                                                      | SmartMi requires the use of a random seed. Generate and store Seed for the requested securityAccessType<br>Transmit SecurityAccess positive response on requestSeed request with Seed as active for the requested securityAccessType. |
|     |           |     | Message length OK <sup>a</sup> .                                                                                          |                                                                                                                                                                                                                                       |
|     |           |     | Optional Preconditions fulfilled                                                                                          |                                                                                                                                                                                                                                       |
| 3   |           | AND | SecurityAccess sendKey received                                                                                           | Unlock security level for subfunction xx.<br>Transmit SecurityAccess positive response on sendKey request.                                                                                                                            |
|     |           |     | Message length OK.                                                                                                        |                                                                                                                                                                                                                                       |
|     |           |     | Key OK.                                                                                                                   |                                                                                                                                                                                                                                       |
| 4   | OR        | AND | SecurityAccess requestSeed received.                                                                                      | Transmit negative response NRC 0x13.                                                                                                                                                                                                  |
|     |           |     | Message Length NOK.                                                                                                       |                                                                                                                                                                                                                                       |
|     |           | AND | SecurityAccess sendKey received.                                                                                          | Transmit negative response NRC 0x24.                                                                                                                                                                                                  |
|     |           |     | SecurityAccess requestSeed received.                                                                                      | Transmit negative response NRC 0x22.                                                                                                                                                                                                  |
|     |           |     | Message length OK.<br>Optional pre-conditions NOT fulfilled <sup>e</sup> .                                                |                                                                                                                                                                                                                                       |
| 5   | OR        | AND | SecurityAccess requestSeed received.                                                                                      | Transmit SecurityAccess positive response with the active stored seed for the requested securityAccessType.                                                                                                                           |
|     |           |     | Static Seed = True.                                                                                                       |                                                                                                                                                                                                                                       |
|     |           |     | requested securityAccessType has an active stored seed.                                                                   |                                                                                                                                                                                                                                       |
|     |           | AND | SecurityAccess requestSeed received.                                                                                      | Generate and store Seed for the requested securityAccessType<br>Transmit SecurityAccess positive response on requestSeed request with Seed as active for the requested securityAccessType.                                            |
|     |           |     | Static Seed = True.<br>requested securityAccessType has no active seed stored (different securityAccessType than before). |                                                                                                                                                                                                                                       |
| 6   |           |     | DiagnosticSessionControl accepted or session timeout occurs.                                                              | Start appropriate diagnostic session.<br>Lock ECU.                                                                                                                                                                                    |
| 7   | OR        |     | SecurityAccess sendKey received.                                                                                          | Transmit negative response NRC 0x24.                                                                                                                                                                                                  |
|     |           | AND | SecurityAccess requestSeed received.                                                                                      | Transmit SecurityAccess positive response with zero seed <sup>d</sup> .                                                                                                                                                               |
|     |           |     | Requested level is unlocked.                                                                                              |                                                                                                                                                                                                                                       |
| 8   | OR        | AND | SecurityAccess requestSeed received.                                                                                      | Transmit SecurityAccess positive response on requestSeed request with Seed as active for the requested securityAccessType.                                                                                                            |
|     |           |     | Requested level is NOT unlocked.                                                                                          |                                                                                                                                                                                                                                       |
|     |           |     | Message length OK.                                                                                                        |                                                                                                                                                                                                                                       |
|     |           |     | Optional pre-conditions fulfilled.                                                                                        |                                                                                                                                                                                                                                       |

| No. | Operation |     | Condition                                                    | Action                                                                                                                                          |
|-----|-----------|-----|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 9   | OR        | AND | SecurityAccess sendKey received.                             | Transmit negative response NRC 0x35.                                                                                                            |
|     |           |     |                                                              |                                                                                                                                                 |
|     |           |     | Message length OK.<br>Key NOK.                               |                                                                                                                                                 |
|     |           | AND | SecurityAccess sendKey received.                             | Transmit negative response NRC 0x24.                                                                                                            |
|     |           |     |                                                              |                                                                                                                                                 |
|     |           | AND | SecurityAccess sendKey received.                             | Transmit negative response NRC 0x13.                                                                                                            |
|     |           |     | Message length NOK.                                          |                                                                                                                                                 |
|     |           | AND | DiagnosticSessionControl accepted or session timeout occurs. | Start appropriate diagnostic session.                                                                                                           |
| 10  | OR        | AND | SecurityAccess requestSeed received.                         | Transmit SecurityAccess positive response with zero seed <sup>e</sup> .                                                                         |
|     |           |     | Requested level is unlocked.                                 |                                                                                                                                                 |
|     |           | AND | SecurityAccess sendKey received.                             | Lock currently unlocked security level. Unlock security level for subfunction xx. Transmit SecurityAccess positive response on sendKey request. |
|     |           |     |                                                              |                                                                                                                                                 |
|     |           |     | Message length OK.<br>Key OK.                                |                                                                                                                                                 |
|     |           | AND | SecurityAccess sendKey received.                             | Transmit negative response NRC 0x35.                                                                                                            |
|     |           |     |                                                              |                                                                                                                                                 |
|     |           |     | Message length OK.<br>Key NOK.                               |                                                                                                                                                 |
|     |           | AND | SecurityAccess sendKey received.                             | Transmit negative response NRC 0x24.                                                                                                            |
|     |           |     |                                                              |                                                                                                                                                 |
|     |           | AND | SecurityAccess sendKey received.                             | Transmit negative response NRC 0x13.                                                                                                            |
|     |           |     | Message length NOK.                                          |                                                                                                                                                 |

- The exact length check can only be done after the evaluation of the sub-function, because the length depends on the sub-function (i.e., length of requestSeed is different from length of sendKey message). The check for the minimum length is done during the general service evaluation process.
- The sendKey sub-function (yy) must be of the expected securityAccessType (the active stored seed is for the corresponding requestSeed securityAccessType, i.e. sendKey securityAccessType – 1).
- Customer specific precondition can be checked (e.g. fingerprint written in this driving cycle, engine not running, and vehicle not moving).
- Once a given security level is unlocked, it shall remain unlocked even after a seed request is received for a different security level until either a new security is completely unlocked or the security access is exited for other reasons (e.g., DiagnosticSessionControl accepted or session timeout occurs).
- Once a given security level is unlocked, it shall remain unlocked even after a seed request is received for a different security level until either a new security is completely unlocked or the security access is exited for other reasons (e.g., diagnosticSessionControl received).
- The sendKey sub-function must be of the expected access type (active stored seed is for sendKey accessType – 1).

### 4.4.3 Request message

#### 4.4.3.1 Request message definition

Request message definition-sub-function = requestSeed

| A_Data byte | Parameter name                                         | Cvt | Hex value            | Mnemonic        |
|-------------|--------------------------------------------------------|-----|----------------------|-----------------|
| #1          | SecurityAccess Request Service Id                      | M   | 27                   | SA              |
| #2          | sub-function = [<br>securityAccessType = requestSeed ] | M   | 01, 03, 05,<br>09,61 | LEV_<br>SAT_RSD |

Request message definition-sub-function = sendKey

| A_Data byte | Parameter name                                       | Cvt | Hex value            | Mnemonic          |
|-------------|------------------------------------------------------|-----|----------------------|-------------------|
| #1          | SecurityAccess Request Service Id                    | M   | 27                   | SA                |
| #2          | sub-function = [<br>securityAccessType = sendKey ]   | M   | 02, 04, 06,<br>0A,62 | LEV_<br>SAT_SK    |
| #3          | securityKey <sup>a</sup> [] = [<br>key#1 (high byte) | M   | 00-FF                | SECKEY_<br>KEY1HB |
| :           | :                                                    | :   | :                    | :                 |
| #n          | key#m (low byte) ]                                   | U   | 00-FF                | KEYmLB            |

- a. The definition of securityKey is refer to < SMG-EEA-4021-安全诊断需求规范>.

#### 4.4.3.2 Request message sub-function parameter \$Level (LEV\_) definition

Table below specifies the sub-function parameter values defined for this service

[suppressPosRspMsgIndicationBit (bit 7) not shown].

The sub-parameter used in the request message of the SecurityAccess service indicates to the node the step in progress for this service. The sub-parameters defined as valid for the request message of this service are indicated in the table below.

The values of the parameter key are not defined in this document.

Request message sub-function parameter definition

| Hex<br>(bit 6-0) | Description                                                 | Level |
|------------------|-------------------------------------------------------------|-------|
| 01               | Off-boardDiagnosticRequestSeed                              | 01    |
| 02               | Off-boardDiagnosticSendKey                                  |       |
| 03               | On-boardDiagnosticRequestSeed/ RemoteDiagnosticsrequestSeed | 03    |
| 04               | On-boardDiagnosticSendKey/ RemoteDiagnosticssendKey         |       |
| 05               | Off-boardFlashRequestSeed                                   | 05    |
| 06               | Off-boardFlashSendKey                                       |       |
| 09               | OTAResponseSeed                                             | 09    |
| 0A               | OTASendKey                                                  |       |
| 61               | SupplierSecurityAccessRequestSeed                           | 61    |
| 62               | SupplierSecurityAccessSendKey                               |       |

#### 4.4.3.3 Request message data parameter definition

The following data parameters are defined for this service.

| Definition                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>securityKey</b><br>The “key” parameter in the request message is the value generated by the security algorithm corresponding to a specific “seed” value. |

#### 4.4.4 Positive response message

##### 4.4.4.1 Positive response message definition

Positive response message definition-sub-function = requestSeed

| A_Data byte | Parameter name                                                        | Cvt | Hex value          | Mnemonic            |
|-------------|-----------------------------------------------------------------------|-----|--------------------|---------------------|
| #1          | SecurityAccess Response Service Id                                    | M   | 67                 | SAPR                |
| #2          | securityAccessType                                                    | M   | 01, 03, 05, 09, 61 | LEV_SAT_SK          |
| #3          | securitySeed*[] = [<br>seed#1 (high byte)<br>:<br>seed#m (low byte) ] | M   | 00-FF              | SECSEED_<br>SEED1HB |
| :           |                                                                       | :   | :                  | :                   |
| #n          |                                                                       | U   | 00-FF              | SEEDmLB             |

- a. The definition of securitySeed is refer to < SMG-EEA-4021-安全诊断需求规范>.

Positive response message definition-sub-function = sendkey

| A_Data byte | Parameter name                     | Cvt | Hex value          | Mnemonic   |
|-------------|------------------------------------|-----|--------------------|------------|
| #1          | SecurityAccess Response Service Id | M   | 67                 | SAPR       |
| #2          | securityAccessType                 | M   | 02, 04, 06, 0A, 62 | LEV_SAT_SK |

##### 4.4.4.2 Positive response message data parameter definition

Response message data parameter definition

| Definition                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>securityAccessType</b><br>This parameter is an echo of bits 6-0 of the sub-function parameter from the request message.                                                                                                                                                                                                                  |
| <b>securitySeed</b><br>The seed parameter is a data value sent by the server and is used by the client when calculating the key needed to access security. The securitySeed data bytes are only present in the response message if the request message was sent with the sub-function set to a value which requests the seed of the server. |

#### 4.4.5 Negative response message & Supported negative response codes (NRC\_)

##### Negative response message definition

| A_Data byte | Parameter name                   | Cvt | Hex value | Mnemonic   |
|-------------|----------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id     | M   | 7F        | SIDRSIDNRQ |
| #2          | SecurityAcces Request Service Id | M   | 27        | SA         |
| #3          | responseCode                     | M   | 00-7F     | NRC_       |

##### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cvt | Mnemonic |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | GR       |
| 12  | <b>subFunctionNotSupported</b><br>Send if the sub-function parameter in the request message is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | SFNS     |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the “identical request message” or “another request message”. The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.<br>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services. | M   | BRR      |
| 22  | <b>conditionsNotCorrect</b><br>This code shall be returned if the criteria for the request SecurityAccess are not met.If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTs, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | U   | CNC      |
| 24  | <b>requestSequenceError</b><br>Send if the “sendKey” sub-function is received without first receiving a “requestSeed” request message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M   | RSE      |
| 35  | <b>invalidKey</b><br>Send if an expected “sendKey” sub-function value is received and the value of the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M   | IK       |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |         |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|
|    | key does not match the server's internally stored/calculated key.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |         |
| 78 | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row. | M | RCRRP   |
| 7E | <b>subFunctionNotSupportedInActiveSession</b><br>This response code indicates that the requested action will not be taken because the server does not support the requested sub-function in the session currently active. Within the programmingSession, negative response code SFNS (subFunctionNotSupported) may optionally be reported instead of negative response code SNFSIAS (SubFunction Not Supported In Active Session). This response code shall only be used when the requested sub-function is known to be supported in another session, otherwise response code SFNS (subFunctionNotSupported) shall be used.                                                                                                                          | U | SFNSIAS |
| 7F | <b>serviceNotSupportedInActiveSession</b><br>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.                                                                                                                                                                                                                                                                                                                                                      | M | SNSIAS  |

## 4.5 CommunicationControl (28 hex) service

The purpose of this service is to switch on/off the transmission and/or the reception of certain messages of (a) server(s) (e.g. application communication messages).

### 4.5.1 Request message

#### 4.5.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                          | Cvt | Hex value | Mnemonic    |
|-------------|-----------------------------------------|-----|-----------|-------------|
| #1          | CommunicationControl Request Service Id | M   | 28        | CC          |
| #2          | sub-function = [ controlType ]          | M   | 00-FF     | LEV_ CTRLTP |
| #3          | communicationType                       | M   | 00-FF     | CTP         |

#### 4.5.1.2 Request message sub-function parameter \$Level (LEV\_) definition

Table below specifies the sub-function parameter values defined for this service

[suppressPosRspMsgIndicationBit (bit 7) not shown].

Request message sub-function parameter definition

| Hex<br>(bit 6-0) | Description                                                                             | Mnemonic     |
|------------------|-----------------------------------------------------------------------------------------|--------------|
| 00               | enableRxAndTx                                                                           | ERXTX        |
| 01               | EnableRxAndDisableTx                                                                    | ERXDTX       |
| 03               | disableRxAndTx<br>Note:sub function 0x03 Shall require a precondition of OFF powermode. | DRXTX        |
| 04 – 3F          | ISOSAEReserved                                                                          | ISOSAERESRVD |
| 40 – 5F          | vehicleManufacturerSpecific                                                             | VMS          |
| 60 – 7E          | systemSupplierSpecific                                                                  | SSS          |
| 7F               | ISOSAEReserved                                                                          | ISOSAERESRVD |

#### 4.5.1.3 Request message data parameter definition

Request message data parameter definition

| Definition                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>communicationType</b><br>This parameter is used to reference the kind of communication to be controlled. The communicationType parameter is a bit-code value which allows control of multiple communication types at the same time (see Below Table for the coding of the communicationType data parameter). |

Definition of communicationType and subnetNumber byte

| Hex<br>(bit 7-0) | Description                                                                                                                                                                                                                                                                                                                                                                               | Cvt |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 01               | normalCommunicationMessages<br>Note: In the receiving node including communication to all connected networks. This only disables the node's communication into the connected networks but not the communication of other nodes on the networks (i.e., receiving node is not responsible to disable communication in each node of the network).                                            | M   |
| 03               | networkManagementCommunicationMessages and normalCommunicationMessages<br>Note: In the receiving node including communication to all connected networks. This only disables the node's communication into the connected networks but not the communication of other nodes on the networks (i.e., receiving node is not responsible to disable communication in each node of the network). | M   |
| F1               | normalCommunicationMessages<br>Note: Disable/Enable network which request is received on (Receiving node (server))                                                                                                                                                                                                                                                                        | U   |
| F3               | networkManagementCommunicationMessages and normalCommunicationMessages<br>Note: Disable/Enable network which request is received on (Receiving node (server))                                                                                                                                                                                                                             | U   |

## 4.5.2 Positive response message

### 4.5.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                           | Cvt | Hex value | Mnemonic |
|-------------|------------------------------------------|-----|-----------|----------|
| #1          | CommunicationControl Response Service Id | M   | 68        | CCPR     |
| #2          | controlType                              | M   | 00-7F     | CTRLTP   |

### 4.5.2.2 Positive response message data parameter definition

Response message data parameter definition

| Definition                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------|
| <b>controlType</b><br>This parameter is an echo of bits 6-0 of the sub-function parameter from the request message. |

## 4.5.3 Negative response message & upported negative response codes (NRC\_)

Negative response message definition

| A_Data byte | Parameter name                          | Cvt | Hex value | Mnemonic   |
|-------------|-----------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id            | M   | 7F        | SIDRSIDNRQ |
| #2          | CommunicationControl Request Service Id | M   | 28        | CC         |
| #3          | responseCode                            | M   | 00-7F     | NRC_       |

Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent. | M   | GR       |
| 12  | <b>subFunctionNotSupported</b><br>Send if the sub-function parameter in the request message is not supported.                                                                                                                                                                                                                                                                                                                                                                                                 | M   | SFNS     |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                           | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might                                                       | M   | BRR      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |        |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
|    | <p>be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.</p> <p>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21. This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p>                                                                                                                                                                                                                                         |   |        |
| 22 | <p><b>conditionsNotCorrect</b></p> <p>Used when the server is in a critical normal mode activity and therefore cannot disable/enable the requested communication type.</p> <p>If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTs, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.</p>                                                                                                                                                                                                                                              | U | CNC    |
| 31 | <p><b>requestOutOfRange</b></p> <p>The server shall use this response code if it detects an error in the communicationType parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M | ROOR   |
| 78 | <p><b>requestCorrectlyReceived-ResponsePending</b></p> <p>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.</p> <p>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.</p> | M | RCRRP  |
| 7F | <p><b>serviceNotSupportedInActiveSession</b></p> <p>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.</p>                                                                                                                                                                                                                                                                                                                                                          | M | SNSIAS |

## 4.6 TesterPresent (3E hex) service

This service is used to indicate to a server (or servers) that a client is still connected to the vehicle and that certain diagnostic services and/or communication that have been previously activated are to remain active.

This service is used to keep one or multiple servers in a diagnostic session other than the defaultSession. This can either be done by transmitting the TesterPresent request message periodically or in case of the absence of other diagnostic services to prevent the server(s) from automatically returning to the defaultSession.

## 4.6.1 Request message

### 4.6.1.1 Request message definition

#### Request message definition

| A_Data byte | Parameter name                     | Cvt | Hex value | Mnemonic      |
|-------------|------------------------------------|-----|-----------|---------------|
| #1          | TesterPresent Request Service Id   | M   | 3E        | TP            |
| #2          | sub-function = [ zeroSubFunction ] | M   | 00/80     | LEV_<br>ZSUBF |

### 4.6.1.2 Request message sub-function parameter \$Level (LEV\_) definition

Table below specifies the sub-function parameter values defined for this service

[suppressPosRspMsgIndicationBit (bit 7) not shown].

#### Request message sub-function parameter definition

| Hex (bit 6-0) | Description                                                                                                                                                           | Mnemonic     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 00            | <b>zeroSubFunction</b><br>This parameter value is used to indicate that no sub-function value beside the suppressPosRspMsgIndicationBit is supported by this service. | ZSUBF        |
| 01 - 7F       | <b>ISOSAEReserved</b>                                                                                                                                                 | ISOSAERESRVD |

## 4.6.2 Positive response message

### 4.6.2.1 Positive response message definition

#### Positive response message definition

| A_Data byte | Parameter name                    | Cvt | Hex value | Mnemonic |
|-------------|-----------------------------------|-----|-----------|----------|
| #1          | TesterPresent Response Service Id | M   | 7E        | TPPR     |
| #2          | zeroSubFunction                   | M   | 00        | ZSUBF    |

### 4.6.2.2 Positive response message data parameter definition

#### Response message data parameter definition

| Definition                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------|
| <b>zeroSubFunction</b><br>This parameter is an echo of bits 6-0 of the sub-function parameter from the request message. |

## 4.6.3 Negative response message & Supported negative response codes (NRC\_)

#### Negative response message definition

| A_Data byte | Parameter name                   | Cvt | Hex value | Mnemonic   |
|-------------|----------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id     | M   | 7F        | SIDRSIDNRQ |
| #2          | TesterPresent Request Service Id | M   | 3E        | TP         |

|    |              |   |       |      |
|----|--------------|---|-------|------|
| #3 | responseCode | M | 00-7F | NRC_ |
|----|--------------|---|-------|------|

#### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cvt | Mnemonic |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | GR       |
| 12  | <b>subFunctionNotSupported</b><br>Send if the sub-function parameter in the request message is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | SFNS     |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.<br>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services. | M   | BRR      |
| 78  | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.                                                                                                                                                                                                                        | M   | RCRRP    |

## 4.7 ReadDataByIdentifier (22 hex) service

The ReadDataByIdentifier service allows the client to request data record values from the server identified by one or more dataIdentifiers.

## 4.7.1 Request message

### 4.7.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                                      | Cvt | Hex value | Mnemonic   |
|-------------|-----------------------------------------------------|-----|-----------|------------|
| #1          | ReadDataByIdentifier Request Service Id             | M   | 22        | RDBI       |
| #2          | dataIdentifier [] #1 = [<br>byte#1 (MSB)<br>byte#2] | M   | 00-FF     | DID_<br>HB |
| #3          |                                                     | M   | 00-FF     | LB         |
| .           | .                                                   | .   | .         | .          |
| .           | .                                                   | .   | .         | .          |
| #n-1        | dataIdentifier [] #m = [<br>byte#1 (MSB)<br>byte#2] | U   | 00-FF     | DID_<br>HB |
| #n          |                                                     | U   | 00-FF     | LB         |

If an ECU supports application specific data identifiers (non corporate standard), then the DID number, the data contents, and other information shall be documented within the ECU <SMG-EEA-4006-诊断数据库 Excel> by supplier .

The tester should not functionally address requests for all ECUs with the results in multiple frame responses unless it can handle sending the flow control frames to each responding node within the allowable network timing parameters.

The maximum number of DIDs that the ECU supports with a single request is 8.

Some DIDs may require security access. Security access is applied to the DID and not the service itself. DIDs may be limited to a single diagnostic session.

When a service 0x22 request is received, the server shall check if any source DID in the request is supported in the current security level and current diagnostic session. If any one source DID in the request is not supported in the current security level or current diagnostic session the server shall send a NRC to the client.

## 4.7.2 Positive response message

### 4.7.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                                      | Cvt | Hex value | Mnemonic        |
|-------------|-----------------------------------------------------|-----|-----------|-----------------|
| #1          | ReadDataByIdentifier Response Service Id            | M   | 62        | RDBIPR          |
| #2          | dataIdentifier [] #1 = [<br>byte#1 (MSB)<br>byte#2] | M   | 00-FF     | DID_<br>HB      |
| #3          |                                                     | M   | 00-FF     | LB              |
| #4          | dataRecord[]#1 = [<br>data#1<br>:<br>:<br>data#k ]  | M   | 00-FF     | DREC_<br>DATA_1 |
| :           |                                                     | :   | :         | :               |
| :           |                                                     | :   | :         | :               |
| #(k-1)+4    |                                                     | U   | 00-FF     | DATA_k          |

|                           |                                                     |             |                     |                                |
|---------------------------|-----------------------------------------------------|-------------|---------------------|--------------------------------|
|                           |                                                     |             |                     |                                |
| :                         | :                                                   | :           | :                   | :                              |
| #n-(o-1)-2<br># n-(o-1)-1 | dataIdentifier [] #m = [<br>byte#1 (MSB)<br>byte#2] | U<br>U      | 00-FF<br>00-FF      | DID_<br>HB<br>LB               |
| # n-(o-1)<br>:<br>#n      | dataRecord[] #m = [<br>data#1<br>:<br>data#o ]      | U<br>:<br>U | 00-FF<br>:<br>00-FF | DREC_<br>DATA_1<br>:<br>DATA_k |

#### 4.7.3 Negative response message & Supported negative response codes (NRC\_)

##### Negative response message definition

| A_Data byte | Parameter name                          | Cvt | Hex value | Mnemonic   |
|-------------|-----------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id            | M   | 7F        | SIDRSIDNRQ |
| #2          | ReadDataByIdentifier Request Service Id | M   | 22        | RDBI       |
| #3          | responseCode                            | M   | 00-7F     | NRC_       |

##### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Cvt | Mnemonic |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                                                                                                                    | M   | GR       |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>This NRC shall be sent if the length of the request message is invalid or the client exceeded the maximum number of dataIdentifiers allowed to be requested at a time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M   | IMLOIF   |
| 14  | <b>responseTooLong</b><br>This NRC shall be sent if the total length of the response message exceeds the limit of the underlying transport protocol (e.g., when multiple DIDs are requested in a single request).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M   | RTL      |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21. | M   | BRR      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |       |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------|
|    | This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |       |
| 22 | <b>conditionsNotCorrect</b><br>This code shall be returned if the criteria for the request <code>ReadDataByIdentifier</code> are not met.<br>If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTS, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.                                                                                                                                                                                                                                                                    | U | CNC   |
| 31 | <b>requestOutOfRange</b><br>This code shall be sent if:<br>none of the requested <code>dataIdentifier</code> values are supported by the device;<br>none of the requested <code>dataIdentifiers</code> are supported in the current session;<br>the requested <code>dynamicDefinedDataIdentifier</code> has not been assigned yet;                                                                                                                                                                                                                                                                                                                                                                                                                   | M | ROOR  |
| 33 | <b>securityAccessDenied</b><br>This code shall be sent if at least one of the <code>dataIdentifiers</code> is secured and the server is not in an unlocked state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M | SAD   |
| 78 | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row. | M | RCRRP |

## 4.8 WriteDataByIdentifier (2E hex) service

The WriteDataByIdentifier service allows the client to write information into the server at an internal location specified by the provided data identifier.

### 4.8.1 Request message

#### 4.8.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                                      | Cvt | Hex value | Mnemonic        |
|-------------|-----------------------------------------------------|-----|-----------|-----------------|
| #1          | WriteDataByIdentifier Request Service Id            | M   | 2E        | WDBI            |
| #2          | dataIdentifier [] #1 = [<br>byte#1 (MSB)<br>byte#2] | M   | 00-FF     | DID_<br>HB      |
| #3          |                                                     | M   | 00-FF     | LB              |
| #4          | dataRecord[] = [<br>data#1<br>:<br>data#m ]         | M   | 00-FF     | DREC_<br>DATA_1 |
| :           |                                                     | :   | :         | :               |
| #m+3        |                                                     | U   | 00-FF     | DATA_m          |

All data received via each service WriteDataByIdentifier(0x2E) shall be written to non-volatile memory within 500 ms and prior to the positive response for the WriteDataByIdentifier(0x2E) request.

Some DIDs may require security access. Security access is applied to the DID and not the service itself.

## 4.8.2 Positive response message

### 4.8.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                            | Cvt | Hex value | Mnemonic   |
|-------------|-------------------------------------------|-----|-----------|------------|
| #1          | WriteDataByIdentifier Response Service Id | M   | 6E        | WDBIPR     |
| #2          | dataIdentifier [] #1 = [ byte#1 (MSB)     | M   | 00-FF     | DID_<br>HB |
| #3          | byte#2]                                   | M   | 00-FF     | LB         |

## 4.8.3 Negative response message & Supported negative response codes (NRC\_)

Negative response message definition

| A_Data byte | Parameter name                           | Cvt | Hex value | Mnemonic   |
|-------------|------------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id             | M   | 7F        | SIDRSIDNRQ |
| #2          | WriteDataByIdentifier Request Service Id | M   | 2E        | WDBI       |
| #3          | responseCode                             | M   | 00-7F     | NRC_       |

Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cvt | Mnemonic |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                          | M   | GR       |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to | M   | BRR      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |        |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
|    | <p>finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.</p> <p>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |   |        |
| 22 | <p><b>conditionsNotCorrect</b></p> <p>This code shall be returned if the criteria for the request WriteDataByIdentifier are not met.If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTS, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.</p>                                                                                                                                                                                                                                                                                        | U | CNC    |
| 31 | <p><b>requestOutOfRange</b></p> <p>This response code shall be sent if:</p> <ol style="list-style-type: none"> <li>1) the dataIdentifier in the request message is not supported in the server or the dataIdentifier is supported for read only purpose (via ReadDataByIdentifier service);</li> <li>2) any data transmitted in the request message after the dataIdentifier is invalid (if applicable to the node).</li> </ol>                                                                                                                                                                                                                                                                                                                                     | M | ROOR   |
| 33 | <p><b>securityAccessDenied</b></p> <p>This code shall be sent if the dataIdentifier, which references a specific address, is secured and the server is not in an unlocked state.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M | SAD    |
| 72 | <p><b>generalProgrammingFailure</b></p> <p>This return code shall be sent if the server detects an error when writing to a memory location.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | M | GPF    |
| 78 | <p><b>requestCorrectlyReceived-ResponsePending</b></p> <p>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.</p> <p>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.</p> | M | RCRRP  |
| 7F | <p><b>serviceNotSupportedInActiveSession</b></p> <p>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.</p>                                                                                                                                                                                                                                                                                                                                                          | M | SNSIAS |

## 4.9 ControlDTCSetting (85 hex) service

The ControlDTCSetting request message can be used to stop the updating of DTC status bits in an individual server or a group of servers.

## 4.9.1 Request message

### 4.9.1.1 Request message definition

#### Request message definition

| A_Data byte | Parameter name                       | Cvt | Hex value | Mnemonic        |
|-------------|--------------------------------------|-----|-----------|-----------------|
| #1          | ControlDTCSetting Request Service Id | M   | 85        | CDTCS           |
| #2          | sub-function = [ DTCSettingType ]    | M   | 00-FF     | LEV_<br>DTCSTP_ |

StartMi does not support the optional 'DTCSettingControlOptionRecord' parameter defined in ISO14229-1 therefore the request message only includes the service identifier and the sub-function parameter.

### 4.9.1.2 Request message sub-function parameter \$Level (LEV\_) definition

#### Request message sub-function parameter definition

| Hex (bit 6-0) | Description                 | Mnemonic     |
|---------------|-----------------------------|--------------|
| 00            | ISOSAEReserved              | ISOSAERESRVD |
| 01            | on                          | ON           |
| 02            | off                         | OFF          |
| 03 – 3F       | ISOSAEReserved              | ISOSAERESRVD |
| 40 – 5F       | vehicleManufacturerSpecific | VMS          |
| 60 – 7E       | systemSupplierSpecific      | SSS          |
| 7F            | ISOSAEReserved              | ISOSAERESRVD |

## 4.9.2 Positive response message

### 4.9.2.1 Positive response message definition

#### Positive response message definition

| A_Data byte | Parameter name                        | Cvt | Hex value | Mnemonic |
|-------------|---------------------------------------|-----|-----------|----------|
| #1          | ControlDTCSetting Response Service Id | M   | C5        | CDTCSPR  |
| #2          | DTCSettingType                        | M   | 00-7F     | DTCSTP   |

## 4.9.3 Negative response message & Supported negative response codes (NRC\_)

#### Negative response message definition

| A_Data byte | Parameter name                       | Cvt | Hex value | Mnemonic   |
|-------------|--------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id         | M   | 7F        | SIDRSIDNRQ |
| #2          | ControlDTCSetting Request Service Id | M   | 85        | CDTCS      |
| #3          | responseCode                         | M   | 00-7F     | NRC_       |

#### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Cvt | Mnemonic |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | M   | GR       |
| 12  | <b>subFunctionNotSupported</b><br>Send if the sub-function parameter in the request message is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | M   | SFNS     |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21. This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services. | M   | BRR      |
| 22  | <b>conditionsNotCorrect</b><br>Used when the server is in a critical normal mode activity and therefore cannot perform the requested DTC control functionality.<br>If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTs, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | U   | CNC      |
| 78  | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.                                                                                                                                                                                                                     | M   | RCRRP    |
| 7F  | <b>serviceNotSupportedInActiveSession</b><br>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | M   | SNSIAS   |

## 4.10 ClearDiagnosticInformation (14 hex) service

The ClearDiagnosticInformation service is used by the client to clear diagnostic information in one or multiple servers' memory.

The server shall send a positive response when the ClearDiagnosticInformation service is completely processed. The server shall send a positive response even if no DTCs are stored.

### 4.10.1 Request message

#### 4.10.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                                                                        | Cvt | Hex value | Mnemonic |
|-------------|---------------------------------------------------------------------------------------|-----|-----------|----------|
| #1          | ClearDiagnosticInformation Request Service Id                                         | M   | 14        | CDTCI    |
| #2          | groupOfDTC [] = [<br>groupOfDTCHighByte<br>groupOfDTCMiddleByte<br>groupOfDTCLowByte] | M   | FF        | HB       |
| #3          |                                                                                       | M   | FF        | MB       |
| #4          |                                                                                       | M   | FF        | LB       |

| Hex    | Parameter "groupOfDTC" Description | Cvt |
|--------|------------------------------------|-----|
| FFFFFF | All Groups (all DTCs)              | M   |

### 4.10.2 Positive response message

#### 4.10.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                                 | Cvt | Hex value | Mnemonic |
|-------------|------------------------------------------------|-----|-----------|----------|
| #1          | ClearDiagnosticInformation Response Service Id | M   | 54        | CDTCIPR  |

### 4.10.3 Negative response message & Supported negative response codes (NRC\_)

Negative response message definition

| A_Data byte | Parameter name                                | Cvt | Hex value | Mnemonic   |
|-------------|-----------------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id                  | M   | 7F        | SIDRSIDNRQ |
| #2          | ClearDiagnosticInformation Request Service Id | M   | 14        | CDTCI      |
| #3          | responseCode                                  | M   | 00-7F     | NRC_       |

Supported negative response codes

| Hex | Description                                                                                               | Cvt | Mnemonic |
|-----|-----------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The | M   | GR       |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |        |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
|    | <p>generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.</p> <p>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |        |
| 13 | <p><b>incorrectMessageLengthOrInvalidFormat</b></p> <p>The length of the message is wrong.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | M | IMLOIF |
| 21 | <p><b>busyRepeatRequest</b></p> <p>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.</p> <p>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.</p> <p>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.</p> <p>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p> | M | BRR    |
| 22 | <p><b>conditionsNotCorrect</b></p> <p>This response code shall be used if internal conditions within the server prevent the clearing of DTC related information stored in the server.</p> <p>If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTS, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | U | CNC    |
| 31 | <p><b>requestOutOfRange</b></p> <p>This return code shall be sent if the specified groupOfDTC parameter is not supported.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M | ROOR   |
| 72 | <p><b>generalProgrammingFailure</b></p> <p>This NRC indicates that the server detected an error when erasing or programming a memory location in the permanent memory device (e.g. Flash Memory).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | M | GPF    |
| 78 | <p><b>requestCorrectlyReceived-ResponsePending</b></p> <p>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.</p> <p>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.</p>                                                                                                                                                                                                                                | M | RCRRP  |
| 7F | <p><b>serviceNotSupportedInActiveSession</b></p> <p>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M | SNSIAS |

## 4.11 ReadDTCInformation (19 hex) service

This service allows a client to read the status of server resident Diagnostic Trouble Code (DTC) information from any server, or group of servers within a vehicle. Unless otherwise required by the particular subfunction, the server shall return all DTC information (e.g., emissions-related and non emissions-related).

### 4.11.1 Request message

#### 4.11.1.1 Request message definition

Request message definition subfunction –reportNumberOfDTCByStatusMask \$01

| A_Data byte | Parameter name                                 | Cvt | Hex value | Mnemonic       |
|-------------|------------------------------------------------|-----|-----------|----------------|
| #1          | ReadDTCInformation Request Service Id          | M   | 19        | RDTCI          |
| #2          | sub-function = [reportNumberOfDTCByStatusMask] | M   | 01        | LEV_<br>RDTCBM |
| #3          | DTCStatusMask                                  | M   | 00-FF     | DTCSM          |

Request message definition subfunction –reportDTCByStatusMask \$02

| A_Data byte | Parameter name                         | Cvt | Hex value | Mnemonic       |
|-------------|----------------------------------------|-----|-----------|----------------|
| #1          | ReadDTCInformation Request Service Id  | M   | 19        | RDTCI          |
| #2          | sub-function = [reportDTCByStatusMask] | M   | 02        | LEV_<br>RDTCBM |
| #3          | DTCStatusMask                          | M   | 00-FF     | DTCSM          |

Request message definition subfunction –reportDTCSnapshotIdentification \$03

| A_Data byte | Parameter name                                   | Cvt | Hex value | Mnemonic        |
|-------------|--------------------------------------------------|-----|-----------|-----------------|
| #1          | ReadDTCInformation Request Service Id            | M   | 19        | RDTCI           |
| #2          | sub-function = [reportDTCSnapshotIdentification] | M   | 03        | LEV_<br>RDTCSSI |

Request message definition subfunction – reportDTCSnapshotRecordByDTCNumber \$04

| A_Data byte | Parameter name                                      | Cvt | Hex value | Mnemonic           |
|-------------|-----------------------------------------------------|-----|-----------|--------------------|
| #1          | ReadDTCInformation Request Service Id               | M   | 19        | RDTCI              |
| #2          | sub-function = [reportDTCSnapshotRecordByDTCNumber] | M   | 04        | LEV_<br>RDTCSSBDTC |
| #3          | DTCMaskRecord [] = [DTCHighByte                     | M   | 00-FF     | DTCMREC_<br>DTCHB  |
| #4          | DTCLowByte                                          | M   | 00-FF     | DTCLB              |
| #5          | DTCFailureTypeByte ]                                | M   | 00-FF     | DTCFTB             |
| #6          | DTCSnapshotRecordNumber                             | M   | 00-FF     | DTCSSRN            |

#### Request message definition subfunction – reportDTCExtDataRecordByDTCNumber \$06

| A_Data byte | Parameter name                                     | Cvt | Hex value | Mnemonic           |
|-------------|----------------------------------------------------|-----|-----------|--------------------|
| #1          | ReadDTCInformation Request Service Id              | M   | 19        | RDTCI              |
| #2          | sub-function = [reportDTCExtDataRecordByDTCNumber] | M   | 06        | LEV_<br>RDTCEDRBDN |
| #3          | DTCMaskRecord [] = [DTCHighByte                    | M   | 00-FF     | DTCMRE_<br>DTCHB   |
| #4          | DTCLowByte                                         | M   | 00-FF     | DTCLB              |
| #5          | DTCFailureTypeByte ]                               | M   | 00-FF     | DTCFTB             |
| #6          | DTC ExtDataRecordNumber                            | M   | 00-FF     | DTCEDRN            |

#### Request message definition subfunction – reportSupportedDTC \$0A

| A_Data byte | Parameter name                        | Cvt | Hex value | Mnemonic        |
|-------------|---------------------------------------|-----|-----------|-----------------|
| #1          | ReadDTCInformation Request Service Id | M   | 19        | RDTCI           |
| #2          | sub-function = [reportSupportedDTC]   | M   | 0A        | LEV_<br>RDTCSSI |

#### 4.11.1.2 Request message sub-function parameter \$Level (LEV\_) definition

##### Request message sub-function parameter definition

| Hex (bit 6-0) | Description                        | Mnemonic     |
|---------------|------------------------------------|--------------|
| 00            | ISOSAEReserved                     | ISOSAERESRVD |
| 01            | reportNumberOfDTCByStatusMask      | RNODTCBSM    |
| 02            | reportDTCByStatusMask              | RDTCSBM      |
| 03            | reportDTCSnapshotIdentification    | RDTCSSI      |
| 04            | reportDTCSnapshotRecordByDTCNumber | RDTCSSBDTC   |
| 06            | reportDTCExtDataRecordByDTCNumber  | RDTCEDRBDN   |
| 0A            | reportSupportedDTC                 | RSUPDTC      |
| 1A – 7F       | ISOSAEReserved                     | ISOSAERESRVD |
| Others        | Reserved                           | RESRVD       |

#### 4.11.2 Positive response message

##### 4.11.2.1 Positive response message definition

##### Positive response message definition- reportNumberOfDTCByStatusMask \$01

| A_Data byte | Parameter name                               | Cvt | Hex value | Mnemonic |
|-------------|----------------------------------------------|-----|-----------|----------|
| #1          | ReadDTCInformation Response Service Id       | M   | 59        | RDTICIPR |
| #2          | reportType = [reportNumberOfDTCByStatusMask] | M   | 01        | RDTCSBM  |
| #3          | DTCStatusAvailabilityMask                    | M   | 00-FF     | DTCSAM   |
| #4          | DTCFormatIdentifier= [                       |     |           | DTCFID_  |

|    |                        |   |       |             |
|----|------------------------|---|-------|-------------|
|    | ISO_14229-1_DTCFormat] | M | 01    | 14229-1DTCF |
| #5 | DTCCCount = [          | M | 00-FF | DTCC_       |
|    | DTCCCountHighByte      | M | 00-FF | DTCCHB      |
| #6 | DTCCCountLowByte]      | M | 00-FF | DTCCLB      |

The DTC format for SmartMi UDS shall use the ISO\_14229-1\_DTCFormat which is defined by SmartMi.

#### Positive response message definition- reportDTCByStatusMask \$02

| A_Data byte | Parameter name                           | Cvt | Hex value | Mnemonic |
|-------------|------------------------------------------|-----|-----------|----------|
| #1          | ReadDTCInformation Response Service Id   | M   | 59        | RDTCIPLR |
| #2          | reportType = [<br>reportDTCByStatusMask] | M   | 02        | RDTCBISM |
| #3          | DTCStatusAvailabilityMask                | M   | 00-FF     | DTCSAM   |
|             | DTCAndStatusRecord[] = [                 |     |           | DTCASR_  |
| #4          | DTCHighByte#1                            | C1  | 00-FF     | DTCHB    |
| #5          | DTCLowByte#1                             | C1  | 00-FF     | DTCLB    |
| #6          | DTCFailureTypeByte#1                     | C1  | 00-FF     | DTCFTB   |
| #7          | statusOfDTC#1                            | C1  | 00-FF     | SODTC    |
| #8          | DTCHighByte#2                            | C2  | 00-FF     | DTCHB    |
| #9          | DTCLowByte#2                             | C2  | 00-FF     | DTCLB    |
| #10         | DTCFailureTypeByte#2                     | C2  | 00-FF     | DTCFTB   |
| #11         | statusOfDTC#2                            | C2  | 00-FF     | SODTC    |
| .           | .                                        | .   | .         | .        |
| .           | .                                        | .   | .         | .        |
| #n-3        | DTCHighByte#m                            | C2  | 00-FF     | DTCHB    |
| #n-2        | DTCLowByte#m                             | C2  | 00-FF     | DTCLB    |
| #n-1        | DTCFailureTypeByte#m                     | C2  | 00-FF     | DTCFTB   |
| #n          | statusOfDTC#m ]                          | C2  | 00-FF     | SODTC    |

C 1: This parameter is only present if DTC information is available to be reported.  
C 2: This parameter is only present if more than one DTC information is available to be reported.

#### Positive response message definition- reportDTCSnapshotIdentification\$03

| A_Data byte | Parameter name                                     | Cvt | Hex value | Mnemonic |
|-------------|----------------------------------------------------|-----|-----------|----------|
| #1          | ReadDTCInformation Response Service Id             | M   | 59        | RDTCIPLR |
| #2          | reportType = [<br>reportDTCSnapshotIdentification] | M   | 03        | RDTCSSI  |
|             | DTCRecord[]#1 = [                                  |     |           | DTCASR_  |
| #3          | DTCHighByte#1                                      | C1  | 00-FF     | DTCHB    |
| #4          | DTCLowByte#1                                       | C1  | 00-FF     | DTCLB    |
| #5          | DTCFailureTypeByte#1]                              | C1  | 00-FF     | DTCFTB   |
| #6          | DTCSnapshotRecordNumber #1                         | C1  | 00-FF     | DTCSSRN  |
| .           | .                                                  | .   | .         | .        |
| .           | .                                                  | .   | .         | .        |
| #n-3        | DTCRecord[]#m = [<br>DTCHighByte#1                 | C2  | 00-FF     | DTCASR_  |
|             |                                                    |     |           | DTCHB    |

|                                                                                                                                          |                            |    |       |         |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----|-------|---------|
| #n-2                                                                                                                                     | DTCLowByte#1               | C2 | 00-FF | DTCLB   |
| #n-1                                                                                                                                     | DTCFailureTypeByte#1]      | C2 | 00-FF | DTCFTB  |
| #n                                                                                                                                       | DTCSnapshotRecordNumber #m | C2 | 00-FF | DTCSSRN |
| C1: the DTCRecord and DTCSnapshotRecordNumber parameter is only present if at least one DTCSnapshot record is available to be reported.  |                            |    |       |         |
| C2: the DTCRecord and DTCSnapshotRecordNumber parameter is only present if more than one DTCSnapshot record is available to be reported. |                            |    |       |         |

Positive response message definition- reportDTCSnapshotRecordByDTCNumber \$04

| A_Data byte | Parameter name                                               | Cvt | Hex value | Mnemonic          |
|-------------|--------------------------------------------------------------|-----|-----------|-------------------|
| #1          | ReadDTCInformation Response Service Id                       | M   | 59        | RDTICIPR          |
| #2          | reportType = [<br>reportDTCSnapshotRecordByDTCNumber]        | M   | 04        | RDTCSBDDTC        |
| #3          | DTCAndStatusRecord[] = [<br>DTCHighByte#1                    | M   | 00-FF     | DTCASR_<br>DTCHB  |
| #4          | DTCLowByte#1                                                 | M   | 00-FF     | DTCLB             |
| #5          | DTCFailureTypeByte#1                                         | M   | 00-FF     | DTCFTB            |
| #6          | statusOfDTC]                                                 | M   | 00-FF     | SODTC             |
| #7          | DTCSnapshotRecordNumber #1                                   | C1  | 00-FF     | DTCSSRN           |
| #8          | DTCSnapshotRecordNumberOfIdentifiers #1                      | C1  | 00-FF     | DTCSSRNI          |
| #9          | DTCSnapshotRecord[] #1 = [<br>dataIdentifier#1 byte #1 (MSB) | C1  | 00-FF     | DTCSSR_<br>DIDB11 |
| #10         | dataIdentifier#1 byte #2 (LSB)                               | C1  | 00-FF     | DIDB12            |
| #11         | snapshotData#1 byte #1                                       | C1  | 00-FF     | SSD11             |
| :           | :                                                            | :   | :         | :                 |
| #11+(p-1)   | snapshotData#1 byte #p                                       | C1  | 00-FF     | SSD1p             |
| :           | :                                                            | :   | :         | :                 |
| #r-(m-1)-2  | dataIdentifier#w byte #1 (MSB)                               | C2  | 00-FF     | DIDB21            |
| #r-(m-1)-1  | dataIdentifier#w byte #2 (LSB)                               | C2  | 00-FF     | DIDB22            |
| #r-(m-1)    | snapshotData#w byte #1                                       | C2  | 00-FF     | SSD21             |
| :           | :                                                            | :   | :         | :                 |
| #r          | snapshotData#w byte #m ]                                     | C2  | 00-FF     | SSD2m             |
| .           | .                                                            | .   | .         | .                 |
| .           | .                                                            | .   | .         | .                 |
| #t          | DTCSnapshotRecordNumber #x                                   | C3  | 00-FF     | DTCSSRN           |
| #t+1        | DTCSnapshotRecordNumberOfIdentifiers #x                      | C3  | 00-FF     | DTCSSRNI          |
| #t+2        | DTCSnapshotRecord[] #x = [<br>dataIdentifier#1 byte #1 (MSB) | C3  | 00-FF     | DTCSSR_<br>DIDB11 |
| #t+3        | dataIdentifier#1 byte #2 (LSB)                               | C3  | 00-FF     | DIDB12            |
| #t+4        | snapshotData#1 byte #1                                       | C3  | 00-FF     | SSD11             |
| :           | :                                                            | :   | :         | :                 |
| #t+4+(p-1)  | snapshotData#1 byte #p                                       | C3  | 00-FF     | SSD1p             |
| :           | :                                                            | :   | :         | :                 |
| #n-(u-1)-2  | dataIdentifier#w byte #1 (MSB)                               | C4  | 00-FF     | DIDB21            |

|            |                                |    |       |        |
|------------|--------------------------------|----|-------|--------|
| #n-(u-1)-1 | dataIdentifier#w byte #2 (LSB) | C4 | 00-FF | DIDB22 |
| #n-(u-1)   | snapshotData#w byte #1         | C4 | 00-FF | SSD21  |
| :          | :                              | :  | :     | :      |
| #n         | snapshotData#w byte #u ]       | C4 | 00-FF | SSD2u  |

C1, the DTCSnapshotRecordNumber and the first dataIdentifier/snapshotData combination in the DTCSnapshotRecord parameter are only present if at least one DTCSnapshot record is available to be reported (DTCSnapshotRecordNumber unequal to FF hex in the request or only one record is available to be reported if DTCSnapshotRecordNumber is set to FF hex in the request).

C2 /C4, there are multiple dataIdentifier/snapshotData combinations allowed to be present in a single DTCSnapshotRecord. This can be the case for the situation where a single dataIdentifier only references an integral part of the data. When the dataIdentifier references a block of data then a single dataIdentifier/snapshotData combination can be used.

C3, the DTCSnapshotRecordNumber and the first dataIdentifier/snapshotData combination in the DTCSnapshotRecord parameter are only present if all records are requested to be reported (DTCSnapshotRecordNumber set to FF hex in the request) and more than one record is available to be reported.

Positive response message definition- reportDTCExtDataRecordByDTCNumber \$06

| A_Data byte | Parameter name                                       | Cvt | Hex value | Mnemonic         |
|-------------|------------------------------------------------------|-----|-----------|------------------|
| #1          | ReadDTCInformation Response Service Id               | M   | 59        | RDTICIPR         |
| #2          | reportType = [<br>reportDTCExtDataRecordByDTCNumber] | M   | 06        | RDTCEDRBDN       |
| #3          | DTCAAndStatusRecord[] = [<br>DTCHighByte#1           | M   | 00-FF     | DTCASR_<br>DTCHB |
| #4          | DTCLowByte#1                                         | M   | 00-FF     | DTCLB            |
| #5          | DTCFailureTypeByte#1                                 | M   | 00-FF     | DTCFTB           |
| #6          | statusOfDTC]                                         | M   | 00-FF     | SODTC            |
| #7          | DTCExtDataRecordNumber#1                             | C1  | 00-FF     | DTCEDRN          |
| #8          | DTCExtDataRecord[] #1 = [<br>extendedData#1 byte #1  | C1  | 00-FF     | DTCSSR_<br>EDD11 |
| :           | :                                                    | :   | :         | :                |
| #7+p        | extendedData#1 byte #p]                              | C1  | 00-FF     | EDD1p            |
| .           | .                                                    | .   | .         | .                |
| .           | .                                                    | .   | .         | .                |
| #8+p        | DTCExtDataRecordNumber#x                             | C2  | 00-FF     | DTCEDRN          |
| #9+p        | DTCExtDataRecord[] #x = [<br>extendedData#x byte #1  | C2  | 00-FF     | EDDx1            |
| :           | :                                                    | :   | :         | :                |
| #8+p+q      | extendedData#x byte #q]                              | C2  | 00-FF     | EDDxp            |

C1: The DTCExtDataRecordNumber and the extendedData in the DTCExtDataRecord parameter are only present if at least one DTCExtDataRecord is available to be reported.

C2: The DTCExtDataRecordNumber and the extendedData in the DTCExtDataRecord parameter are only present if all records are requested to be reported (DTCExtDataRecordNumber set to 0xFE or 0xFF in the request) and more than one record is available to be reported.

Positive response message definition- reportSupportedDTCs \$0A

| A_Data byte | Parameter name | Cvt | Hex value | Mnemonic |
|-------------|----------------|-----|-----------|----------|
|-------------|----------------|-----|-----------|----------|

|      |                                        |    |       |          |
|------|----------------------------------------|----|-------|----------|
| #1   | ReadDTCInformation Response Service Id | M  | 59    | RDTCIPLR |
| #2   | reportType = [<br>reportSupportedDTCs] | M  | 0A    | RSUPDTC  |
| #3   | DTCStatusAvailabilityMask              | M  | 00-FF | DTCSAM   |
|      | DTCAndStatusRecord[] = [               |    |       | DTCASR_  |
| #4   | DTCHighByte#1                          | C1 | 00-FF | DTCHB    |
| #5   | DTCLowByte#1                           | C1 | 00-FF | DTCLB    |
| #6   | DTCFailureTypeByte#1                   | C1 | 00-FF | DTCFTB   |
| #7   | statusOfDTC#1                          | C1 | 00-FF | SODTC    |
| #8   | DTCHighByte#2                          | C2 | 00-FF | DTCHB    |
| #9   | DTCLowByte#2                           | C2 | 00-FF | DTCLB    |
| #10  | DTCFailureTypeByte#2                   | C2 | 00-FF | DTCFTB   |
| #11  | statusOfDTC#2                          | C2 | 00-FF | SODTC    |
| .    | .                                      | .  | .     | .        |
| .    | .                                      | .  | .     | .        |
| #n-3 | DTCHighByte#m                          | C2 | 00-FF | DTCHB    |
| #n-2 | DTCLowByte#m                           | C2 | 00-FF | DTCLB    |
| #n-1 | DTCFailureTypeByte#m                   | C2 | 00-FF | DTCFTB   |
| #n   | statusOfDTC#m ]                        | C2 | 00-FF | SODTC    |

C 1: This parameter is only present if DTC information is available to be reported.  
C 2: This parameter is only present if more than one DTC information is available to be reported.

#### 4.11.3 Negative response message & Supported negative response codes (NRC\_)

##### Negative response message definition

| A_Data byte | Parameter name                        | Cvt | Hex value | Mnemonic   |
|-------------|---------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id          | M   | 7F        | SIDRSIDNRQ |
| #2          | ReadDTCInformation Request Service Id | M   | 19        | RDBPI      |
| #3          | responseCode                          | M   | 00-7F     | NRC_       |

##### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent. | M   | GR       |
| 12  | <b>subFunctionNotSupported</b><br>Send if the sub-function parameter in the request message is not supported.                                                                                                                                                                                                                                                                                                                                                                                                 | M   | SFNS     |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                           | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested                                                                                                                                                                                                                                                                                                                                                                                               | M   | BRR      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |         |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|
|    | <p>operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.</p> <p>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.</p> <p>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.</p> <p>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p> |   |         |
| 31 | <p><b>requestOutOfRange</b><br/>This code is returned if:</p> <ol style="list-style-type: none"> <li>the client specified a DTCMaskRecord that was not recognized by the server;</li> <li>the client specified an invalid DTCSnapshotRecordNumber/DTCExtDataRecordNumber.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M | ROOR    |
| 78 | <p><b>requestCorrectlyReceived-ResponsePending</b><br/>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.</p> <p>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.</p>                                                                                                               | M | RCRRP   |
| 7E | <p><b>subFunctionNotSupportedInActiveSession</b><br/>This response code indicates that the requested action will not be taken because the server does not support the requested sub-function in the session currently active. Within the programmingSession, negative response code SFNS (subFunctionNotSupported) may optionally be reported instead of negative response code SNFSIAS (subFunctionNotSupportedInActiveSession). This response code shall only be used when the requested sub-function is known to be supported in another session, otherwise response code SFNS (subFunctionNotSupported) shall be used.</p>                                                                                                                                                                                                                                                 | M | SFNSIAS |
| 7F | <p><b>serviceNotSupportedInActiveSession</b><br/>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M | SNSIAS  |

#### 4.12 InputOutputControlByIdentifier (2F hex) service

The InputOutputControlByIdentifier service is used by the client to substitute a value for an input signal, internal server function and/or force control to a value for an output (actuator) of an electronic system. In general, this service is used for relatively simple (e.g., static) input substitution / output control whereas routineControl is used if more complex input substitution / output control is necessary.

## 4.12.1 Request message

### 4.12.1.1 Request message definition

#### Request message definition

| A_Data byte | Parameter name                                                                                            | Cvt | Hex value | Mnemonic                  |
|-------------|-----------------------------------------------------------------------------------------------------------|-----|-----------|---------------------------|
| #1          | InputOutputControlByIdentifier Request Service Id                                                         | M   | 2F        | IOCBI                     |
| #2          | dataIdentifier#1[] = [<br>byte#1 (MSB)<br>byte#2]                                                         | M   | 00-FF     | IOI_<br>B1                |
| #3          |                                                                                                           | M   | 00-FF     | B2                        |
| #4          | ControlOptionRecord[] = [<br>inputOutputControlParameter#1<br>ControlState#1<br>.<br>.<br>ControlState#m] | M1  | 00-FF     | COR_<br>IOCP_<br>CS_<br>. |
| #5          |                                                                                                           | C1  | 00-FF     | CS_<br>.                  |
| .           |                                                                                                           | .   | .         | .                         |
| #4+(m-1)    |                                                                                                           | C1  | 00-FF     | CS_<br>.                  |
| #4+m        | controlEnableMaskRecord[] = [<br>controlMask#1<br>.<br>.<br>controlMask#r]                                | C2  | 00-FF     | CEM_<br>CM_<br>.          |
| .           |                                                                                                           | .   | .         | .                         |
| #4+m+(r-1)  |                                                                                                           | C2  | 00-FF     | CM_<br>.                  |

M1: InputOutputControlParameter shall be implemented as defined in following table.

C1: The presence of this parameter depends on the dataIdentifier and the inputOutputControlParameter.

C2: The presence of this parameter depends on the dataIdentifier (see controlEnableMaskRecord definition) and vehicle manufacturer requirements.

The tool may take control of multiple outputs simultaneously with a single request message or by sending multiple device control service messages.

Certain combinations of outputs may not be allowed to be controlled at the same time due to safety or to avoid product damage. Any such restrictions shall be negotiated between the DRE, and shall be written down.

It shall be ensured that the actual output or input state can be retrieved by using the service

ReadDataByIdentifier(0x22) with the same DID.

#### InputOutputControlParameter

| Hex<br>(bit 6-0) | Description                                                                                                                                                                                                                                                                                                                                    | Cvt | Mnemonic |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00               | returnControlToECU<br>This value shall indicate to the server that the client no longer has control over the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier.<br>Number of controlState bytes in request: 0<br>Number of controlState bytes in possible response: depends on the dataIdentifier | M   | RCTECU   |
| 01               | resetToDefault<br>This value shall indicate to the server that it is requested to reset the input signal, internal parameter or output signal referenced by the                                                                                                                                                                                | U   | RTD      |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------|
|       | inputOutputLocalIdentifier to its default state.<br>Number of controlState bytes in request: 0<br>Number of controlState bytes in possible response: depends on the dataIdentifier                                                                                                                                                                                                                                                                                                                                                                     |   |              |
| 02    | freezeCurrentState<br>This value shall indicate to the server that it is requested to freeze the current state of the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier.<br>Number of controlState bytes in request: 0<br>Number of controlState bytes in possible response: depends on the dataIdentifier                                                                                                                                                                                                | U | FCS          |
| 03    | shortTermAdjustment<br>This value shall indicate to the server that it is requested to adjust the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier in RAM to the value(s) included in the controlOption parameter(s) (e.g. set Idle Air Control Valve to a specific step number, set pulse width of valve to a specific value/duty cycle).<br>Number of controlState bytes in request: depends on the dataIdentifier<br>Number of controlState bytes in possible response: depends on the dataIdentifier | M | STA          |
| 04-FF | ISOSAEReserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M | ISOSAERESRVD |

#### 4.12.1.2 Request message data parameter definition

The following data parameters are defined for this service.

| Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>controlOptionRecord</b><br>The controlOptionRecord consists of one or multiple bytes (inputOutputControlParameter and controlState#1 to controlState#m).<br>The controlOptionRecord parameter details shall be implemented as defined in 'InputOutputControlParameter' table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>controlEnableMaskRecord</b><br>The ControlEnableMask of each dataIdentifier consists of one or multiple bytes (controlMask#1 to controlMask#r, same length with the DID which be controlled). The ControlEnableMask shall only be supported when the inputOutputControlParameter is used and the dataIdentifier has a parameter type of bit-mapped. There shall be one bit in the ControlEnableMask corresponding to each bit of individual parameter defined within the dataIdentifier.<br>The value of each bit shall determine whether the corresponding parameter in the dataIdentifier will be affected by the request. A bit value of '0' in the ControlEnableMask shall represent that the corresponding parameter is not affected by this request and a bit value of '1' shall represent that the corresponding parameter is affected by this request. The most significant bit of ControlMask#1 shall correspond to the first parameter in the ControlState starting at the most significant bit of ControlState#1, the second most significant bit of ControlMask#1 shall correspond to the second parameter in the ControlState, and continuing on in this fashion utilizing as many ControlMask bytes as necessary to mask all parameters.<br>For example, the least significant bit of ControlMask#2 would correspond to the 16th parameter in the controlState. For bit-mapped dataIdentifiers, unsupported bits shall also have a corresponding bit in the ControlEnableMask so that the position of the mask bit of every parameter in the ControlEnableMask shall exactly match the position of the corresponding parameter in the controlState. |

## 4.12.2 Positive response message

### 4.12.2.1 Positive response message definition

#### Positive response message definition

| A_Data byte | Parameter name                                                                                             | Cvt | Hex value | Mnemonic                  |
|-------------|------------------------------------------------------------------------------------------------------------|-----|-----------|---------------------------|
| #1          | InputOutputControlByIdentifier Response Service Id                                                         | M   | 6F        | IOCBIPR                   |
| #2          | dataIdentifier [] = [<br>byte#1 (MSB)<br>byte#2]                                                           | M   | 00-FF     | IOI_<br>B1                |
| #3          |                                                                                                            | M   | 00-FF     | B2                        |
| #4          | controlStatusRecord [] = [<br>inputOutputControlParameter#1<br>ControlState#1<br>.<br>.<br>ControlState#m] | M   | 00-FF     | COR_<br>IOCP_<br>CS_<br>. |
| #5          |                                                                                                            | C1  |           | CS_<br>.                  |
| .           |                                                                                                            | .   |           | .                         |
| #4+(m-1)    |                                                                                                            | C1  |           | CS_<br>.                  |

C1: The presence of this parameter depends on the dataIdentifier and the inputOutputControlParameter

### 4.12.2.2 Positive response message data parameter definition

The following data parameters are defined for this service.

| Definition                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>controlStatusRecord</b><br>The controlState parameter consists of multiple bytes (InputOutputControlParameter and controlState#1 to controlState#m) which include e.g. feedback data. The controlStatusRecord parameter details shall be implemented as defined in 'InputOutputControlParameter' table. |

## 4.12.3 Negative response message & Supported negative response codes (NRC\_)

#### Negative response message definition

| A_Data byte | Parameter name                                    | Cvt | Hex value | Mnemonic   |
|-------------|---------------------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id                      | M   | 7F        | SIDRSIDNRQ |
| #2          | InputOutputControlByIdentifier Request Service Id | M   | 2F        | IOCBIR     |
| #3          | responseCode                                      | M   | 00-7F     | NRC_       |

#### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of | M   | GR       |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |        |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
|    | times, the NRC 0x10 should be sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |        |
| 13 | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M | IMLOIF |
| 21 | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21. This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services. | M | BRR    |
| 22 | <b>conditionsNotCorrect</b><br>This code shall be returned if the criteria for the request InputOutputControl are not met.<br>If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTS, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | U | CNC    |
| 31 | <b>requestOutOfRange</b><br>This NRC shall be sent if:<br>--- the requested dataIdentifier value is not supported by the device;<br>--- the value contained in the inputOutputControlParameter is invalid (see definition of inputOutputControlParameter);<br>--- one or multiple of the applicable controlState values of the controlOptionRecord record are invalid;<br>--- the combination of bits enabling control in the ControlEnableMaskRecord is not supported by the device;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M | ROOR   |
| 33 | <b>securityAccessDenied</b><br>This code shall be returned if a client sends a request with a valid secure dataIdentifier and the server's security feature is currently active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M | SAD    |
| 78 | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.                                                                                                                                                                                                                     | M | RCRRP  |
| 7F | <b>serviceNotSupportedInActiveSession</b><br>This response code indicates that the requested action will not be taken because the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M | SNSIAS |

|  |                                                                                                                                                                                                                                                               |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  | server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used. |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

### 4.13 RoutineControl (31 hex) service

The RoutineControl service is used by the client to execute a defined sequence of steps and obtain any relevant results. There is a lot of flexibility with this service, but typical usage may include functionality such as erasing memory, resetting or learning adaptive data, running a self-test, overriding the normal server control strategy, and controlling a server value to change over time including predefined sequences (e.g., close convertible roof) to name a few. In general, when used to control outputs this service is used for more complex type control whereas inputOutputControlByIdentifier is used for relatively simple (e.g., static) output control.

#### 4.13.1 Request message

##### 4.13.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                                                                               | Cvt | Hex value | Mnemonic       |
|-------------|----------------------------------------------------------------------------------------------|-----|-----------|----------------|
| #1          | RoutineControl Request Service Id                                                            | M   | 31        | RC             |
| #2          | sub-function = [ routineControlType ]                                                        | M   | 00-FF     | LEV_<br>RCTP_  |
| #3          | routineIdentifier [] = [ byte#1 (MSB)<br>byte#2]                                             | M   | 00-FF     | RI_<br>B1      |
| #4          |                                                                                              | M   | 00-FF     | B2             |
| #5          | routineControlOptionRecord[] = [ routineControlOption#1<br>.<br>.<br>routineControlOption#m] | C/U | 00-FF     | RCEOR_<br>RCO_ |
| .           |                                                                                              | .   | .         | .              |
| .           |                                                                                              | .   | .         | .              |
| #n          |                                                                                              | C/U | 00-FF     | RCO_           |

The presence of the C parameter is user-optional for sub-function parameter startRoutine and stopRoutine.

##### 4.13.1.2 Request message sub-function parameter \$Level (LEV\_) definition

The sub-function parameters are used by this service to select the control of the routine. Explanations and usage of the possible levels are detailed below [suppressPosRspMsgIndicationBit (bit 7) not shown].

Request message sub-function definition

| Hex (bit 6-0) | Description           | Mnemonic     |
|---------------|-----------------------|--------------|
| 00            | ISOSAEReserved        | ISOSAERESRVD |
| 01            | startRoutine          | STR          |
| 02            | stopRoutine           | STPR         |
| 03            | requestRoutineResults | RRR          |

|       |                |              |
|-------|----------------|--------------|
| 04-7F | ISOSAEReserved | ISOSAERESRVD |
|-------|----------------|--------------|

#### 4.13.1.3 Request message data parameter definition

The following data parameters are defined for this service.

| Definition                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>routineIdentifier</b><br>This parameter identifies a server local routine and is out of the range of defined dataIdentifiers                                                                                                                                                                                                                                            |
| <b>routineControlOptionRecord</b><br>This parameter record contains either:<br>— routine entry option parameters, which optionally specify start conditions of the routine (e.g. timeToRun, startUpVariables, etc.); or<br>— routine exit option parameters which optionally specify stop conditions of the routine (e.g.timeToExpireBeforeRoutineStops, variables, etc.). |

Some RIDs may require security access. Security access is applied to the RID and not the service itself.

Once active, the routine application is responsible for monitoring all conditions for the continued execution of the routine. If any conditions are not met during execution then the routine application shall terminate the routine.

### 4.13.2 Positive response message

#### 4.13.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                                                              | Cvt | Hex value | Mnemonic    |
|-------------|-----------------------------------------------------------------------------|-----|-----------|-------------|
| #1          | RoutineControl Response Service Id                                          | M   | 71        | RCPR        |
| #2          | routineControlType                                                          | M   | 00-7F     | RCTP_       |
| #3          | routineIdentifier [] = [<br>byte#1 (MSB)<br>byte#2]                         | M   | 00-FF     | RI_<br>B1   |
| #4          |                                                                             | M   | 00-FF     | B2          |
| #5          | routineStatusRecord[] = [<br>routineStatus#1<br>.<br>.<br>routineStatus#m ] | U   | 00-FF     | RSR_<br>RS_ |
| .           |                                                                             | .   | .         | .           |
| .           |                                                                             | .   | .         | .           |
| #n          |                                                                             | U   | 00-FF     | RS_         |

U: The RoutineStatusByte #m is only to be included in the routineStatusRecord [] if specified for the routineIdentifier (RID) by SmartMi.

#### 4.13.2.2 Positive response message data parameter definition

Response message data parameter definition

| Definition                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------|
| <b>routineControlType</b><br>This parameter is an echo of bits 6-0 of the sub-function parameter from the request message. |
| <b>routineIdentifier</b><br>This parameter is an echo of the routineIdentifier from the request message.                   |

#### routineStatusRecord

This parameter record is used to give to the client either:

- additional information about the status of the server following the start of the routine; or
- additional information about the status of the server after the routine has been stopped (e.g. totalRunTime, results generated by the routine before stopped, etc.); or
- results (exit status information) of the routine which has been stopped previously in the server.

### 4.13.3 Negative response message & Supported negative response codes (NRC\_)

#### Negative response message definition

| A_Data byte | Parameter name                    | Cvt | Hex value | Mnemonic   |
|-------------|-----------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id      | S   | 7F        | SIDRSIDNRQ |
| #2          | RoutineControl Request Service Id | S   | 31        | RC         |
| #3          | responseCode                      | M   | 00-7F     | NRC_       |

#### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Cvt | Mnemonic |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                              | M   | GR       |
| 12  | <b>subFunctionNotSupported</b><br>Send if the sub-function parameter in the request message is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                              | M   | SFNS     |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M   | IMLOIF   |
| 22  | <b>conditionsNotCorrect</b><br>This code shall be returned if the criteria for the request RoutineControl are not met. If a node supports response code \$22, the reason(s) to generate this response code shall be documented in the CTS, SSTS, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.                                                                                       | U   | CNC      |
| 24  | <b>requestSequenceError</b><br>This NRC shall be returned if<br>--- the routine is currently active and can not be restarted when the 'startRoutine' sub-function is received (it is up to the vehicle manufacturer whether a given routine can be restarted while active),<br>--- the routine is not currently active when the 'stopRoutine' sub-function is received,<br>--- routine results are not available when the 'requestRoutineResults' sub-function is received (e.g., the requested routineIdentifier has never been started). | M   | RSE      |
| 31  | <b>requestOutOfRange</b><br>This code shall be returned if:<br>1) the server does not support the requested routineIdentifier;<br>2) the user optional routineControlOptionRecord contains invalid data for the                                                                                                                                                                                                                                                                                                                            | M   | ROOR     |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |       |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------|
|    | requested routineIdentifier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |       |
| 33 | <b>securityAccessDenied</b><br>This code shall be sent if this code is returned if a client sends a request with a valid secure routineIdentifier and the server's security feature is currently active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | M | SAD   |
| 72 | <b>generalProgrammingFailure</b><br>This return code shall be sent if the server detects an error when performing a routine, which accesses server internal memory. An example is when the routine erases or programmes a certain memory location in the permanent memory device (e.g. Flash Memory) and the access to that memory location fails.                                                                                                                                                                                                                                                                                                                                                                                                   | M | GPF   |
| 78 | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row. | M | RCRRP |

#### 4.14 RequestDownload (34 hex) service

The requestDownload service is used by the client to initiate a data transfer from the client to the server (download).

##### 4.14.1 Request message

###### 4.14.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                                                     | Cvt | Hex value | Mnemonic  |
|-------------|--------------------------------------------------------------------|-----|-----------|-----------|
| #1          | RequestDownload Request Service Id                                 | M   | 34        | RD        |
| #2          | dataFormatIdentifier                                               | M   | 00-FF     | DFI_      |
| #3          | addressAndLengthFormatIdentifier                                   | M   | 00-FF     | ALFID     |
| #4          | memoryAddress[] = [<br>byte#1 (MSB)<br>byte#2<br>byte#3<br>byte#4] | M   | 00-FF     | MA_<br>B1 |
| #5          |                                                                    | M   | 00-FF     | B2        |
| #6          |                                                                    | M   | 00-FF     | B3        |
| #7          |                                                                    | M   | 00-FF     | B4        |
| #8          | memorySize [] = [<br>byte#1 (MSB)<br>byte#2<br>byte#3<br>byte#4]   | M   | 00-FF     | MS_<br>B1 |
| #9          |                                                                    | M   | 00-FF     | B2        |
| #10         |                                                                    | M   | 00-FF     | B3        |
| #11         |                                                                    | M   | 00-FF     | B4        |

###### 4.14.1.2 Request message data parameter definition

The following data parameters are defined for this service.

|            |
|------------|
| Definition |
|------------|

#### dataFormatIdentifier

This data parameter is a one-byte value with each nibble encoded separately. The high nibble specifies the “compressionMethod” and the low nibble specifies the “encryptingMethod”. The value 00 hex specifies that no compressionMethod nor encryptingMethod is used. Values other than 00 hex are vehicle-manufacturer-specific.

#### addressAndLengthFormatIdentifier

This parameter is a one byte value with each nibble encoded separately. SmartMi supports only 4 byte lengths and address sizes in the request message. If the ECU uses less than 4 bytes for addressing or length then the Most Significant Byte(s) are ignored by the ECU

bit 7 - 4: Length (number of bytes) of the memorySize parameter required to be 4 (0100b)

bit 3 - 0: Length (number of bytes) of the memoryAddress parameter required to be 4 (0100b)

#### memoryAddress

The parameter memoryAddress is the starting address of the server memory to which the data is to be written. The number of bytes used for this address is defined by the low nibble (bit 3 - 0) of the addressFormatIdentifier. Byte#m in the memoryAddress parameter is always the least significant byte of the address being referenced in the server. The most significant byte of the address can be used as a memoryIdentifier.

An example of the use of a memoryIdentifier would be a dual processor server with 16-bit addressing and memory address overlap (when a given address is valid for either processor but yields a different physical memory device or when internal and external flash is used). In this case, an otherwise unused byte within the memoryAddress parameter can be specified as a memoryIdentifier used to select the desired memory device. Usage of this functionality shall be as defined by vehicle manufacturer/system supplier.

#### memorySize

This parameter shall be used by the server to compare the uncompressed memory size with the total amount of data transferred during the TransferData service. This increases the programming security. The number of bytes used for this size is defined by the high nibble (bit 7 - 4) of the addressFormatIdentifier.

## 4.14.2 Positive response message

### 4.14.2.1 Positive response message definition

#### Positive response message definition

| A_Data byte | Parameter name                      | Cvt | Hex value | Mnemonic |
|-------------|-------------------------------------|-----|-----------|----------|
| #1          | RequestDownload Response Service Id | M   | 74        | RDPR     |
| #2          | lengthFormatIdentifier              | M   | 00-F0     | LFID     |
| #3          | maxNumberOfBlockLength [] = [       | M   | 00-FF     | MNROB_   |
| .           | byte#1 (MSB)                        | .   | .         | B1       |
| .           | .                                   | .   | .         | .        |
| #n          | byte#m ]                            | M   | 00-FF     | Bm       |

### 4.14.2.2 Positive response message data parameter definition

#### Response message data parameter definition

| Definition                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>lengthFormatIdentifier</b><br>This parameter is a one-byte value with each nibble encoded separately:<br>bit 7 - 4: length (number of bytes) of the maxNumberOfBlockLength parameter;<br>bit 3 - 0: reserved by document, to be set to 0 hex. |

The format of this parameter is compatible to the format of the addressAndLengthFormatIdentifier parameter contained in the request message, except that the lower nibble has to be set to 0 hex.

#### maxNumberOfBlockLength

This parameter is used by the requestDownload positive response message to inform the client how many data bytes (maxNumberOfBlockLength) shall be included in each TransferData request message from the client. This length reflects the complete message length, including the service identifier and the data parameters present in the TransferData request message. This parameter allows the client to adapt to the receive buffer size of the server before it starts transferring data to the server.

### 4.14.3 Negative response message & Supported negative response codes (NRC\_)

#### Negative response message definition

| A_Data byte | Parameter name                     | Cvt | Hex value | Mnemonic   |
|-------------|------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id       | M   | 7F        | SIDRSIDNRQ |
| #2          | RequestDownload Request Service Id | M   | 34        | RD         |
| #3          | responseCode                       | M   | 00-7F     | NRC_       |

#### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cvt | Mnemonic |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   | GR       |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.<br>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services. | M   | BRR      |
| 22  | <b>conditionsNotCorrect</b><br>This return code shall be sent if a server receives a request for this service while in the process of receiving a download of a software or calibration module. This could occur if there is a data size mismatch between the server and the client during the download of a module.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | U   | CNC      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |        |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
|    | If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTs, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.                                                                                                                                                                                                                                                                                                                                                                                                                 |   |        |
| 31 | <b>requestOutOfRange</b><br>This return code shall be sent if<br>1) the specified dataFormatIdentifier is not valid,<br>2) the specified addressAndLengthFormatIdentifier is not valid, or<br>3) the specified memoryAddress/memorySize is not valid.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M | ROOR   |
| 33 | <b>securityAccessDenied</b><br>This return code shall be sent if the server is secure (for servers that support the SecurityAccess service) when a request for this service has been received.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | M | SAD    |
| 70 | <b>uploadDownloadNotAccepted</b><br>This NRC indicates that an attempt to download to a server's memory cannot be accomplished due to some fault conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M | UDNA   |
| 78 | <b>requestCorrectlyReceived-ResponsePending</b><br>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row. | M | RCRRP  |
| 7F | <b>serviceNotSupportedInActiveSession</b><br>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.                                                                                                                                                                                                                                                                                                                                                      | M | SNSIAS |

## 4.15 RequestUpload (0x35) service

The RequestUpload service is used by the client to initiate a data transfer from the server to the client (upload). After the server has received the requestUpload request message the server shall take all necessary actions to send data before it sends a positive response message.

### 4.15.1 Request message

#### 4.15.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                   | Cvt | Hex value | Mnemonic |
|-------------|----------------------------------|-----|-----------|----------|
| #1          | TransferData Request Service Id  | M   | 35        | RU       |
| #2          | dataFormatIdentifier             | M   | 00-FF     | DFI_     |
| #3          | addressAndLengthFormatIdentifier | M   | 00-FF     | ALFID    |

|          |                      |    |       |     |
|----------|----------------------|----|-------|-----|
| #4       | memoryAddress [] = [ | M  | 00-FF | MA_ |
| .        | byte#1(MSB)          | .  | .     | B1  |
| .        | .                    | .  | .     | .   |
| #(m-1)+4 | byte#m ]             | U1 | 00-FF | Bm  |
| #n-(k-1) | memorySize [] = [    | M  | 00-FF | MS_ |
| .        | byte#1(MSB)          | .  | .     | B1  |
| .        | .                    | .  | .     | .   |
| #n       | byte#k ]             | U2 | 00-FF | Bk  |

U1: The presence of this parameter depends on address length information parameter of the addressAndLengthFormatIdentifier

U2: The presence of this parameter depends on the memory size length information of the addressAndLengthFormatIdentifier.

#### 4.15.1.2 Request message data parameter definition

The following data parameters are defined for this service.

| Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>dataFormatIdentifier</b><br>This data-parameter is a one byte value with each nibble encoded separately. The high nibble specifies the "compressionMethod", and the low nibble specifies the "encryptingMethod". The value 0x00 specifies that neither compressionMethod nor encryptingMethod is used. Only value 0x00 shall be used.                                                                                                                               |
| <b>addressAndLengthFormatIdentifier</b><br>This parameter is a one byte value with each nibble encoded separately:<br>bit 7 - 4: Length (number of bytes) of the memorySize parameter<br>bit 3 - 0: Length (number of bytes) of the memoryAddress parameter                                                                                                                                                                                                            |
| <b>memoryAddress</b><br>The parameter memoryAddress is the starting address of server memory from which data is to be retrieved. The number of bytes used for this address is defined by the low nibble (bit 3 - 0) of the addressAndLengthFormatIdentifier. Byte#m in the memoryAddress parameter is always the least significant byte of the address being referenced in the server. The most significant byte(s) of the address can be used as a memory identifier. |
| <b>memorySize</b><br>This parameter shall be used by the server to compare the memory size with the total amount of data transferred during the TransferData service. This increases the programming security. The number of bytes used for this size is defined by the high nibble (bit 4) of the addressAndLengthFormatIdentifier.                                                                                                                                   |

## 4.15.2 Positive response message

### 4.15.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                    | Cvt | Hex value | Mnemonic |
|-------------|-----------------------------------|-----|-----------|----------|
| #1          | RequestUpload Response Service Id | M   | 75        | RDPR     |
| #2          | lengthFormatIdentifier            | M   | 00-F0     | LFID     |
|             | maxNumberOfBlockLength [] = [     |     |           | MNROB_   |

|    |              |   |       |    |
|----|--------------|---|-------|----|
| #3 | byte#1 (MSB) | M | 00-FF | B1 |
| .  | .            | . | .     | .  |
| .  | .            | . | .     | .  |
| #n | byte#m ]     | M | 00-FF | Bm |

#### 4.15.2.2 Positive response message data parameter definition

##### Response message data parameter definition

| Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>lengthFormatIdentifier</b><br>This parameter is a one byte value with each nibble encoded separately:<br>— bit 7 - 4: Length (number of bytes) of the maxNumberOfBlockLength parameter;<br>— bit 3 - 0: reserved by document, to be set to 0x0;<br>The format of this parameter is compatible to the format of the addressAndLengthFormatIdentifier parameter contained in the request message, except that the lower nibble has to be set to 0x0.                                                                                                                                                                                                                                                                                                                                    |
| <b>maxNumberOfBlockLength</b><br>This parameter is used by the requestUpload positive response message to inform the client how many data bytes shall be included in each TransferData positive response message from the server. This length reflects the complete message length, including the service identifier and the data-parameters present in the TransferData positive response message. This parameter allows the client to adapt to the send buffer size of the server before the server starts transferring data to the client. A client is required to accept transferData responses that are equal in length to the reported maxNumberOfBlockLength.<br>NOTE The last transferData response within a given block may be required to be less than maxNumberOfBlockLength. |

#### 4.15.3 Negative response message & Supported negative response codes (NRC\_)

##### Negative response message definition

| A_Data byte | Parameter name                     | Cvt | Hex value | Mnemonic   |
|-------------|------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id       | M   | 7F        | SIDRSIDNRQ |
| #2          | RequestDownload Request Service Id | M   | 34        | RD         |
| #3          | responseCode                       | M   | 00-7F     | NRC_       |

##### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent. | M   | GR       |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                           | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation                                                                                                                                                      | M   | BRR      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |        |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
|    | <p>documents.</p> <p>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.</p> <p>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.</p> <p>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p>                                                                                                                               |   |        |
| 22 | <p><b>conditionsNotCorrect</b></p> <p>This return code shall be sent if a server receives a request for this service while in the process of receiving a download of a software or calibration module. This could occur if there is a data size mismatch between the server and the client during the download of a module.</p> <p>If a node supports response code \$22, the reason(s) to generate this response code <i>shall</i> be documented in the CTS, SSTS, or supplemental diagnostic specification referenced by either of the two preceding documents, and <i>shall</i> not conflict with any of the other negative response codes defined for this service.</p>                                                                                         | U | CNC    |
| 31 | <p><b>requestOutOfRange</b></p> <p>This return code shall be sent if</p> <ol style="list-style-type: none"> <li>1) the specified dataFormatIdentifier is not valid,</li> <li>2) the specified addressAndLengthFormatIdentifier is not valid, or</li> <li>3) the specified memoryAddress/memorySize is not valid.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                         | M | ROOR   |
| 33 | <p><b>securityAccessDenied</b></p> <p>This return code shall be sent if the server is secure (for servers that support the SecurityAccess service) when a request for this service has been received.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M | SAD    |
| 70 | <p><b>uploadDownloadNotAccepted</b></p> <p>This NRC indicates that an attempt to download to a server's memory cannot be accomplished due to some fault conditions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | M | UDNA   |
| 78 | <p><b>requestCorrectlyReceived-ResponsePending</b></p> <p>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.</p> <p>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.</p> | M | RCRRP  |
| 7F | <p><b>serviceNotSupportedInActiveSession</b></p> <p>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.</p>                                                                                                                                                                                                                                                                                                                                                          | M | SNSIAS |

## 4.16 TransferData (36 hex) service

The TransferData service is used by the client to transfer data either from the client to the server (download).

### 4.16.1 Request message

#### 4.16.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                  | Cvt | Hex value | Mnemonic |
|-------------|---------------------------------|-----|-----------|----------|
| #1          | TransferData Request Service Id | M   | 36        | TD       |
| #2          | blockSequenceCounter            | M   | 00-FF     | BSC      |
| #3          | transferDataRecord [] = [       | M   | 00-FF     | TRPR_    |
| .           | Data#1                          |     |           | TRTP_    |
| .           | .                               |     |           | .        |
| #n          | Data #m ]                       | U   | 00-FF     | TRTP_    |

#### 4.16.1.2 Request message data parameter definition

The following data parameters are defined for this service.

| Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>blockSequenceCounter</b></p> <p>The blockSequenceCounter parameter value starts at 01 hex with the first TransferData request that follows the RequestDownload (34 hex) service. Its value is incremented by 1 for each subsequent TransferData request. At the value of FF hex, the blockSequenceCounter rolls over and starts at 00 hex with the next TransferData request message.</p> <p>Example use cases:</p> <ol style="list-style-type: none"> <li>If a TransferData request to download data is correctly received and processed in the server but the positive response message does not reach the client, then the client would determine an application layer timeout and would repeat the same request (including the same blockSequenceCounter). The server would receive the repeated TransferData request and could determine based on the included blockSequenceCounter that this TransferData request is repeated. The server would send the positive response message immediately without writing the data once again into its memory.</li> <li>If the TransferData request to download data is not received correctly in the server, then the server would not send a positive response message. The client would determine an application layer timeout and would repeat the same request (including the same blockSequenceCounter). The server would receive the repeated TransferData request and could determine based on the included blockSequenceCounter that this is a new TransferData. The server would process the service and would send the positive response message.</li> <li>If a TransferData request to upload data is correctly received and processed in the server but the positive response message does not reach the client, then the client would determine an application layer timeout and would repeat the same request (including the same blockSequenceCounter). The server would receive the repeated TransferData request and could determine based on the included blockSequenceCounter that this TransferData request is repeated. The server would send the positive response message immediately, accessing the previously provided data once again in its memory.</li> <li>If the TransferData request to upload data is not received correctly in the server, then the server would not send a positive response message. The client would determine an application layer timeout and would repeat the same request (including the same blockSequenceCounter). The server would receive the repeated TransferData request and could determine based on the</li> </ol> |

included blockSequenceCounter that this is a new TransferData. The server would process the service and would send the positive response message.

## 4.16.2 Positive response message

### 4.16.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                      | Cvt | Hex value | Mnemonic |
|-------------|-------------------------------------|-----|-----------|----------|
| #1          | RequestDownload Response Service Id | M   | 76        | RDPR     |
| #2          | blockSequenceCounter                | M   | 00-FF     | BSC      |

## 4.16.3 Negative response message & Supported negative response codes (NRC\_)

Negative response message definition

| A_Data byte | Parameter name                  | Cvt | Hex value | Mnemonic   |
|-------------|---------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id    | M   | 7F        | SIDRSIDNRQ |
| #2          | TransferData Request Service Id | M   | 36        | TD         |
| #3          | responseCode                    | M   | 00-7F     | NRC_       |

Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Cvt | Mnemonic |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent.                                                                                                                                                                                                                    | M   | GR       |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong (e.g. message length does not meet the requirements of the maxNumberOfBlockLength parameter returned in the positive response to requestDownload).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b><br>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.<br>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.<br>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21. | M   | BRR      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |        |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
|    | <p>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |        |
| 24 | <p><b>requestSequenceError</b><br/>The server shall use this response code:</p> <ol style="list-style-type: none"> <li>1) if the RequestDownload service is not active when a request for this service is received;</li> <li>2) if the RequestDownload is active, but the server has already received all data as determined by the memorySize parameter in the active RequestDownload service.</li> </ol> <p>The repetition of a TransferData request message with a blockSequenceCounter equal to the one included in the previous TransferData request message shall be accepted by the server.</p>                                                                                                                                                        | M | CNC    |
| 31 | <p><b>requestOutOfRange</b><br/>This return code shall be sent if the transferRequestParameterRecord contains additional control parameters (e.g. additional address information) and this control information is invalid.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M | ROOR   |
| 33 | <p><b>securityAccessDenied</b><br/>This return code shall be sent if the server is secure (for servers that support the SecurityAccess service) when a request for this service has been received.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M | SAD    |
| 71 | <p><b>transferDataSuspended</b><br/>This return code shall be sent if:</p> <ol style="list-style-type: none"> <li>1) the response code indicates that a data transfer operation was halted due to a fault;</li> <li>2) the download module length does not meet the requirements of the memorySize parameter sent in the request message of the requestDownload service.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                           | M | TDS    |
| 72 | <p><b>generalProgrammingFailure</b><br/>This return code shall be sent if the server detects an error when erasing or programming a memory location in the permanent memory device (e.g. Flash Memory) during the download of data.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | M | GPF    |
| 73 | <p><b>wrongBlockSequenceCounter</b><br/>This return code shall be sent if the server detects an error in the sequence of the blockSequenceCounter.<br/>The repetition of a TransferData request message with a blockSequenceCounter equal to the one included in the previous TransferData request message shall be accepted by the server.</p>                                                                                                                                                                                                                                                                                                                                                                                                               | M | WBSC   |
| 78 | <p><b>requestCorrectlyReceived-ResponsePending</b><br/>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.<br/>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.</p> | M | RCRRP  |
| 7F | <p><b>serviceNotSupportedInActiveSession</b><br/>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.</p>                                                                                                                                                                                                                                                                                                                                                       | M | SNSIAS |

|       |                                                                                                                                                                                                                                                                  |   |         |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|
| 92/93 | <b>voltageTooHigh/voltageTooLow</b><br>This return code shall be sent, as applicable, if the voltage measured at the primary power pin of the server is out of the acceptable range for downloading data into the server's permanent memory (e.g. Flash Memory). | M | VTH/VTL |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|

## 4.17 RequestTransferExit (37 hex) service

This service is used by the client to terminate a data transfer between client and server.

### 4.17.1 Request message

#### 4.17.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                         | Cvt | Hex value | Mnemonic |
|-------------|----------------------------------------|-----|-----------|----------|
| #1          | RequestTransferExit Request Service Id | M   | 37        | RTE      |

### 4.17.2 Positive response message

#### 4.17.2.1 Positive response message definition

Positive response message definition

| A_Data byte | Parameter name                          | Cvt | Hex value | Mnemonic |
|-------------|-----------------------------------------|-----|-----------|----------|
| #1          | RequestTransferExit Response Service Id | M   | 77        | RTEPR    |

### 4.17.3 Negative response message & Supported negative response codes (NRC\_)

Negative response message definition

| A_Data byte | Parameter name                         | Cvt | Hex value | Mnemonic   |
|-------------|----------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id           | S   | 7F        | SIDRSIDNRQ |
| #2          | RequestTransferExit Request Service Id | S   | 37        | RTE        |
| #3          | responseCode                           | M   | 00-7F     | NRC_       |

Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent. | M   | GR       |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                           | M   | IMLOIF   |
| 21  | <b>busyRepeatRequest</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M   | BRR      |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |        |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
|    | <p>This NRC indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.</p> <p>EXAMPLE In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC 0x21 while a different client finishes a diagnostic task.</p> <p>If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC 0x78 shall be used instead of NRC 0x21.</p> <p>This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p> |   |        |
| 24 | <p><b>requestSequenceError</b></p> <p>This return code shall be sent if:</p> <ol style="list-style-type: none"> <li>1) the programming process is not completed when a request for this service is received;</li> <li>2) the RequestDownload service is not active.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | M | CNC    |
| 33 | <p><b>securityAccessDenied</b></p> <p>This return code shall be sent if the server is secure (for servers that support the SecurityAccess service) when a request for this service has been received.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | M | SAD    |
| 72 | <p><b>generalProgrammingFailure</b></p> <p>This NRC shall be returned if the server detects an error when finalizing the data transfer between the client and server (e.g., via an integrity check).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M | GPF    |
| 78 | <p><b>requestCorrectlyReceived-ResponsePending</b></p> <p>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a responsecode different from this.</p> <p>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. The NRC 78 can only be sent 8 times in a row.</p>                                                                                                                                                                                                | M | RCRRP  |
| 7F | <p><b>serviceNotSupportedInActiveSession</b></p> <p>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | M | SNSIAS |

#### 4.18 RequestFileTransfer (38 hex) service

The requestFileTransfer service is used by the client to initiate a file data transfer from either the client to the server or from the server to the client (download or upload). Additionally, this service has capabilities to

retrieve information about the file system.

## 4.18.1 Request message

### 4.18.1.1 Request message definition

Request message definition

| A_Data byte | Parameter name                         | Cvt  | Hex value | Mnemonic |
|-------------|----------------------------------------|------|-----------|----------|
| #1          | requestFileTransfer Request Service Id | M    | 38        | RFT      |
| #2          | modeOfOperation                        | M    | 01-06     | MOOP     |
| #3          | filePathAndNameLength [ byte#1 (MSB)   | M    | 00-FF     | FPL_B1   |
| #4          | byte#2] (LSB)                          | M    | 00-FF     | FPL_B2   |
| #5          | filePathAndName = [ byte#1 (MSB)       | M    | 00-FF     | FP_B1    |
| :           | :                                      | :    | :         | :        |
| #5+n-1      | byte#n ]                               | C1   | 00-FF     | FP_Bn    |
| #5+n        | dataFormatIdentifier                   | C2   | 00-FF     | DFI_     |
| #5+n+1      | fileSizeParameterLength                | C2   | 00-FF     | FSL      |
| #5+n+2      | fileSizeUnCompressed= [ byte#1 (MSB)   | C2   | 00-FF     | FSUC_B1  |
| :           | :                                      | :    | :         | :        |
| #5+n+2+k-1  | byte#k ]                               | C2,3 | 00-FF     | FSUC_Bk  |
| #5+n+2+k    | fileSizeCompressed= [ byte#1 (MSB)     | C2   | 00-FF     | FSC_B1   |
| :           | :                                      | :    | :         | :        |
| #5+n+1+2k   | byte#k ]                               | C2,3 | 00-FF     | FSC_Bk   |

C1: The length (number of bytes) of this message parameter is defined by the filePathAndNameLength parameter.

C2: The presence of these parameters depends on the modeOfOperation parameter.

C3: The length (number of bytes) of this message parameter is defined by the fileSizeParameterLength.

### 4.18.1.2 Request message data parameter definition

The following data parameters are defined for this service.

| Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>modeOfOperation</b><br>This data-parameter defines the type of operation to be applied to the file or directory indicated in the filePathAndName parameter. <ul style="list-style-type: none"> <li>➤ <b>ISO/SAE reserved(0x00):</b> This value is reserved by this document for future definition.</li> <li>➤ <b>AddFile(0x01):</b> This value shall be used to add the file (download) defined in the filePathAndName parameter.</li> <li>➤ <b>DeleteFile(0x02):</b> This value shall be used to delete the file defined in the filePathAndName parameter.</li> <li>➤ <b>ReplaceFile(0x03):</b> This value shall be used to replace the file (download) defined in the filePathAndName parameter. If the file is not stored at the location the file shall be added.</li> <li>➤ <b>ReadFile(0x04):</b> This value shall be used to read the file (upload) at the location defined by the filePathAndName parameter.</li> </ul> |

- **ReadDir(0x05):** This value shall be used to read the directory defined in the filePathAndName parameter. This value implies that the request does not include a fileName.
- **ResumeFile(0x06):** This value shall be used to resume downloading the file defined in the filePathAndName parameter at the returned filePosition indicator. The file specified in the filePathAndName must already exist in the ECU's file system.
- **ISO/SAE reserved:** This value is reserved by this document for future definition.

#### **filePathAndNameLength**

Defines the length in byte for the parameter filePath.

#### **filePathAndName**

Defines the file system location of the server where the file which shall be added, deleted, replaced or read from depending on the parameter modeOfOperation parameter.

In addition this parameter includes the file name of the file which shall be added, deleted, replaced or read as part of the file path. If the modeOfOperation parameter equals 0x05 (ReadDir), this parameter indicates the directory to be read.

Each byte of this parameter shall be encoded in ASCII format.

#### **dataFormatIdentifier**

This data-parameter is a one-byte value with each nibble encoded separately. The high nibble specifies the "compressionMethod", and the low nibble specifies the "encryptingMethod". The value 0x00 specifies that neither compressionMethod nor encryptingMethod is used. Values other than 0x00 are vehicle manufacturer specific.

If the modeOfOperation parameter equals to 0x02 (DeleteFile) and 0x05 (ReadDir) this parameter shall not be included in the request message.

#### **fileSizeParameterLength**

Defines the length in bytes for both parameters fileSizeUncompressed and fileSizeCompressed.

If the modeOfOperation parameter equals to 0x02 (DeleteFile), 0x04 (ReadFile) or 0x05 (ReadDir) this parameter shall not be included in the request message.

If the modeOfOperation parameter equals to 0x02 (DeleteFile) this parameter shall not be included in the response message.

If the modeOfOperation parameter equals to 0x05 (ReadDir) the value of this parameter shall be equal to 0x00 in the response message.

#### **fileSizeUncompressed**

Defines the size of the uncompressed file in bytes.

If the modeOfOperation parameter equals 0x02 (DeleteFile), 0x04 (ReadFile) or 0x05 (ReadDir) this parameter shall not be included in the request message.

#### **fileSizeCompressed**

Defines the size of the compressed file in bytes.

In the request message: If an uncompressed file is transferred all bytes of this parameter shall be set to the size information used in the parameter fileSizeUncompressed.

If the modeOfOperation parameter equals to 0x02 (DeleteFile), 0x04 (ReadFile) or 0x05 (ReadDir) this parameter shall not be included in the request message.

In the response message: If the modeOfOperation parameter equals to 0x01 (AddFile), 0x02 (DeleteFile, 0x03 (ReplaceFile) ) or 0x05 (ReadDir) this parameter shall not be included in the response message.

## 4.18.2 Positive response message

### 4.18.2.1 Positive response message definition

#### Positive response message definition

| A_Data byte | Parameter name                          | Cvt  | Hex value   | Mnemonic  |
|-------------|-----------------------------------------|------|-------------|-----------|
| #1          | requestFileTransfer Response Service Id | M    | 78          | RFTPR     |
| #2          | modeOfOperation                         | M    | 01-06       | MOOP      |
| #3          | lengthFormatIdentifier                  | C1   | 0x00 – 0xFF | LFID      |
| #4          | maxNumberOfBlockLength= [               | C1,2 | 0x00 – 0xFF | MNROB_    |
| :           | byte#1 (MSB)                            | :    | :           | B1        |
| #4+(m-1)    | :                                       | :    | :           | :         |
|             | byte#m ]                                | C1,2 | 0x00 – 0xFF | Bm        |
| #4+m        | dataFormatIdentifier                    | C1   | 0x00 – 0xFF | DFI_      |
| #4+m+1      | fileSizeOrDirInfoParameterLength [      | C1   | 0x00 – 0xFF | FSDIL_B1  |
| #4+m+2      | byte#1 (MSB)                            | C1   | 0x00 – 0xFF | FSDIL_B2  |
|             | byte#2 (LSB)]                           |      |             |           |
| #4+m+3      | fileSizeUncompressedOrDirInfoLength= [  | C1,3 | 0x00 – 0xFF | FSUDIL_B1 |
| :           | byte#1 (MSB)                            | :    | :           | :         |
| #4+m+3+k-1  | :                                       | :    | :           | :         |
|             | byte#k ]                                | C1,3 | 0x00 – 0xFF | FSUDIL_Bk |
| #4+m+3+k    | fileSizeCompressed= [                   | C1,3 | 0x00 – 0xFF | FSC_B1    |
| :           | byte#1 (MSB)                            | :    | :           | :         |
| #4+m+3+2k-1 | :                                       | :    | :           | :         |
|             | byte#k ]                                | C1,3 | 0x00 – 0xFF | FSC_Bk    |

C1: The presence of these parameters depends on the modeOfOperation parameter.

C2: The length (number of bytes) of this message parameter is defined by the fileSizeOrDirInfoParameterLength parameter.

C3: The length (number of bytes) of this message parameter is defined by the lengthFormatIdentifier parameter.

### 4.18.2.2 Response message data parameter definition

The following data parameters are defined for this service.

| Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>lengthFormatIdentifier</b><br>Defines the length (number of bytes) of the maxNumberOfBlockLength parameter.<br>If the modeOfOperation parameter equals to 0x02 (DeleteFile) this parameter shall be not be included in the response message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>maxNumberOfBlockLength</b><br>This parameter is used by the requestFileTransfer positive response message to inform the client how many data bytes (maxNumberOfBlockLength) to include in each TransferData request message from the client or how many data bytes the server will include in a TransferData positive response when uploading data. This length reflects the complete message length, including the service identifier and the data parameters present in the TransferData request message or positive response message. This parameter allows either the client to adapt to the receive buffer size of the server before it starts transferring data to the server or to indicate how many data bytes will be included in each TransferData positive response in the event that data is uploaded. A server is required to accept transferData requests that are equal in length to its reported maxNumberOfBlockLength. It is server specific what transferData request lengths less than maxNumberOfBlockLength are accepted (if any). |

NOTE The last transferData request within a given block may be required to be less than maxNumberOfBlockLength. It is not allowed for a server to write additional data bytes (i.e., pad bytes) not contained within the transferData message (either in a compressed or uncompressed format), as this would affect the memory address of where the subsequent transferData request data would be written.

If the modeOfOperation parameter equals to 0x02 (DeleteFile) this parameter shall be not be included in the response message.

#### fileSizeOrDirInfoParameterLength

Defines the length in bytes for both parameters fileSizeUncompressedOrDirInfoLength and fileSizeCompressed.

If the modeOfOperation parameter equals to 0x01 (AddFile), 0x02 (DeleteFile) or 0x03 (ReplaceFile) this parameter shall not be included in the response message.

#### fileSizeUncompressedOrDirInfoLength

Defines the size of the uncompressed file to be uploaded or the length of the directory information to be read in bytes.

If the modeOfOperation parameter equals to 0x01 (AddFile), 0x02 (DeleteFile) or 0x03 (ReplaceFile) this parameter shall not be included in the response message.

### 4.18.3 Negative response message & Supported negative response codes (NRC\_)

#### Negative response message definition

| A_Data byte | Parameter name                         | Cvt | Hex value | Mnemonic   |
|-------------|----------------------------------------|-----|-----------|------------|
| #1          | Negative Response Service Id           | S   | 7F        | SIDRSIDNRQ |
| #2          | requestFileTransfer Request Service Id | S   | 38        | RTE        |
| #3          | responseCode                           | M   | 00-7F     | NRC_       |

#### Supported negative response codes

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 10  | <b>generalReject</b><br>This NRC indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation. At no means shall this NRC be a general replacement for the response codes defined in this document.<br>If the number of consecutive NRC 0x78 is sent more than the maximum number of times, the NRC 0x10 should be sent. | M   | GR       |
| 13  | <b>incorrectMessageLengthOrInvalidFormat</b><br>The length of the message is wrong.                                                                                                                                                                                                                                                                                                                                                                                                                           | M   | IMLOIF   |
| 22  | <b>conditionsNotCorrect</b><br>This NRC shall be returned if a server receives a request for this service while in the process of downloading or uploading data or other conditions to be able to execute this service are not met.                                                                                                                                                                                                                                                                           | M   | CNC      |
| 31  | <b>requestOutOfRange</b><br>This return code shall be sent if: <ul style="list-style-type: none"> <li>➤ The specified dataFormatIdentifier is not valid</li> <li>➤ The specified modeOfOperation is not valid</li> <li>➤ The specified fileSizeParameterLength is not valid</li> <li>➤ The specified filePathAndNameLength is not valid</li> </ul>                                                                                                                                                            | M   | BRR      |

|    |                                                                                                                                                                                                                    |   |      |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|
|    | <ul style="list-style-type: none"> <li>➤ The specified fileSizeUncompressed is not valid</li> <li>➤ The specified fileSizeCompressed is not valid</li> <li>➤ The specified filePathAndName is not valid</li> </ul> |   |      |
| 33 | <b>securityAccessDenied</b><br>This return code shall be sent if the server is secure (for server's that support the SecurityAccess service) when a request for this service has been received.                    | M | SAD  |
| 70 | <b>uploadDownloadNotAccepted</b><br>This response code indicates that an attempt to download to a server's memory cannot be accomplished due to some fault conditions.                                             | M | UDNA |

## 5 Diagnostic Error Storage

### 5.1 Time Message

The TCP shall send time message to GW, and GW shall send time message to each ECU.

Each ECU can consume time message to perform corresponding functions.

Each ECU shall record the TIME into non-volatile memory before each sleep.

If the TCP time message is lost, Each ECU calculate the time using its internal clock (crystal oscillator or RTC) , until the new time message has be received again.

The time message described in following table.

| Message Name | Message ID | Period | Signal          | Start Bit | Length (Bits) | Byte Order | Data Type | Range | Conversion                                                                                                      |
|--------------|------------|--------|-----------------|-----------|---------------|------------|-----------|-------|-----------------------------------------------------------------------------------------------------------------|
| TCP_Time     | 0x460      | 1000ms | TimeMod         | 48        | 1             | Motorola   | unsigned  | 0~1   | 0x0:24<br>0x1:12                                                                                                |
|              |            |        | TimeIndcn       | 49        | 1             | Motorola   | unsigned  | 0~1   | 0x0:am<br>0x1:pm                                                                                                |
|              |            |        | Time_Hour       | 35        | 5             | Motorola   | unsigned  | 0~23  | 0x00-0x17:Valid hour<br>0x18-0x1E:Reserved<br>0x1F:Invalid                                                      |
|              |            |        | Time_Minutes    | 40        | 6             | Motorola   | unsigned  | 0~59  | 0x00-0x3B:Valid minute<br>0x3C-0x3E:Reserved<br>0x3F:Invalid                                                    |
|              |            |        | Time_Second     | 50        | 6             | Motorola   | unsigned  | 0~59  | 0x00-0x3B:Valid second<br>0x3C-0x3E:Reserved<br>0x3F:Invalid                                                    |
|              |            |        | Time_Year_Left  | 12        | 4             | Motorola   | unsigned  | 0~99  | 0x0-0x9:Valid year(tens place)<br>0xA-0xE:Reserved<br>0xF:Invalid<br>hundreds place is 0, thousands place is 2. |
|              |            |        | Time_Year_Right | 8         | 4             | Motorola   | unsigned  | 0~99  | 0x0-0x9:Valid year(ones place)<br>0xA-0xE:Reserved<br>0xF:Invalid<br>hundreds place is 0,                       |

|  |  |  |            |    |   |          |          |      |                                                        |
|--|--|--|------------|----|---|----------|----------|------|--------------------------------------------------------|
|  |  |  |            |    |   |          |          |      | thousands place is 2.                                  |
|  |  |  | Time_Month | 20 | 4 | Motorola | unsigned | 0~12 | 0x0:Invalid<br>0x1-0xC:Valid month<br>0xD-0xF:Reserved |
|  |  |  | Time_Day   | 24 | 5 | Motorola | unsigned | 0~1F | 0x0:Invalid<br>0x1-0x1F:Valid day                      |

## 5.2 Diagnostic Fault Detection and DTC Clear/Setting Requirement

### 5.2.1 Diagnostic Fault Detection

If a detected fault can be regarded as potential hazard, failsoft action shall be performed to ensure a safe behavior of the system.

The RAM check, ROM/EEPROM/Flash check and I/O check shall be performed but not limited during initialization or shutdown self-test.

The self-test operation refers to all tests performed once the ECU is running in operational software.

All self-test shall be performed without any activation done.

All ECUs shall continuously perform self-test while in an active state.

All ECUs shall be designed so that diagnostics can be performed on all discrete inputs and outputs connected to the ECU. If multiple ECUs are connected to the same circuit they shall set the same DTC + FTB combination.

The total test time for initialization and a complete run of all self-tests shall not take longer than 2 seconds. The initialization self-test shall not interfere with communication startup unless there is a fault detected and it is considered to be severe enough to inhibit the startup of the ECU itself.

### 5.2.2 Operation Cycle

The operation cycle where the fault detection algorithm is active shall be specified for each DTC + FTB combination.

An Operation Cycle defines the start and end conditions for a Test to run.

Each fault detection algorithm is only active (test running) in a particular operation cycle.

Possible operation cycles include; driving cycle, engine warm up cycle, ignition on/off cycle, power up/power down cycle, operation active/passive cycle, accumulated operating time, ECU initialization/shutdown or other defined cycles.

This document supports several common Operation cycles. If the supplier chooses another operation cycle for a/some DTC(s), the supplier shall provide the definition (start and end) of the operation cycle.

- Ignition on/off cycle (Ignition cycle): A transition in Vehicle Power Mode from Off to Run to Off irrespective of any other power modes in between.

- Wake/Sleep cycle (Wake cycle): A transition in NM state from sleep mode to network mode to sleep mode irrespective of any other power modes in between.
- Power up/power down cycle (Power cycle): A transition in power state from Power-down to Power-up to Power-down irrespective of any other power modes in between.

### 5.2.3 Content of Error Storage

Each DTC + FTB combination related error storage shall be implemented including below content.

| No. | Description             | Size                                 |
|-----|-------------------------|--------------------------------------|
| 1   | Diagnostic Trouble Code | 3 Bytes                              |
| 2   | DTC status              | 1 Bytes                              |
| 3   | DTC Severity            | 1 Bytes                              |
| 4   | DTC Extended Data       | Refer to 'DTC Extended Data' session |
| 5   | DTC Snapshot            | Refer to 'DTC Snapshot' session      |

#### 5.2.3.1 Diagnostic Trouble Code

A DTC is defined as a failure indicator used to direct repair personnel to the most likely suspect circuit, component, system, and the general location of the diagnostic malfunction. The DTC number shall have 3 bytes value.

##### 5.2.3.1.1 DTC Failure Type

The DTC failure Type is the low byte of DTC, contains DTC Failure Category and Sub Type of a base DTC. It represents the type of fault in the circuit or system.

| High Nibble(bit7-4)  | Low Nibble(bit3-0)   |
|----------------------|----------------------|
| DTC Failure Category | DTC Failure Sub Type |

The detail refer to Append-Failure Type

#### 5.2.3.2 DTC Status

DTC Status Bit

| Bit | Description                        | Cvt<br>(Enhanced) | Mnemonic |
|-----|------------------------------------|-------------------|----------|
| 7   | warningIndicatorRequested          | C1                | WIR      |
| 6   | testNotCompletedThisOperationCycle | M                 | TNCTOC   |
| 5   | testFailedSinceLastClear           | M                 | TFSLC    |
| 4   | testNotCompletedSinceLastClear     | M                 | TNCSLC   |
| 3   | confirmedDTC                       | M                 | CDTC     |
| 2   | pendingDTC                         | C2                | PDTC     |
| 1   | testFailedThisOperationCycle       | U                 | TFTOC    |
| 0   | testFailed                         | M                 | TF       |

**C1:** Required for any ECU that requests / activates an indication when a DTC is set.

|                                                                                   |                                                          |         |                |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|----------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |                |
|                                                                                   | V23                                                      | Rev 3.3 | Page 86 of 146 |

|                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C2:</b> Required for any ECU that support DTCs that requires fails over multiple operation cycles before the DTC becomes confirmed.<br><b>U:</b> user optional |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### **Test Failed (Bit 0):**

The DTC Status: Test Failed shall be implemented for all DTCs.

This bit shall indicate the result of the most recently performed test.

The value of this bit shall be retained across operation cycles.

The value of this bit shall be initialized to 0 for a first time power up of the ECU.

The value of this bit shall be initialized to 0 after the ECU has been programmed.

The value of this bit shall be set to 1 when the most recent completed Test reports Failed.

The value of this bit shall be set to 0 when the most recently completed Test reports Passed.

The value of this bit shall be set to 0 on a call to Clear Diagnostic Information.

The value of this bit shall remain unchanged when the conditions for Automatic Clearing of DTC Information have been met (value must = 0 for conditions to be met).

#### **Test Failed This Operation Cycle (Bit 1):**

The DTC Status: Test Failed This Operation Cycle support is optional. When this status is not supported, the bit shall be reported as 0. When this status is supported, the bit shall be implemented as follows.

This bit shall indicate whether or not a diagnostic test has reported a failed result at any time during the current operation cycle.

The value of this bit shall be initialized to 0 at the beginning of an operation cycle.

The value of this bit shall be initialized to 0 for a first time power up of the ECU.

The value of this bit shall be initialized to 0 after the ECU has been programmed.

The value of this bit shall be set to 1 when the Test reports failed.

The value of this bit shall be set to 0 when the Test does not report Failed during the current operation cycle.

The value of this bit shall be set to 0 on a call to Clear Diagnostic Information.

The value of this bit shall remain unchanged when the conditions for Automatic Clearing of DTC Information have been met (value must = 0 for conditions to be met).

#### **Pending DTC (Bit 2):**

The DTC Status: Pending DTC shall be implemented for all DTCs that require Test Fails over multiple operation cycles before the DTC becomes confirmed, otherwise support is optional. When this status is not supported, the bit shall be reported as 0. When this status is supported, the bit shall be implemented as follows.

This bit shall indicate whether or not a diagnostic test has reported a Failed result at any time during the current or the last completed operation cycle (if the current operating cycle is still active).

The value of this bit shall only be updated if the test runs and completes. If the test did not complete during the

|                                                                                   |                                                          |         |                |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|----------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |                |
|                                                                                   | V23                                                      | Rev 3.3 | Page 87 of 146 |

current operation cycle, the status bit shall not be changed.

The value of this bit shall be initialized to 0 for a first time power up of the ECU.

The value of this bit shall be initialized to 0 after the ECU has been programmed.

The value of this bit shall be set to 1 when the Test reports a Failed result at least once during the current operation cycle.

The value of this bit shall be set to 0 when the Test reports a Passed result at least once and does not report a failed result by the end of the current operation cycle.

The value of this bit shall be set to 0 on a call to Clear Diagnostic Information.

The value of this bit shall remain unchanged when the conditions for Automatic Clearing of DTC Information have been met (value must = 0 for conditions to be met).

#### **Confirmed DTC (Bit 3):**

The DTC Status: Confirmed DTC shall be implemented for all DTCs.

This bit shall indicate whether a malfunction was detected enough times to warrant that the DTC is stored in long-term memory (i.e., Pending DTC has been set = 1 one or more consecutive operation cycles, depending on the Confirmation Threshold).

The value of this bit shall be retained across operation cycles.

The value of this bit shall be initialized to 0 for a first time power up of the ECU.

The value of this bit shall be initialized to 0 after the ECU has been programmed.

The value of this bit shall be set to 1 when the Confirmation Threshold for Test Failed results has been reached.

Once set, the value of this bit shall remain unchanged until a call to Clear Diagnostic Information or the conditions for Automatic Clearing of DTC Information have been met.

The value of this bit shall be set to 0 on a call to Clear Diagnostic Information.

The value of this bit shall be set to 0 when the conditions for Automatic Clearing of DTC Information have been met.

#### **Test Not Completed Since Last Clear (Bit 4):**

The DTC Status: Test Not Completed Since Last Clear shall be implemented for all DTCs.

This bit shall indicate whether the Test has ever run and completed since the last time a call was made to Clear Diagnostic Information.

The value of this bit shall be retained across operation cycles.

The value of this bit shall be initialized to 1 for a first time power up of the ECU.

The value of this bit shall be initialized to 1 after the ECU has been programmed.

The value of this bit shall be set to 0 when the Test has returned either a Passed or Failed result.

The value of this bit shall remain unchanged when the Test does not report a Passed or Failed result during the

|                                                                                   |                                                          |         |                |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|----------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |                |
|                                                                                   | V23                                                      | Rev 3.3 | Page 88 of 146 |

current operation cycle.

The value of this bit shall be set to 1 on a call to Clear Diagnostic Information.

The value of this bit shall remain unchanged when the conditions for Automatic Clearing of DTC Information have been met (value must = 0 for conditions to be met).

#### **Test Failed Since Last Clear (Bit 5):**

The DTC Status: Test Failed Since Last Clear shall be implemented for all DTCs.

This bit shall indicate whether a DTC test has completed with a Failed result since the last time a call was made to Clear Diagnostic Information.

The value of this bit shall be retained across operation cycles.

The value of this bit shall be initialized to 0 for a first time power up of the ECU.

The value of this bit shall be initialized to 0 after the ECU has been programmed.

The value of this bit shall be set to 1 when the Test has returned a Failed result.

The value of this bit shall remain unchanged when the Test does not report a Failed result during the current operation cycle.

The value of this bit shall be set to 0 on a call to Clear Diagnostic Information.

The value of this bit shall be set to 0 when the conditions for Automatic Clearing of DTC Information have been met.

#### **Test Not Completed This Operation Cycle (Bit 6):**

The DTC Status: Test Not Completed This Operation Cycle shall be implemented for all DTCs.

This bit shall indicate whether or not a diagnostic test has ever run and completed during the current operation cycle (or completed during the current operation cycle after the last time a call was made to Clear Diagnostic Information).

The value of this bit shall be initialized to 1 at the beginning of an operation cycle.

The value of this bit shall be initialized to 1 for a first time power up of the ECU.

The value of this bit shall be initialized to 1 after the ECU has been programmed.

The value of this bit shall be set to 0 when the Test has returned either a Passed or Failed result.

The value of this bit shall remain unchanged when the Test does complete during the current operation cycle.

The value of this bit shall be set to 1 on a call to Clear Diagnostic Information.

The value of this bit shall remain unchanged when the conditions for Automatic Clearing of DTC Information have been met.

#### **Warning Indicator Requested (Bit 7):**

The DTC Status: Warning Indicator Requested shall be implemented for all DTCs supported by an ECU that controls a warning indicator (lamp, LED, DIC message, chime, etc.).

This bit shall indicate whether or not the ECU is requesting the warning indicator to be active.

The value of this bit shall be initialized to 0 for a first time power up of the ECU.

The value of this bit shall be initialized to 0 after the ECU has been programmed.

The value of this bit shall be set to 1 when the Confirmation Threshold for Test Failed results has been reached and a warning indicator is supported for the DTC.

The value of this bit shall be set to 0 on a call to Clear Diagnostic Information.

The value of this bit shall remain unchanged when the conditions for Automatic Clearing of DTC Information have been met (value must = 0 for conditions to be met).

### 5.2.3.3 DTC Severity

The severity of a DTC indicates how serious the failure is, and the value must be specified from the system developer. The severity could also be used as the priority of a DTC. The value is bigger, and the priority is higher. It's allowed to overwrite a registered error if the storage is full and the new DTC with a higher priority occurs.

The severity information is reported in a 1-byte value. Only the upper 3 bits(bit7-5) of the 1-byte value are used to represent the DTC severity information. The remaining bits(bit 4-0) shall be set to zero(0).

| DTC Severity               |       |       |                     |       |       |       |       |
|----------------------------|-------|-------|---------------------|-------|-------|-------|-------|
| Bit 7                      | Bit 6 | Bit 5 | Bit 4               | Bit 3 | Bit 2 | Bit 1 | Bit 0 |
| 3 bit severity information |       |       | Reserved(padding 0) |       |       |       |       |

The following bit codes shall be applied for the 3-bit DTC severity information value.

| Bit 7-5 | Description                                                                                          |
|---------|------------------------------------------------------------------------------------------------------|
| 000b    | noSeverityAvailable<br>There is no severity information available.                                   |
| 001b    | maintenanceOnly<br>This value indicates that the failure requires maintenance only.                  |
| 010b    | checkAtNextHalt<br>This value indicates to the failure requires a check of the vehicle at next halt. |
| 100b    | checkImmediately<br>This value indicates to the failure requires an immediate check of the vehicle.  |

The DTC severity information value shall be defined for each DTC.

### 5.2.3.4 DTC Extended Data

DTC Extended Data which consists of extended status information associated with a DTC contains DTC parameter values, which have been identified at the time of the request by 0x19 0x06. The following extended data record number shall be supported.

| Extended Data Record | Parameter | Bytes | Cvt |
|----------------------|-----------|-------|-----|
|----------------------|-----------|-------|-----|

| Number |                           |   |   |
|--------|---------------------------|---|---|
| 01     | DTC Severity              | 1 | M |
| 02     | DTC occurrence counter    | 1 | M |
| 03     | Self-Aging counter        | 1 | M |
| 04     | DTC first occurrence time | 7 | M |

The extended data record, once stored, shall not update the ‘DTC first occurrence time’ if the DTC test fails again.

The extended data record, once stored, shall update the occurrence counter if the DTC test fails again.

The extended data record shall update the aging cycle counter if the DTC test passes.

#### 5.2.3.4.1 DTC Occurrence Counter

The DTC occurrence counter is increased by 1 after a specified qualification time and the state of a failure changed from passive to active. The counter indicates counts number of operation cycles in which "testFailed" has been reported. Below is the definition for the counter values.

ECU shall not increment the DTC Occurrence Counter anymore, if it has reached its maximum value 255.

| Counter | Description                                           |
|---------|-------------------------------------------------------|
| 0       | Init value                                            |
| 1-254   | Shows how often errors occurs during operation cycles |
| 255     | Maximum value                                         |

#### 5.2.3.4.2 Self-Aging counter

Self-Aging is defined as the following process:

- A failure condition matures and a fault record(DTC) is stored
- The failure condition does not mature for a certain period of time(operation cycles)
- The fault record is removed from NVM without a diagnostic request 0x14

#### Change Self\_Aging Counter

- First Entry

The Self-aging counter value shall initially be set to zero(0) at the instance of an active DTC.

- DTC status change from active to inactive

The Self aging counter value shall be set to one (1) after completing an operation cycle in which test did not fail .

- After operation cycle

The Self-aging counter is incremented after each Self-Aging cycle which Error is inactive if value is between 1 and 254.

If the self-aging counter value reaches its maximum value(255) and the DTC still remains inactive, the value shall remain latched at 255.

| Counter | Description                    |
|---------|--------------------------------|
| 0       | Init value and error is active |
| 1-254   | Error is inactive              |
| 255     | Maximum Counter cycle          |

For each DTC, after the Self-aging counter equal to specified number(1-254), this DTC shall be deleted from NVM. If no specified number is defined, then the DTC will be deleted after 40 operation cycle.

#### 5.2.3.4.3 DTC first occurrence time

| Parameter | Type |
|-----------|------|
| Year_H    | BCD  |
| Year_L    | BCD  |
| Month     | BCD  |
| Day       | BCD  |
| Hour      | BCD  |
| Minute    | BCD  |
| Second    | BCD  |

#### 5.2.3.5 DTC Snapshots

When a failure has been reported, the ECU shall examine the previously stored snapshot records to see if any of the DTCs already captured match the DTC that is currently failing. If a matching DTC is found the data contained within that snapshot record will be updated.

If no matching DTC record is found among the DTC records stored in the snapshot record buffer, then the DTC and data will be stored in the next snapshot record position within the buffer. If the buffer is full, then the oldest data will be overwritten with the current failing DTC.

The following snapshot data record is mandatory for all DTCs. User can add others and write in <SMG-EEA-4006-诊断数据库Excel>.

| DIDs | Description      | Length | Data Type | Unit | Cvt            | Transform       |
|------|------------------|--------|-----------|------|----------------|-----------------|
| D001 | Vehicle Speed    | 2Byte  | Unsigned  | kph  | M              | $N=E*0.05625$   |
| D002 | FMCU ActualSpeed | 2Byte  | Unsigned  | rpm  | C <sub>1</sub> | $N=E*100-16000$ |
| D003 | RMCU ActualSpeed | 2Byte  | Unsigned  | rpm  | C <sub>1</sub> | $N=E*100-16000$ |
| D004 | Odometer         | 3Byte  | Unsigned  | Km   | M              | $N=E*1$         |

|      |                      |       |          |   |   |         |
|------|----------------------|-------|----------|---|---|---------|
| D005 | DTC OccurrenceTime   | 7Byte | BCD      | / | M |         |
| D006 | Power supply voltage | 1Byte | Unsigned | V | M | N=E*0.1 |

C1: Only for the vehicle with the rear/former motor.

#### 5.2.4 DTC Primary Memory

The fault memory is called primary memory.

##### 5.2.4.1 Layout

The layout of DTC primary memory uses FIFO(First in and first out) structure.

##### 5.2.4.2 Size

The size of DTCs that store in the primary memory should be 20. Include the snapshot data,extended data,DTC status and so on.

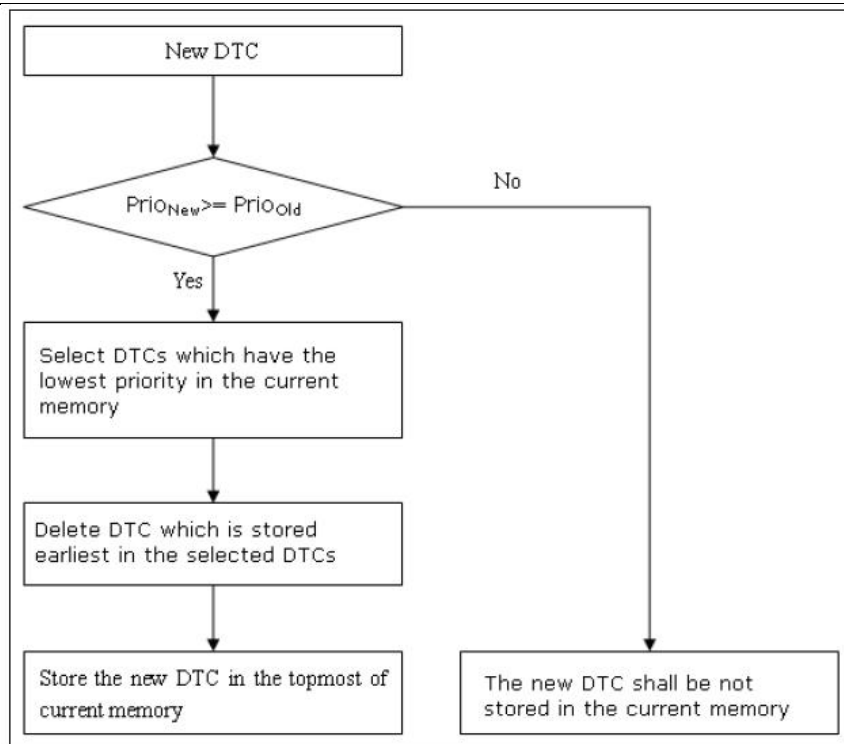
##### 5.2.4.3 Primary Memory Management

➤ Primary memory space isn't full

If the primary memory space isn't full, the new DTC shall be stored in the topmost position of primary memory. The newest DTC should record on the top position and the existing DTCs should move one position down.

➤ Primary memory space is full

The following figure shows if the primary memory space is full, the new DTC shall be stored according to categories shown in the following flowchart.



### 5.2.5 DTC\_Active Message

Every ECUs which is connected to a CAN network and supports diagnostic Service \$19: Read DTC Information shall support DTC\_Active Message.

The DTC\_Triggered message is a message transmitted by an ECU when a DTC transitions to a confirmed status or causes a warning indication in order to provide the notification.

The DTC\_Triggered shall be transmitted to ICM. ICM's own DTC does not need to be sent and should be packaged directly.

Each ECU shall maintain an internal DTC\_Active status flag for each DTC that contains DTCs for which a DTC\_Active message has been transmitted. This is to ensure that the DTC\_Active message is only transmitted when the status of a DTC transitions to the confirmed state. This flag or array shall be stored in non-volatile memory. The DTC\_Active status flag is independent of the diagnostics portion of the ECU and does not interfere with the normal setting or clearing of DTCs. Also, the flag does not interfere with the normal diagnostic communication and reporting to a test tool.

When the DTC status confirmed bit becomes “True” and if the DTC\_Active status flag equals to “False”, the ECU shall transmit the DTC\_Active message. Once this message has been transmitted for the DTC, the specific DTC code shall not cause the message to be transmitted again until the algorithm for the DTC has passed and failed again and the DTC\_Active status flag for the DTC has been set to “False”.

When a supported DTC transitions to a confirmed state (status bit 3 transitions from 0 to 1) and the ECU is not in the Communication Active state, the transmission of the DTC\_Active message shall be delayed until the ECU is in the Communication Active state.

After the application layer has updated the data for the DTC\_Active message, the ECU shall wait a delay of 1500ms before updating the data with information for a different current DTC.

Two messages may be transmitted with the same values due to the periodic transmit rate of 1000ms. This may not always occur due to the asynchronous nature of when the DTC transitions to a confirmed state in relationship to when the message is transmitted. The delay will ensure that the DTC\_Active message is transmitted at least once before the trigger bit set to “True”.

The trigger status flag shall be set to “True” after the DTC\_Triggered message indicating the current DTC has been transmitted. The trigger status flag shall be set to “False” per the following:

- The flag shall be set to “False” after completing an ignition cycle for which the DTC algorithm has passed and not failed. This will limit the transmission of the DTC\_Triggered message for intermittent DTCs within an ignition cycle.
- The flag shall be set to “False” upon receipt of the service \$14.

## 5.2.6 DTC Active Message Transmit Requirement

### 5.2.6.1 The DTC\_Active Message Structure

The structure of DTC trigger message as the following tables show:

| Description:                                                                                                | Short Desc        | Length(Bit) |
|-------------------------------------------------------------------------------------------------------------|-------------------|-------------|
| Diagnostic Trouble Code Information Extended                                                                | DTCInfo           | 56          |
| Unused 1                                                                                                    | DTCIUnused1       | 1           |
| Unused 2                                                                                                    | DTCIUnused2       | 1           |
| Unused 3                                                                                                    | DTCIUnused3       | 1           |
| Unused 4                                                                                                    | DTCIUnused4       | 1           |
| Unused 5                                                                                                    | DTCIUnused5       | 1           |
| Unused 6                                                                                                    | DTCIUnused6       | 1           |
| Warning Indicator Triggered<br>(The Warning Indicator Triggered bit shall equal the DTC_Active status flag) | DTCI_WrnIndTrggrd | 1           |
| Diagnostic Trouble Code Triggered                                                                           | DTCI_DTCTriggered | 1           |
| Diagnostic Trouble Code Source<br>(This shall be equal to the Diagnostic Address of the transmitting ECU.)  | DTCI_DTCSource    | 8           |
| Diagnostic Trouble Code Number                                                                              | DTCI_DTCNumber    | 16          |
| Diagnostic Trouble Code Failure Type                                                                        | DTCI_DTCFailType  | 8           |
| Warning Indicator Requested Status                                                                          | DTCI_WIR          | 1           |
| Test Not Completed This Operation Cycle                                                                     | DTCI_TNCTOC       | 1           |
| Test Failed Since Last Clear                                                                                | DTCI_TFSLC        | 1           |

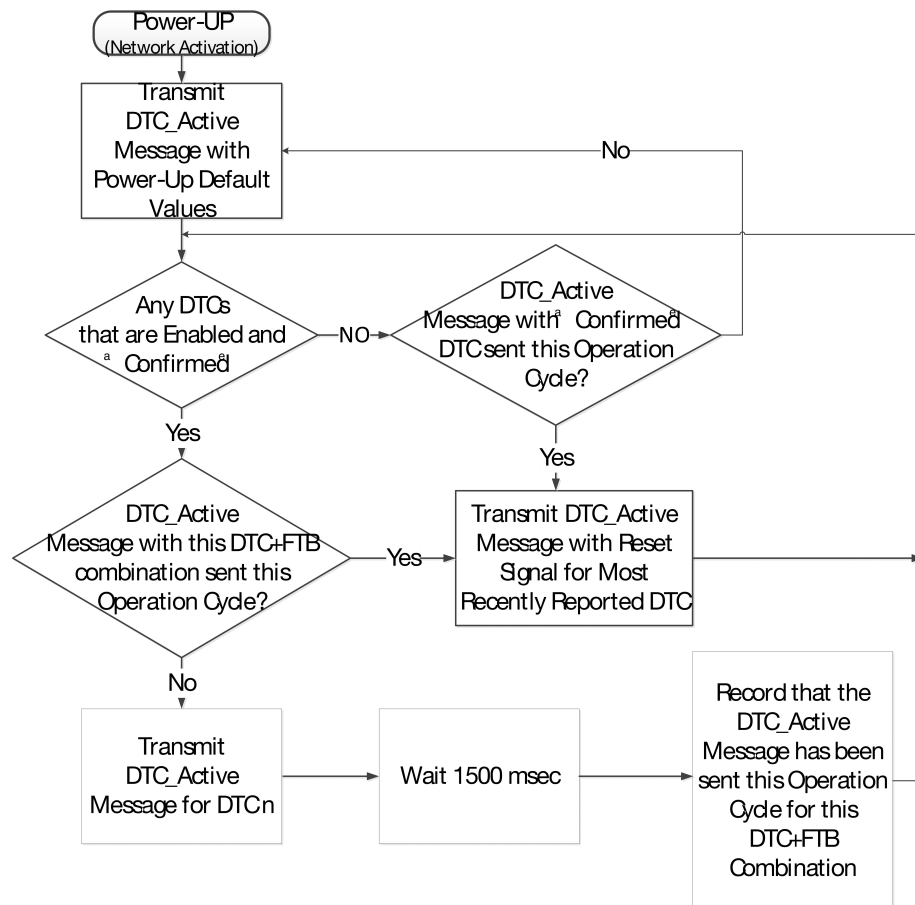
|                                     |              |   |
|-------------------------------------|--------------|---|
| Test Not Completed Since Last Clear | DTCI_TNCSLC  | 1 |
| Confirmed DTC                       | DTCI_CDTC    | 1 |
| Pending DTC                         | DTCI_PDTC    | 1 |
| Test Failed This Operation Cycle    | DTCI_TFTOC   | 1 |
| Test failed                         | DTCI_TF      | 1 |
| DTC severity                        | DTC_Severity | 8 |

### 5.2.6.2 The DTC\_Active Message Transmit Process

The ECU shall transmit the DTC\_Active message per the requirements in the database. The power up default values for the message shall be as the following table. The message shall be transmitted periodically as defined in the database with these values until a DTC becomes confirmed.

| Name                                 | Default Value    |
|--------------------------------------|------------------|
| Warning Indicator Triggered          | False            |
| Diagnostic Trouble Code Triggered    | False            |
| Diagnostic Trouble Code Source       | Source ID of ECU |
| Diagnostic Trouble Code Number       | \$0000           |
| Diagnostic Trouble Code Status Byte  | \$00             |
| Diagnostic Trouble Code Failure Type | \$00             |
| DTC severity                         | \$00             |

The following chart shows that the DTC\_Active message transmit-process, the details refer to the following steps.



- 1) When the DTC status Confirmed bit becomes “True” (status bit 3 transitions from 0 to 1) and Communication is active and the particular DTC + FTB combination has not already been transmitted in the current operation cycle, the ECU shall transmit the DTC\_Active message with the following data:

| Name                                | Current Value                         |
|-------------------------------------|---------------------------------------|
| Warning Indicator Triggered         | (As required for that DTC type)       |
| Diagnostic Trouble Code Triggered   | True                                  |
| Diagnostic Trouble Code Source      | Source ID of ECU                      |
| Diagnostic Trouble Code Number      | the diagnostic trouble code           |
| Diagnostic Trouble Code Status Byte | the associated diagnostic status byte |
| DTC severity                        | the associated DTC severity           |

- 2) The ECU shall wait 1500ms to ensure the DTC\_Active message is transmitted with the values in the previous step.
- 3) The ECU shall disable transmission of the DTC\_Active message with that particular DTC + FTB combination for the duration of the current operation cycle.

- 4) The ECU shall transmit the DTC\_Active message again with the following values to reset the triggers.

Note that this does not affect the status of the DTC. If more than one DTC becomes confirmed at the same time, then the DTC\_Active message with the DTC\_Active reset shall be sent after all of the confirmed DTCs have been reported. This will ensure timely reporting of the DTCs.

| Name                                 | Current Value                          |
|--------------------------------------|----------------------------------------|
| Warning Indicator Triggered          | False                                  |
| Diagnostic Trouble Code Triggered    | False                                  |
| Diagnostic Trouble Code Source       | Source ID of ECU                       |
| Diagnostic Trouble Code Number       | the diagnostic trouble code            |
| Diagnostic Trouble Code Status Byte  | the associated diagnostic status byte  |
| Diagnostic Trouble Code Failure Type | the associated diagnostic failure type |
| DTC severity                         | the associated DTC severity            |

- 5) The ECU shall transmit the next DTC\_Active as defined in the previous consecutive steps if there is another confirmed DTC. Otherwise, it shall continue to transmit the DTC\_Active message with the Diagnostic Trouble Code Triggered signal equal to “False” and the other fields containing the data for the last reported confirmed DTC.

### 5.2.6.3 Example

While the vehicle is running, a Brake System ECU confirms two DTC + FTB combinations; C0036 00 and C0041 00. Neither DTC + FTB combination has been transmitted in the current operation cycle. The ECU transmits the DTC Triggered message per the following sequence.

Note: The message is transmitted once every 1 second per the requirements in the Message Database.

t = t1 (sec):    \$xxx \$00 \$28 \$00 \$00 \$00 \$00 \$00    /\* No confirmed DTCs.\*/

t = t1 + 1: \$xxx \$00 \$28 \$00 \$00 \$00 \$00 \$00    /\* No confirmed DTCs.\*/

t = t1 + 1.5:    No message transmitted.    /\* DTCs C0036 00 and C0041 00 are confirmed.\*/

t = t1 + 2: \$xxx \$03 \$28 \$40 \$36 \$00 \$AF \$00    /\* DTC C0036 00 is reported.\*/

t = t1 + 3: \$xxx \$03 \$28 \$40 \$36 \$00 \$AF \$00    /\* Message sent again while waiting 1.5 sec.\*/

t = t1 + 3.5: No message transmitted.    /\* DTC Trigger disabled for C0036 00 for operation cycle duration.\*/

t = t1 + 4: \$xxx \$03 \$28 \$40 \$41 \$00 \$AF \$00    /\* DTC C0041 00 is reported.\*/

t = t1 + 5: \$xxx \$03 \$28 \$40 \$41 \$00 \$AF \$00    /\* Message sent again while waiting 1.5 sec.\*/

t = t1 + 5.5: No message transmitted.    /\*DTC Trigger disabled for C0041 00 for operation cycle duration.\*/

t = t1 + 6: \$xxx \$00 \$28 \$40 \$41 \$00 \$AF \$00    /\* Reset trigger transmitted. \*/

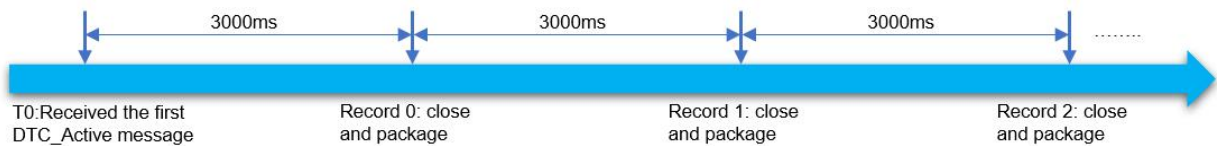
t = t1 + 7: \$xxx \$00 \$28 \$40 \$41 \$00 \$AF \$00 /\* Message sent per periodic requirements. \*/

t = t1 + 8+: \$xxx \$00 \$28 \$40 \$41 \$00 \$AF \$00 /\* Message sent periodically with these values until another DTC becomes current.\*/

5.2.6.4 The Operation of ICM

The ECU (ICM) containing the central repository function shall provide sufficient non-volatile memory to store the 20 most recent data records (Data Records 0 to 19 recorded as a push down stack (PDS)).

A new data record shall be stored if greater than 3000ms have elapsed since the receipt of the previous DTC\_Active message.



The data records shall be stored chronologically, in ascending order with Data Record 0 containing the most recently recorded data record.

The ECU (ICM) containing the central repository function shall receive all DTC\_Active messages that are transmitted in the vehicle.

The receipt of a DTC\_Active message with the Diagnostic Trouble Code Triggered bit set to one (1) shall result in. A new data record stored in the central repository, an addition to the data record currently being recorded, removal of an existing record, or no further action.

A new DTC and relevant data shall be stored for the first DTC\_Active message received since the beginning of the ignition cycle.

An addition to the DTC and relevant data currently being recorded shall be made if less than or equal to 3000ms have elapsed since the receipt of the DTC\_Active message that caused the current data record to be recorded.

An existing DTC and relevant data shall be deleted(less than or equal to 3000ms) when the current DTC\_Active message contains the same DTC + FTB combination and DTC Source as the existing record. The DTC Status may be different between the existing DTC and relevant data and the new DTC and relevant data.

No further action shall be taken when nine or more DTC Triggered messages are received in a 3000ms period.

When the maximum allowable number of data records has been stored in the PDS and the storing criteria are again met, the PDS overflows and the oldest data record shall be deleted to allow for storage of the new data record.

When the ignition is from ON to OFF, the ICM shall send the PDS data to the TSP. and the clear the PDS. All

this work shall be complete in 1 min. If the cellular or WIFI is off, only clear the PDS.

The data stored in the data record shall be the last known values received in the current operation cycle. All unknown data values and all unused bits and bytes shall be set to 0.

Data records in the central repository shall include the environmental data / vehicle operating conditions as the following table shown:

| Byte(s) | Bit(s) | Signal                                | Comment                                                                            |
|---------|--------|---------------------------------------|------------------------------------------------------------------------------------|
| 0-1     |        | DTC code                              | DTC and relevant data                                                              |
| 2       |        | Failure Type Byte                     |                                                                                    |
| 3       |        | DTC Status                            |                                                                                    |
| 4       |        | DTC Source(ECU Address)               |                                                                                    |
| 5       |        | DTC severity                          |                                                                                    |
| 6-11    |        | Time                                  | Format reference message 0x460                                                     |
| 10      |        | Accelerator Actual Position           | $N=E*1(\%)$ ;when received the first DTC_Active Message in this Record             |
| 11      |        | Battery Voltage                       | $N=E*0.1(V)$ ;when received the first DTC_Active Message in this Record            |
| 12      |        | Brake Pedal Position                  | $N=E*1(\%)$ ;when received the first DTC_Active Message in this Record             |
| 13      |        | CMA Temp Max                          | $N=E*1-40 (^{\circ}C)$ ;when received the first DTC_Active Message in this Record  |
| 14      |        | CMA Temp Min                          | $N=E*1-40 (^{\circ}C)$ ;when received the first DTC_Active Message in this Record  |
| 15      |        | Outside Air Temperature               | $N=E*1-40 (^{\circ}C)$ ;when received the first DTC_Active Message in this Record  |
| 16      | 0      | Ready                                 | 1=Ready; 0=Not Ready;when received the first DTC_Active Message in this Record     |
|         | 1-7    | Not used                              | Set 0                                                                              |
| 17-20   | 31     | Positioning System Elevation Validity |                                                                                    |
|         | 0-30   | Positioning System Elevation          |                                                                                    |
| 21-24   | 31     | Positioning System Latitude Validity  |                                                                                    |
|         | 0-30   | Positioning System Latitude           |                                                                                    |
| 25-28   | 31     | Positioning System Longitude Validity |                                                                                    |
|         | 0-30   | Positioning System Longitude          |                                                                                    |
| 29-30   |        | Steering Wheel Angle                  | $N=E*0.125 (^{\circ})$ ; when received the first DTC_Active Message in this Record |
| 31-34   |        | Vehicle Odometer                      |                                                                                    |
| 35-36   | 15     | Vehicle Speed Validity                |                                                                                    |
|         | 0-14   | Vehicle Speed                         | $N=E*0.05625(km/h)$ ; when received the first DTC_Active Message in this Record    |
| 37      | 7      | Windshield Wash Wipe Active           |                                                                                    |
|         | 0-6    | Not used                              | Set 0                                                                              |
| 38      | 7      | System Power Mode Validity            |                                                                                    |
|         | 4-6    | System Power Mode                     |                                                                                    |
|         | 0-3    | Not used                              | Set 0                                                                              |

|       |  |                                |                                          |
|-------|--|--------------------------------|------------------------------------------|
| 39-44 |  | Not used                       | Set 0                                    |
| 45-59 |  | Additional DTC_Active Message1 | This data shall be set to 0, if not used |
| 50-54 |  | Additional DTC_Active Message2 | This data shall be set to 0, if not used |
| 55-59 |  | Additional DTC_Active Message3 | This data shall be set to 0, if not used |
| 60-64 |  | Additional DTC_Active Message4 | This data shall be set to 0, if not used |
| 65-69 |  | Additional DTC_Active Message5 | This data shall be set to 0, if not used |
| 70-74 |  | Additional DTC_Active Message6 | This data shall be set to 0, if not used |
| 75-79 |  | Additional DTC_Active Message7 | This data shall be set to 0, if not used |

### 5.2.7 Clear DTC

In general, there are two methods for clear DTC information, the details of each method as the following shown.

The clearing of DTC information shall be possible with the diagnostic service \$14. A request to clear diagnostic information shall deactivate any remedial actions taken due to fault detection and clear the DTC Status bits (DTC Status byte set to \$50), the Extended Data Records, the Snapshot Records, along with any other DTC related data such as flags, counters, timers, etc., specific to each DTC and Failure Type combination.

The ECU shall implement an “Automatic Clearing of DTC Information” function for each DTC. There shall be an automatic clearing of DTC information method implemented for each ECU. Each ECU shall implement an aging cycle counter for each DTC.

Automatic Clearing of DTC Information shall occur when the aging cycle counter has incremented to the 40.

## 6 The Vehicle Learn Processing

A learn is any procedure performed by an ECU on an input or output that establishes a valid operating value or range of operating values which is then stored in long term memory.

### 6.1 Learn Processing Category

Learns may be initiated automatically or manually (triggered learns).

While automatic learns are preferred, only learns that take less than 5 seconds to complete after all pre-conditions for performing the learn have been met, do not pose a safety concern to someone in or near the vehicle, do not impact the safe operation of the vehicle, and do not require any operator intervention shall be implemented as automatic learns. If more time is needed, this shall be negotiated with the representatives from ME, Electrical Systems Engineering and VSE.

|                                                                                   |                                                          |         |                 |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|-----------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |                 |
|                                                                                   | V23                                                      | Rev 3.3 | Page 101 of 146 |

### 6.1.1 Automatic Learns

Automatic learns shall be initiated by the ECU when it determines that a learned value is not present and that all pre-conditions for performing the learn process have been met.

The ECU shall keep trying to learn while the pre-conditions are met and the learn process has not been successfully completed.

All automatic learns shall also support an unlearn function using the UDS Routine Control service (\$31). This function may either allow each learned value to be unlearned individually or unlearn all learned values at once for the case where the learns should all be relearned at the same time.

### 6.1.2 Manual Learns

Manually learns (triggered learns) shall be initiated using the UDS Routine Control service (\$31) to start the learn procedure.

## 6.2 Learn Processing Requirements

All ECUs which need to implement the learn process also shall meet the following requirements but not limited:

- ME Learn Processing Timer: The overall time to execute the learn process for a single ECU in the plant should not exceed 15s. If more time is needed, this shall be negotiated with the representatives from ME, Electrical Systems Engineering and VSIE.
- All learns, their required conditions, and the estimated time to completion shall be documented in the <SMG-EEA-4006-诊断数据库Excel> .
- Independent learns supported by an ECU shall have the capability to run simultaneously.
- All learns must be discussed with and agreed to by manufacturing and service engineering.

Note: In cases where a reasonably long time is needed, an “all self-tests completed” flag should also be discussed with ME.

### 6.2.1 Learn Status

A DID shall exist that communicates the status of the learn state (Not Learned, In Progress, Learned, and Learn Failed). The status “Learned” shall only be indicated after the learned data has been transferred to permanent memory.

Note: For the case where there are multiple learns that must be performed at the same time, a single DID indicating the overall status of the learn process is permitted.

For each automatic learn supported by an ECU, a DID shall be supported that indicates the reason why the learn process failed.

For each triggered learn supported by an ECU, a routine result shall be supported that indicates the reason why

the learn process failed.

If a learn fails for an identifiable hardware fault, a DTC shall be stored to indicate the condition.

### 6.3 Learn with OTA

In the OTA process, it is forbidden to erase the completed learned unless the ECU can learn automatically.

## 7 Appendix A-ECU Address

| No | ECU     | ECU Name                             | ECU Address |
|----|---------|--------------------------------------|-------------|
| 1  | VCCM    | Vehicle Central Control Module       | 0x10        |
| 2  | MCU_F   | Motor Control Unit_Front             | 0x42        |
| 3  | MCU_R   | Motor Control Unit_Rear              | 0x43        |
| 4  | IPU     | Integrated Power unit                | 0x40        |
| 5  | BMS     | Battery Management System            | 0x41        |
| 6  | IEBM    | Integrated Electronic Brake Module   | 0x50        |
| 7  | EPS     | Electronic Power Steering            | 0x51        |
| 8  | IDM     | Intelligent Driving Module           | 0x60        |
| 9  | SRR_FL  | Short Range Radar_Front Left         | 0x61        |
| 10 | SRR_FR  | Short Range Radar_Front Right        | 0x62        |
| 11 | SRR_RL  | Short Range Radar_Rear Left          | 0x63        |
| 12 | SRR_RR  | Short Range Radar_Rear Right         | 0x64        |
| 13 | LRR_F   | Long Range Radar_Front               | 0x65        |
| 14 | ICM     | Intelligent Cockpit Module           | 0x80        |
| 15 | ZCU_L   | Zone Control Unit_Left               | 0x11        |
| 16 | SDM     | Sensing & Diagnostic Module (Airbag) | 0x12        |
| 17 | SRC     | Sunroof Controller                   | 0x05        |
| 18 | ELS_L   | Exterior Light System_Left           | 0x06        |
| 19 | ELS_R   | Exterior Light System_Right          | 0x07        |
| 20 | DSP     | Door Switch Panel                    | 0x08        |
| 21 | SWS_L   | Steering Wheel Switch_Left           | 0x09        |
| 22 | SWS_R   | Steering Wheel Switch_Rear           | 0x0A        |
| 23 | RLS/LSS | Rain Light Sensor/Light Sun Sensor   | 0x02        |
| 24 | EBS     | Electrical Battery Sensor            | 0x03        |
| 25 | TCP     | Telematics Connected Platform        | 0x30        |
| 26 | ETC     | Electronic Toll Collection           | 0x31        |
| 27 | DKM     | Data key Module                      | 0x32        |
| 28 | BLE_L   | Bluetooth Low Energy_Left            | 0x33        |
| 29 | BLE_R   | Bluetooth Low Energy_Right           | 0x34        |

|    |        |                                                           |      |
|----|--------|-----------------------------------------------------------|------|
| 30 | BLE_B  | Bluetooth Low Energy_Back                                 | 0x35 |
| 31 | NFC_ST | NFC_Start                                                 | 0x36 |
| 32 | LBM    | lumber                                                    | 0x0B |
| 33 | EXV_HP | Heat Pump Electronic expansion valve                      | 0x71 |
| 34 | EXV_B  | Battery Electronic expansion valve                        | 0x72 |
| 35 | EWP_B  | Battery Electronic Water Pump                             | 0x73 |
| 36 | EWP_FD | Front Driver Electronic Water Pump                        | 0x74 |
| 37 | EWP_H  | Heating Electronic Water Pump                             | 0x75 |
| 38 | CWV    | Coolant X Ways Valve                                      | 0x76 |
| 39 | WPTC   | Battery Water Positive Temperature Coefficient Thermistor | 0x77 |
| 40 | EAS    | Electric Air Conditioning Compressor                      | 0x78 |
| 41 | ETRS   | Electric transmission range selection                     | 0x04 |
| 42 | EXV_H  | HVAC Electronic Expansion valve                           | 0x7C |
| 43 | ISP    | IP switch panel                                           | 0x0C |

Note: Functional request OBD:18DB33F1; Physical request OBD: 18 DA XX F1; OBD response: 18 DA F1  
XX; XX is ECU Address.

The example of ECU CAN(FD) diagnostic Address;

| ECU  | Tester | CAN(FD)         |                       |                    |
|------|--------|-----------------|-----------------------|--------------------|
|      |        | PhysicalAddress | Non Periodic response | Functional request |
| XX   | YY     | 14DAXXYY        | 14DAYYXX              | 10DBFEYY           |
| VCCM | F1     | 14DA10F1        | 14DAF110              | 10DBFEF1           |
|      | F2     | 14DA10F2        | 14DAF210              | 10DBFEF2           |
|      | F4     | 14DA10F4        | 14DAF410              | 10DBFEF4           |
|      | F5     | 14DA10F5        | 14DAF510              | 10DBFEF5           |
|      | F6     | 14DA10F6        | 14DAF610              | 10DBFEF6           |

The ECU DoIP logicalAddress;

| Network      |    |    |              |                |    |   |   | Diagnostic Address |   |   |   |   |   |   |   |
|--------------|----|----|--------------|----------------|----|---|---|--------------------|---|---|---|---|---|---|---|
| Network Type |    |    | Message Type | Network Number |    |   |   |                    |   |   |   |   |   |   |   |
| 15           | 14 | 13 | 12           | 11             | 10 | 9 | 8 | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|              |    |    |              |                |    |   |   | XX=01...FF         |   |   |   |   |   |   |   |

XX=01...FF

Network Type: 0x0

Message Type: This bit is used to identify a response as being non-periodic or periodic, i.e. USDT or UUDT.Enumerations: 0x0=non-periodic, 0x1=periodic

Network Number: 0x0

Diagnostic Address: ECU Address

The Functional request logical address is 0x0FFE.

The Physical request logical address is 0x0FXX;XX is client ID.

For Example:

| EthernetE<br>CU | Tester | DoIP logical Address |                       |                    |
|-----------------|--------|----------------------|-----------------------|--------------------|
|                 |        | PhysicalAddress      | Non Periodic response | Functional request |
| XX              | YY     | 0FYY 00XX            | 00XX 0FYY             | 0FYY 0FFE          |
| VCCM            | F2     | 0FF2 0010            | 0010 0FF2             | 0FF2 0FFE          |
|                 | F4     | 0FF4 0010            | 0010 0FF4             | 0FF4 0FFE          |
|                 | F5     | 0FF5 0010            | 0010 0FF5             | 0FF5 0FFE          |
|                 | F6     | 0FF6 0010            | 0010 0FF6             | 0FF6 0FFE          |

For LIN ECU, if used UDS. The NAD of the Lin slave is the LIN ECU Address. And note, the Lin Master shall router the diagnostic message and operate the Client addresses priority.

## 8 Appendix B-Corporate Standard DIDs

### 8.1 Data Identifier (DID) Ranges

The following table lists all possible DID numbers. The DID NUM of the ECU shall be assigned by the SmartMi diagnostic engineer.

| DID NUM       | Diagnostic Identifier Name                       | Comments                                                                                                                                         |
|---------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xC000–0xC2FF | System Supplier Routine DIDs                     | This range of values is for System Supplier Routine DIDs use                                                                                     |
| 0xF010–0xF1EF | SmartMi Corporate Common DIDs                    | This range will be used for Corporate Common DID.                                                                                                |
| 0xF1F0–0xF1FF | Identification Option – System Supplier Specific | This range of values shall be used for system supplier specific server/vehicle system identification options.                                    |
| 0xFA10–0xFA18 | Event Data Retrieval (EDR) Data Identifiers      | This range of values is reserved for Event Data Retrieval (EDR) Data Identifiers                                                                 |
| 0xFA19–0xFAFF | Safety System Data Identifiers                   | This range of values represent safety system related DIDs.                                                                                       |
| 0xFD00–0xFEFF | System Supplier Specific                         | This range of values shall be used to reference system-supplier-specific record data identifiers and input/output identifiers within the server. |
| 0xFF00        | eraseMemory                                      | eraseMemory                                                                                                                                      |
| Others        | Reserved                                         | System Supplier shall not use                                                                                                                    |

### 8.2 Read and Write DIDs

Below table shows the overview of the SmartMi Cmmmon DID implementation. Cvt definition as below:

M: Mandatory

O: Optional

C: Conditional

C1: Applicable for ECUs with Vehicle Configuration

C2: Applicable for ECUs with ECU Configuration

C3: Applicable for ECUs with Separate Released Calibration

C4: Ethernet supported ECU

C5: Special Case

C6: Ethernet and switch supported ECU

NA: Not Applicable

| DID# | Description                                                                  | Application SW | Bootloader SW | Comment |
|------|------------------------------------------------------------------------------|----------------|---------------|---------|
| F085 | Power Moding State                                                           | M              | NA            |         |
| F087 | Vehicle Mode Status                                                          | O              | NA            |         |
| F0A9 | XCP CANID Persistence flag                                                   | O              | NA            |         |
| F0B3 | Supplier component TraceabilityCharacters                                    | M              | NA            |         |
| F0B4 | Manufacturing Traceability Characters                                        | M              | NA            |         |
| D004 | Vehicle Odometer                                                             | O              | NA            |         |
| F0F0 | Programmed State                                                             | M              | M             |         |
| F0F4 | Safe Bypass                                                                  | M              | NA            |         |
| F0FE | Total Vehicle Operation Time                                                 | M              | NA            |         |
| F186 | Active Diagnostic Session DataIdentifier                                     | M              | M             |         |
| F187 | Vehicle Manufacturer Spare Part Number                                       | M              | M             |         |
| F189 | Vehicle Manufacturer ECU SW Part Number and Version Number<br>Version Number | M              | M             |         |
| F1C1 | Vehicle Manufacturer Boot SW Part Number and Version Number                  | M              | M             |         |
| F1C2 | Vehicle Manufacturer APP SW Part Number and Version Number                   | M              | M             |         |
| F1C3 | Vehicle Manufacturer Cal SW Part Number and Version Number                   | C3             | C3            |         |
| F18A | System Supplier Identifier                                                   | M              | NA            |         |
| F18B | ECU Manufacturing Date                                                       | M              | NA            |         |
| F18C | ECU Serial Number                                                            | M              | M             |         |
| F190 | Vehicle Identification Number                                                | M              | NA            |         |
| F193 | System Supplier ECU Hardware Version Number                                  | O              | O             |         |
| F194 | System Supplier ECU Software Part Number                                     | O              | O             |         |
| F195 | System Supplier ECU Software Version Number                                  | O              | O             |         |
| F197 | ECU Name                                                                     | M              | M             |         |
| F198 | Repair Shop Code Or Tester Serial Number                                     | M              | NA            |         |
| F199 | Programming Date                                                             | M              | M             |         |
| F19E | ECU Diagnostic Database Version                                              | M              | NA            |         |
| F1A2 | ECU Matrix Version of the ECU product                                        | M              | NA            |         |
| F1B0 | ECU Address                                                                  | M              | M             |         |
| F1B1 | ECU PRID                                                                     | M              | M             |         |
| F1B2 | Log File Information                                                         | O              | O             |         |
| F1E5 | Link Status                                                                  | C4             | NA            |         |
| F1E6 | 100BASE-T1 SQI threshold                                                     | C4             | NA            |         |
| F1E7 | 1000BASE-T1 SQI threshold                                                    | C4             | NA            |         |
| F1E8 | SQI                                                                          | C4             | NA            |         |
| F1E9 | Polarity detection                                                           | C4             | NA            |         |
| F1EB | Switch counter                                                               | C4             | NA            |         |
| F1EC | MAC address                                                                  | C4             | C4            |         |
| F1D3 | Signal lines short/open circuits Number                                      | C4             | NA            |         |
| F1D5 | Signal lines current status                                                  | C4             | NA            |         |
| F1D6 | Switch ARL table                                                             | C6             | NA            |         |
| F1D7 | MASTER/SLAVE status                                                          | C4             | NA            |         |
| F1D8 | Test mode status                                                             | C4             | NA            |         |
| F100 | Vehicle Configuration                                                        | C1             | NA            |         |
| F101 | ECU Configuration                                                            | C2             | NA            |         |

### 8.2.1 DID 0xF085: Power Moding State

| Data Identifier | DID Name                                                                                                                                                                                                                                                                             | R/W                | Sessions          | Security Levels | Data Type | Len    | Appl or Boot |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|-----------------|-----------|--------|--------------|
| 0xF085          | Power Moding State                                                                                                                                                                                                                                                                   | Read               | Default, Extended | None            | Enum      | 1 Byte | App          |
| Description     | PowerModingState; The value shall be send in network by PowerModeMaster, and other ECU shall receive it.<br>0x00: OFF<br>0x01: ON<br>0x02: Ready<br>0xFF: Power Moding is Invalidity<br>0x03 - 0xFE: Reserved<br><b>Note:</b> The Power Moding State shall be supported by all ECUs. |                    |                   |                 |           |        |              |
|                 | Position                                                                                                                                                                                                                                                                             | Byte 0             |                   |                 |           |        |              |
|                 | Parmeter                                                                                                                                                                                                                                                                             | Power Moding State |                   |                 |           |        |              |
|                 | Default value                                                                                                                                                                                                                                                                        | 0x00(Off)          |                   |                 |           |        |              |

### 8.2.2 DID 0xF087: Vehicle Mode Status

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                                                                            | R/W                 | Sessions          | Security Levels | Data Type | Len    | Appl or Boot |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------|-----------------|-----------|--------|--------------|
| 0xF087          | Vehicle Mode Status                                                                                                                                                                                                                                                                                                                                                 | Read                | Default, Extended | None            | Enum      | 1 Byte | App          |
|                 |                                                                                                                                                                                                                                                                                                                                                                     | Write               | Extended          | 1,3             |           |        |              |
| Description     | This DID contains the current vehicle mode state of the ECU. The value shall be send in network by VCCM, and other ECUs shall receive it. Only VCCM(GW) support "Write".<br>0x00: Normal<br>0x01: Transport Mode<br>0x02: Factory Mode<br>0x03: Storage Mode<br>Others: Reserved<br><b>Note:</b> This DID shall be supported by an ECU which supports Vehicle Mode. |                     |                   |                 |           |        |              |
|                 | Position                                                                                                                                                                                                                                                                                                                                                            | Byte 0              |                   |                 |           |        |              |
|                 | Parmeter                                                                                                                                                                                                                                                                                                                                                            | Vehicle Mode Status |                   |                 |           |        |              |
|                 | Default value                                                                                                                                                                                                                                                                                                                                                       | 0x00(Normal)        |                   |                 |           |        |              |

### 8.2.3 DID 0xF0A9: XCP CANID Persistence flag

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W               | Sessions          | Security Levels | Data Type | Len    | Appl or Boot |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|-----------------|-----------|--------|--------------|
| 0xF0A9          | XCP CANID Persistence flag                                                                                                                                                                                                                                                                                                                                                                                                | Read              | Default, Extended | None            | Enum      | 1 Byte | App          |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                           | Write             | Extended          | 1,3             |           |        |              |
| Description     | XCP CANID Persistence flag provides a means to permit XCP CAN ID network traffic from OBD Connector CAN networks.The default value is 0x00.<br>0x00: The routing of XCP on the OBD interface is disabled.<br>0x01: The routing of XCP on the OBD interface is enabled.<br>Others: Reserved<br><b>Note:</b> This DID is a requirement only for VCCM. If the VCCM does not have this feature, the DID may not be supported. |                   |                   |                 |           |        |              |
|                 | Byte Position                                                                                                                                                                                                                                                                                                                                                                                                             | Byte 0            |                   |                 |           |        |              |
|                 | Parameter                                                                                                                                                                                                                                                                                                                                                                                                                 | Persistence Value |                   |                 |           |        |              |
|                 | Hex value                                                                                                                                                                                                                                                                                                                                                                                                                 | 0x00(Enable)      |                   |                 |           |        |              |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                   |                 |           |        |              |

### 8.2.4 DID 0xF0B3: Supplier component Traceability Characters

| Data Identifier | DID Name                                                                                                                                                                          | R/W  | Sessions          | Security Levels | Data Type       | Len     | Appl or Boot |    |    |    |    |    |     |    |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|-----------------|-----------------|---------|--------------|----|----|----|----|----|-----|----|
| 0xF0B3          | Supplier component Traceability Characters                                                                                                                                        | Read | Default, Extended | None            | Unicode & ASCII | 20 Byte | App          |    |    |    |    |    |     |    |
|                 |                                                                                                                                                                                   |      |                   |                 |                 |         |              |    |    |    |    |    |     |    |
| Description     | Byte 0- Byte 7(Unicode): Production of the city                                                                                                                                   |      |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Byte 8 (ASCII): Line Identification;The Line Identifier is a 1 byte field that is used to indicate the component assembly line.                                                   |      |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Byte 9 (ASCII): Shift Identification; This is a one byte field that identifies the shift that assembled the component.                                                            |      |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Byte 10- Byte19: Traceability Number; This value is a supplier assigned serial, lot or batch number. Designed by the supplier. The value shall be filled with 0x30, if not using. |      |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Note: This DID shall be supported by all ECUs. Shall be injected by the supplier's production line.                                                                               |      |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Position (Byte)                                                                                                                                                                   | 0    | 1                 | 2               | 3               | 4       | 5            | 6  | 7  | 8  | 9  | 10 | ... | 19 |
|                 | Traceability(example)                                                                                                                                                             | 北    |                   | 京               |                 | 0       | 0            | 0  | 0  | A  | A  | 0  |     | 0  |
|                 | Hex value                                                                                                                                                                         | 53   | 17                | 4E              | AC              | 00      | 00           | 00 | 00 | 41 | 41 | 30 |     | 30 |

### 8.2.5 DID 0xF0B4: Manufacturing Traceability Characters

| Data Identifier | DID Name                                                                                                                                                                   | R/W   | Sessions          | Security Levels | Data Type       | Len     | Appl or Boot |    |    |    |    |    |     |    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------|-----------------|-----------------|---------|--------------|----|----|----|----|----|-----|----|
| 0xF0B4          | Manufacturing Traceability Characters                                                                                                                                      | Read  | Default, Extended | None            | Unicode & ASCII | 20 Byte | App          |    |    |    |    |    |     |    |
|                 |                                                                                                                                                                            | write | Extended          | 1,3             |                 |         |              |    |    |    |    |    |     |    |
| Description     | Byte 0- Byte 7(Unicode): Production of the city                                                                                                                            |       |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Byte 8 (ASCII): Line Identification;The Line Identifier is a 1 byte field that is used to indicate the component assembly line.                                            |       |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Byte 9 (ASCII): Shift Identification; This is a one byte field that identifies the shift that assembled the component.                                                     |       |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Byte 10- Byte19: Traceability Number; This value is a OEM assigned serial, lot or batch number. Designed by the OEM ME. The value shall be filled with 0x30, if not using. |       |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Note: This DID shall be supported by VCCM(GW).                                                                                                                             |       |                   |                 |                 |         |              |    |    |    |    |    |     |    |
|                 | Position (Byte)                                                                                                                                                            | 0     | 1                 | 2               | 3               | 4       | 5            | 6  | 7  | 8  | 9  | 10 | ... | 19 |
|                 | Traceability(example)                                                                                                                                                      | 北     |                   | 京               |                 | 0       |              | 0  |    | A  | A  | 0  |     | 0  |
|                 | Hex value                                                                                                                                                                  | 53    | 17                | 4E              | AC              | 00      | 30           | 00 | 30 | 41 | 41 | 30 |     | 30 |

### 8.2.6 DID 0xD004: Vehicle Odometer

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                                                                                                  | R/W                | Sessions          | Security Levels | Data Type | Len    | Appl or Boot |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|-----------------|-----------|--------|--------------|
| 0xD004          | Vehicle Odometer                                                                                                                                                                                                                                                                                                                                                                          | Read               | Default, Extended | None            | Unsigned  | 3 Byte | App          |
|                 |                                                                                                                                                                                                                                                                                                                                                                                           |                    |                   |                 |           |        |              |
| Description     | This DID contains the Odometer value as stored by the ECU. The odometer value shall be stored with a scaling of E=N.<br>This value may come from the network or from the result of its own calculation.<br>Note: This DID is only required by ECUs that support the odometer function (at least two ECUs need support odometer recording and have a mechanism for backing up each other). |                    |                   |                 |           |        |              |
|                 | Position                                                                                                                                                                                                                                                                                                                                                                                  | 1                  |                   | 2               |           | 3      |              |
|                 | ECU Odometer Value(example)                                                                                                                                                                                                                                                                                                                                                               | N=20000, E=20000km |                   |                 |           |        |              |
|                 | Hex value                                                                                                                                                                                                                                                                                                                                                                                 | 00                 |                   | 4E              |           | 20     |              |
|                 |                                                                                                                                                                                                                                                                                                                                                                                           |                    |                   |                 |           |        |              |

### 8.2.7 DID 0xF0F0: Programmed State

| Data Identifier | DID Name                                                                                                                                                                                                                                                                        | R/W     | Sessions                      | Security Levels | Data Type | Len         | Appl or Boot |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------|-----------------|-----------|-------------|--------------|
| 0xF0F0          | ProgrammedState                                                                                                                                                                                                                                                                 | Read    | Default, Programming Extended | NA              | Hex/ ENM  | 2 – 2n Byte | APP& Boot    |
|                 |                                                                                                                                                                                                                                                                                 | Write   | Only in boot Extended         | 05,09           |           |             |              |
| Description     | The Programmed State provides a reliable mechanism for a Tester to determine if a logical partition is programmed with a valid content (i.e. Application software or calibration data). Each logical partition has a dedicated ProgrammedState that can be read using this DID. |         |                               |                 |           |             |              |
|                 | Byte 0~1:                                                                                                                                                                                                                                                                       |         |                               |                 |           |             |              |
|                 | Byte0:File Type, Refer to “文件类型表” in <QSMG-EEA-6002-Utility 需求规范>.                                                                                                                                                                                                              |         |                               |                 |           |             |              |
|                 | Byte1: 0x00 programming is valid<br>0x01 programming is invalid                                                                                                                                                                                                                 |         |                               |                 |           |             |              |
|                 | Byte 2n-2 ~ 2n-1:<br>n is the number of Program File.<br>Note: This DID shall be supported only by ECUs which implement <QSMG-EEA-6002-Utility 需求规范> or <QSMG-EEA-4011-BootLoader 需求规范> requirements.                                                                           |         |                               |                 |           |             |              |
|                 | Position                                                                                                                                                                                                                                                                        | Byte 0  | Byte 1                        | ...             |           | ...         |              |
|                 | Parameter                                                                                                                                                                                                                                                                       | APP SW1 | programming is valid          | ...             |           | ...         |              |
|                 | Hex value                                                                                                                                                                                                                                                                       | 01      | 00                            | ...             |           | ...         |              |

### 8.2.8 DID 0xF0F4: Safe Bypass

| Data Identifier | DID Name                                                                                             | R/W          | Sessions          | Security Levels | Data Type | Len | Appl or Boot |
|-----------------|------------------------------------------------------------------------------------------------------|--------------|-------------------|-----------------|-----------|-----|--------------|
| 0xF0F4          | Safe Bypass                                                                                          | Read         | Default, Extended | NA              | ENM       | 2   | APP          |
|                 |                                                                                                      | Write        | Extended          | 1,3             |           |     |              |
| Description     | The Safe Bypass is Used during debugging. It should be removed or left unwritable during production. |              |                   |                 |           |     |              |
|                 | Byte 0: E2E Bypass<br>0x00: E2E enable<br>0x01: E2E disable                                          |              |                   |                 |           |     |              |
|                 | Byte 1: SecOC Bypass<br>0x00: SecOC enable<br>0x01: SecOC disable                                    |              |                   |                 |           |     |              |
|                 | Position                                                                                             | Byte 0       | Byte 1            |                 |           |     |              |
| Parameter       | E2E enable                                                                                           | SecOC enable |                   |                 |           |     |              |
| Default value   | 00                                                                                                   | 00           |                   |                 |           |     |              |

### 8.2.9 DID 0xF0FE: Total Vehicle Operation Time

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W        | Sessions         | Security Levels | Data Type | Len                        | Appl or Boot  |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|-----------------|-----------|----------------------------|---------------|
| 0xF0FE          | Total Vehicle Operation Time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Read       | Default Extended | None            | Hex       | 5 byte                     | App           |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Write      | Extended         | 1,3             |           |                            |               |
| Description     | Total Vehicle Operation Time: Total Vehicle Operation Time is a running time counter that is incremented whenever any partial network on the vehicle is active.<br>Vehicle Operation Time - Total is the accumulated value over the operational life of the vehicle. Each ECU shall support and calculate it.<br><br>Total Vehicle Operation Time: $E = N * 100ms$<br><br>Other ECUs receive this value from VCCM.<br>Other ECUs shall store the latest valid value received in permanent memory during power-down, only when the stored value is less than the currently received value.<br><br>Note: This DID shall be supported by all ECUs. Only VCCM supports write. |            |                  |                 |           |                            |               |
|                 | Signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Start Byte | Start Bit        | Length (Bits)   | Data Type | Range                      | Conversion    |
|                 | Total Vehicle Operation Time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0          | 7                | 40              | USN       | 0.0 - 1.099511627775E14 ms | $E = N * 100$ |

### 8.2.10 DID 0xF186: Active Diagnostic Session Data Identifier

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                                                                            | R/W                   | Sessions                       | Security Levels | Data Type | Len    | Appl or Boot |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------|-----------------|-----------|--------|--------------|
| 0xF186          | ActiveDiagnosticSessionDataIdentifier                                                                                                                                                                                                                                                                                                                               | Read                  | Default<br>Program<br>Extended | None            | Enum      | 1 Byte | App&Boot     |
|                 |                                                                                                                                                                                                                                                                                                                                                                     |                       |                                |                 |           |        |              |
| Description     | This DID shall be used to report the active diagnostic session in the server. The values are defined by the following parameters:<br>0x00= Reserved<br>0x01= DefaultSession<br>0x02= ProgrammingSession<br>0x03= ExtendedDiagnosticSession<br>0x60= SupplierDiagnosticSession<br>0x05 - 0x7F= Reserved<br><br><b>Note:</b> This DID shall be supported by all ECUs. |                       |                                |                 |           |        |              |
|                 | Byte Position                                                                                                                                                                                                                                                                                                                                                       | Byte 0                |                                |                 |           |        |              |
|                 | Parameter                                                                                                                                                                                                                                                                                                                                                           | DiagnosticSessionType |                                |                 |           |        |              |
|                 | Default Value                                                                                                                                                                                                                                                                                                                                                       | 0x01 (DefaultSession) |                                |                 |           |        |              |

### 8.2.11 DID 0xF187: VehicleManufacturerSparePart Number

| Data Identifier    | DID Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W    | Sessions                 | Security Levels | Data Type | Len    | Appl or Boot |               |        |        |        |        |        |        |        |        |        |                    |   |   |   |   |   |   |   |   |   |           |      |      |      |      |      |      |      |      |      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------|-----------------|-----------|--------|--------------|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------------------|---|---|---|---|---|---|---|---|---|-----------|------|------|------|------|------|------|------|------|------|
| 0xF187             | VehicleManufacturerSparePart Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Read   | Default Program Extended | None            | ASCII     | 9 Byte | App&B oot    |               |        |        |        |        |        |        |        |        |        |                    |   |   |   |   |   |   |   |   |   |           |      |      |      |      |      |      |      |      |      |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                          |                 |           |        |              |               |        |        |        |        |        |        |        |        |        |                    |   |   |   |   |   |   |   |   |   |           |      |      |      |      |      |      |      |      |      |
| Description        | <div>This DID shall be used to report the part number of the entire ECU. The values are from BOM. The following figure describes the encoding rules. This value should be provided by DRE.</div> <div><div>零部件编码规则：由四部分构成，初步定为9位字母数字组合。</div><div><div><div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div></div><div><div>四位流水数字，由系统按照1-9999的顺序自动生成</div><div>2位小类分类号</div><div>2位大类分类号，原则上使用现行在用。例如：电池-11，电机-12，电控-13，充电-14。</div><div>企业产品类别编码，固定为V。</div></div></div></div><div>Note: This DID shall be supported by all ECUs.</div><table><tr><th>Byte Position</th><th>Byte 0</th><th>Byte 1</th><th>Byte 2</th><th>Byte 3</th><th>Byte 4</th><th>Byte 5</th><th>Byte 6</th><th>Byte 7</th><th>Byte 8</th></tr><tr><td>Parameter(example)</td><td>V</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>Hex Value</td><td>0x56</td><td>0x31</td><td>0x31</td><td>0x30</td><td>0x31</td><td>0x30</td><td>0x30</td><td>0x30</td><td>0x31</td></tr></table></div> |        |                          |                 |           |        |              | Byte Position | Byte 0 | Byte 1 | Byte 2 | Byte 3 | Byte 4 | Byte 5 | Byte 6 | Byte 7 | Byte 8 | Parameter(example) | V | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | Hex Value | 0x56 | 0x31 | 0x31 | 0x30 | 0x31 | 0x30 | 0x30 | 0x30 | 0x31 |
| Byte Position      | Byte 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Byte 1 | Byte 2                   | Byte 3          | Byte 4    | Byte 5 | Byte 6       | Byte 7        | Byte 8 |        |        |        |        |        |        |        |        |                    |   |   |   |   |   |   |   |   |   |           |      |      |      |      |      |      |      |      |      |
| Parameter(example) | V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1      | 1                        | 0               | 1         | 0      | 0            | 0             | 1      |        |        |        |        |        |        |        |        |                    |   |   |   |   |   |   |   |   |   |           |      |      |      |      |      |      |      |      |      |
| Hex Value          | 0x56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0x31   | 0x31                     | 0x30            | 0x31      | 0x30   | 0x30         | 0x30          | 0x31   |        |        |        |        |        |        |        |        |                    |   |   |   |   |   |   |   |   |   |           |      |      |      |      |      |      |      |      |      |

8.2.12 **DID 0xF189: Vehicle Manufacturer ECU SW Part Number and Version Number**

| Data Identifier     | DID Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W  | Sessions         | Security Levels | Data Type | Len     | Appl or Boot |               |    |    |    |    |    |    |    |   |   |   |    |    |    |    |    |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------|-----------------|-----------|---------|--------------|---------------|----|----|----|----|----|----|----|---|---|---|----|----|----|----|----|---------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0xF189              | Vehicle Manufacturer ECU SW Part Number and Version Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Read | Default Extended | None            | ASCII     | 15 Byte | App&Boot     |               |    |    |    |    |    |    |    |   |   |   |    |    |    |    |    |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Description         | This DID shall be used to report the part number of the entire ECU SW.The following figure describes the encoding rules. This value should be provided by DRE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |                  |                 |           |         |              |               |    |    |    |    |    |    |    |   |   |   |    |    |    |    |    |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|                     | <div>3.2.4软硬件编码规则——软件总成</div> <div>硬件放在零部件号中进行管理。软件总成向下游生产系统进行传递，作为生产车间刷写程序。软件总成由4部分构成，共有15位。其编码规则如下：</div> <div><div><div><div></div><div>s</div><div>-</div><div><div></div><div></div><div></div><div></div></div></div><div><div><div></div><div></div><div></div><div></div></div><div>四位顺序码代表版本号</div></div><div><div><div></div><div></div><div></div><div></div></div><div>固定位：一位下划线</div></div><div><div><div></div><div></div><div></div><div></div></div><div>固定位：S，代表们软件编码</div></div><div><div><div></div><div></div><div></div><div></div></div><div>九位零部件编码</div></div></div></div>                                                             |      |                  |                 |           |         |              |               |    |    |    |    |    |    |    |   |   |   |    |    |    |    |    |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|                     | <div>Note: This DID shall be supported by all ECUs.</div> <table><tr><td>Byte Position</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td></tr><tr><td>Parameter(e xample)</td><td>V</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>S</td><td>_</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>Hex Value</td><td>56</td><td>31</td><td>31</td><td>30</td><td>31</td><td>30</td><td>30</td><td>30</td><td>31</td><td>53</td><td>5F</td><td>30</td><td>30</td><td>30</td><td>31</td></tr></table> |      |                  |                 |           |         |              | Byte Position | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | Parameter(e xample) | V | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | S | _ | 0 | 0 | 0 | 1 | Hex Value | 56 | 31 | 31 | 30 | 31 | 30 | 30 | 30 | 31 | 53 | 5F | 30 | 30 | 30 |
| Byte Position       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1    | 2                | 3               | 4         | 5       | 6            | 7             | 8  | 9  | 10 | 11 | 12 | 13 | 14 |   |   |   |    |    |    |    |    |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Parameter(e xample) | V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1    | 1                | 0               | 1         | 0       | 0            | 0             | 1  | S  | _  | 0  | 0  | 0  | 1  |   |   |   |    |    |    |    |    |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Hex Value           | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 31   | 31               | 30              | 31        | 30      | 30           | 30            | 31 | 53 | 5F | 30 | 30 | 30 | 31 |   |   |   |    |    |    |    |    |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

8.2.13 DID 0xF1C1: Vehicle Manufacturer Boot SW Part Number and Version Number

| Data Identifier                                       | DID Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W  | Sessions                    | Security Levels | Data Type | Len       | Appl or Boot |    |    |    |    |    |    |    |    |    |    |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------|-----------------|-----------|-----------|--------------|----|----|----|----|----|----|----|----|----|----|
| 0xF1C1                                                | Vehicle Manufacturer Boot SW Part Number and Version Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Read | Default ProgrammingExtended | None            | ASCII     | 17*n Byte | App&Boot     |    |    |    |    |    |    |    |    |    |    |
|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |                             |                 |           |           |              |    |    |    |    |    |    |    |    |    |    |
| Description                                           | This DID shall be used to report the part number of the ECU Boot SW. The following figure describes the encoding rules. This value should be provided by DRE.<br>n is the number of Boot SW.<br><div><div><div></div><div>S</div><div></div><div></div><div></div><div>-</div><div></div><div></div><div></div></div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div><div>版本号，用三位顺序码表示</div><div>固定位：一位下划线</div><div>FileType,文件类型；参考&lt;QSMG-EEA-6002-Utility需求规范&gt;文件类型表。</div><div>软件大类区分码，A:APP SW；B:Boot；C:Cal。</div><div>固定位：S，代表们软件编码</div><div>九位零部件编码</div></div> |      |                             |                 |           |           |              |    |    |    |    |    |    |    |    |    |    |
| <b>Note:</b> This DID shall be supported by all ECUs. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |                             |                 |           |           |              |    |    |    |    |    |    |    |    |    |    |
| Byte Position                                         | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1    | 2                           | 3               | 4         | 5         | 6            | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| Parameter(example)                                    | V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1    | 1                           | 0               | 1         | 0         | 0            | 0  | 1  | S  | B  | 0  | 1  | _  | 0  | 0  | 1  |
| Hex Value                                             | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 31   | 31                          | 30              | 31        | 30        | 30           | 30 | 31 | 53 | 42 | 30 | 31 | 5F | 30 | 30 | 31 |

8.2.14 DID 0xF1C2: Vehicle Manufacturer APP SW Part Number and Version Number

| Data Identifier                                                                                                   | DID Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W  | Sessions            | Security Levels | Data Type | Len          | Appl or Boot |    |    |    |    |    |    |    |    |    |    |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------|-----------------|-----------|--------------|--------------|----|----|----|----|----|----|----|----|----|----|
| 0xF1C2                                                                                                            | Vehicle Manufacturer APP SW Part Number and Version Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Read | Default<br>Extended | None            | ASCII     | 16*n<br>Byte | App&Boot     |    |    |    |    |    |    |    |    |    |    |
| Description                                                                                                       | <p>This DID shall be used to report the part number of the ECU APP SW. The following figure describes the encoding rules. This value should be provided by DRE.<br/>n is the number of APP SW.</p> <div><div><div></div><div>S</div><div></div><div></div><div></div><div>-</div><div></div><div></div><div></div></div><div><div>九位零部件编码</div><div>固定位：S，代表们软件编码</div><div>软件大类区分码，A:APP SW；<br/>B:Boot；C:Cal。</div><div>FileType,文件类型；参考&lt;QSMG-EEA-6002-Utility需求规范&gt;文件类型表。</div><div>固定位：一位下划线</div><div>版本号，用三位顺序码表示</div></div></div> |      |                     |                 |           |              |              |    |    |    |    |    |    |    |    |    |    |
| <p>If there is no APP, Boot should return all 0x00.<br/><b>Note:</b> This DID shall be supported by all ECUs.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |                     |                 |           |              |              |    |    |    |    |    |    |    |    |    |    |
| Byte Position                                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1    | 2                   | 3               | 4         | 5            | 6            | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| Parameter(example)                                                                                                | V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1    | 1                   | 0               | 1         | 0            | 0            | 0  | 1  | S  | A  | 1  | 0  | _  | 0  | 0  | 1  |
| Hex Value                                                                                                         | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 31   | 31                  | 30              | 31        | 30           | 30           | 30 | 31 | 53 | 41 | 31 | 30 | 5F | 30 | 30 | 31 |

### 8.2.15 DID 0xF1C3: Vehicle Manufacturer Cal SW Part Number and Version Number

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R/W  | Sessions            | Security Levels | Data Type | Len          | Appl or Boot |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------|-----------------|-----------|--------------|--------------|
| 0xF1C3          | Vehicle Manufacturer Cal SW Part Number and Version Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Read | Default<br>Extended | None            | ASCII     | 17*n<br>Byte | App&Boot     |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |                     |                 |           |              |              |
| Description     | This DID shall be used to report the part number of the ECU Cal SW. The following figure describes the encoding rules.<br>This value should be provided by DRE.<br>n is the number of Cal SW.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |                     |                 |           |              |              |
|                 | <div><div><div><div></div><div>S</div><div></div><div></div><div></div><div>-</div><div></div><div></div><div></div></div><div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div></div></div> |      |                     |                 |           |              |              |

### 8.2.16 DID 0xF18A: System Supplier Identifier

| Data Identifier | DID Name                                                                                                            | R/W  | Sessions         | Security Levels | Data Type | Len | Appl or Boot |
|-----------------|---------------------------------------------------------------------------------------------------------------------|------|------------------|-----------------|-----------|-----|--------------|
| 0xF18A          | System Supplier Identifier                                                                                          | Read | Default Extended | None            | ASCII     | TBD | App          |
|                 |                                                                                                                     |      |                  |                 |           |     |              |
| Description     | Code for each supplier, provided by purchasing department.<br><b>Note:</b> This DID shall be supported by all ECUs. |      |                  |                 |           |     |              |
|                 | Byte Position                                                                                                       | TBD  |                  |                 |           |     |              |
|                 | Parameter                                                                                                           | TBD  |                  |                 |           |     |              |
|                 | Default Value                                                                                                       | TBD  |                  |                 |           |     |              |

### 8.2.17 DID 0xF18B: ECU Manufacturing Date

| Data Identifier | DID Name                                                                                                                                                                                      | R/W    | Sessions         | Security Levels | Data Type | Len      | Appl or Boot |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------|-----------------|-----------|----------|--------------|
| 0xF18B          | ECU Manufacturing Date                                                                                                                                                                        | Read   | Default Extended | None            | BCD       | 5 Byte   | App          |
|                 |                                                                                                                                                                                               | Write  | Extended         | 1               |           |          |              |
| Description     | This value needs to be written in the OEM production line.<br>Byte 0~1: Year<br>Byte 2: Month<br>Byte 3: Day<br>Byte 4: Hour(24)<br><br><b>Note:</b> This DID shall be supported by all ECUs. |        |                  |                 |           |          |              |
|                 | Byte Position                                                                                                                                                                                 | Byte 0 | Byte 1           | Byte 2          | Byte 3    | Byte 4   |              |
|                 | Parameter                                                                                                                                                                                     | Year   |                  | Month           | Day       | Hour(24) |              |
|                 | Default Value                                                                                                                                                                                 | 20     | 24               | 06              | 30        | 15       |              |
|                 |                                                                                                                                                                                               |        |                  |                 |           |          |              |

### 8.2.18 DID 0xF18C: ECU Serial Number

| Data Identifier | DID Name                                                                                                                                                                                        | R/W  | Sessions                    | Security Levels | Data Type | Len     | Appl or Boot |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------|-----------------|-----------|---------|--------------|
| 0xF18C          | ECU Serial Number                                                                                                                                                                               | Read | Default ProgrammingExtended | None            | TBD       | 16 Byte | App&Boot     |
|                 |                                                                                                                                                                                                 |      |                             |                 |           |         |              |
| Description     | It is designed and implemented by the supplier and injected by the supplier's production line. Requires different values for each ECU.<br><b>Note:</b> This DID shall be supported by all ECUs. |      |                             |                 |           |         |              |
|                 | Byte Position                                                                                                                                                                                   |      |                             |                 |           |         |              |
|                 | Parameter                                                                                                                                                                                       |      |                             |                 |           |         |              |
|                 | Default Value                                                                                                                                                                                   |      |                             |                 |           |         |              |

### 8.2.19 DID 0xF190: Vehicle Identification Number

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                      | R/W              | Sessions         | Security Levels | Data Type | Len     | Appl or Boot |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|-----------------|-----------|---------|--------------|
| 0xF190          | Vehicle Identification Number                                                                                                                                                                                                                                                                                 | Read             | Default Extended | None            | ASCII     | 17 Byte | App          |
|                 |                                                                                                                                                                                                                                                                                                               | Write            | Extended         | 1,3             |           |         |              |
| Description     | This DID contains the Vehicle Identification Number. The data content shall be filled by the OEM manufacturer. If the VIN has not yet been programmed into the ECU permanent memory, those locations shall be either “\$00” or “\$FF”. All 17 digits must be programmed in order for the ECU to use this DID. |                  |                  |                 |           |         |              |
|                 | <b>Note:</b> This DID shall be supported by all ECUs.                                                                                                                                                                                                                                                         |                  |                  |                 |           |         |              |
|                 | Byte Position                                                                                                                                                                                                                                                                                                 | Byte1 – Byte17   |                  |                 |           |         |              |
|                 | Parameter                                                                                                                                                                                                                                                                                                     | VIN              |                  |                 |           |         |              |
|                 | Default value                                                                                                                                                                                                                                                                                                 | All \$00 or \$FF |                  |                 |           |         |              |

### 8.2.20 DID 0xF193: System Supplier ECU Hardware Version Number

It is up to the supplier to decide whether to use the DID, as well as to determine its content, but the format of the DID must be ASCII format.

### 8.2.21 DID 0xF195: System Supplier ECU Software Version Number

It is up to the supplier to decide whether to use the DID, as well as to determine its content, but the format of the DID must be ASCII format.

### 8.2.22 DID 0xF197: ECU Name

| Data Identifier | DID Name                                                                                                    |       | R/W   | Sessions                     | Security Levels | Data Type | Len    | Appl or Boot |       |
|-----------------|-------------------------------------------------------------------------------------------------------------|-------|-------|------------------------------|-----------------|-----------|--------|--------------|-------|
| 0xF197          | ECU Name                                                                                                    |       | Read  | Default Programming Extended | None            | ASCII     | 8 Byte | App&Boot     |       |
|                 |                                                                                                             |       |       |                              |                 |           |        |              |       |
| Description     | This DID description the name of ECU. Supplier can get the name from <QSMG-EEA-4001-UDS 诊断服务规范> Appendix A. |       |       |                              |                 |           |        |              |       |
|                 | Note: This DID shall be supported by all ECUs.                                                              |       |       |                              |                 |           |        |              |       |
|                 | Byte Position                                                                                               | Byte0 | Byte1 | Byte2                        | Byte3           | Byte4     | Byte5  | Byte6        | Byte7 |
|                 | Parameter(Example)                                                                                          | V     | C     | C                            | M               |           |        |              |       |
|                 | value                                                                                                       | 56    | 43    | 43                           | 4D              | 20        | 20     | 20           | 20    |

### 8.2.23 DID 0xF198: Repair Shop Code Or Tester Serial Number

| Data Identifier | DID Name                                                                                                                                                                                                                                 | R/W   | Sessions             | Security Levels | Data Type | Len     | Appl or Boot |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------|-----------------|-----------|---------|--------------|
| 0xF198          | Repair Shop Code Or Tester Serial Number                                                                                                                                                                                                 | Read  | Default Extended     | None            | ASCII     | 10 Byte | App          |
|                 |                                                                                                                                                                                                                                          | Write | Extended programming | 1,3             |           |         |              |
| Description     | This DID contains the RepairShopCodeOrTesterSerialNumber which identifies the dealer’s code or the tester’s serial number. This ASCII string is programmed into the ECU’s memory during the last programming session at the dealer site. |       |                      |                 |           |         |              |
|                 | Note: This DID shall be supported by all ECUs.                                                                                                                                                                                           |       |                      |                 |           |         |              |
|                 | Byte Position                                                                                                                                                                                                                            |       | Byte1 – Byte10       |                 |           |         |              |
|                 | Parameter                                                                                                                                                                                                                                |       | RepairShopCode       |                 |           |         |              |
|                 | Default value                                                                                                                                                                                                                            |       | All 0s               |                 |           |         |              |

### 8.2.24 DID 0xF199: Programming Date

| Data Identifier | DID Name                                                                                                                                                                                  | R/W    | Sessions             | Security Levels | Data Type | Len      | Appl or Boot |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------|-----------------|-----------|----------|--------------|
| 0xF199          | Programming Date                                                                                                                                                                          | Read   | Default Extended     | None            | BCD       | 5 Byte   | App&B<br>oot |
|                 |                                                                                                                                                                                           | Write  | Extended programming | 1,5             |           |          |              |
| Description     | This value needs to be written in the OEM production line.<br>Byte 0~1: Year<br>Byte 2: Month<br>Byte 3: Day<br>Byte 4: Hour(24)<br><b>Note:</b> This DID shall be supported by all ECUs. |        |                      |                 |           |          |              |
|                 | Byte Position                                                                                                                                                                             | Byte 0 | Byte 1               | Byte 2          | Byte 3    | Byte 4   |              |
|                 | Parameter                                                                                                                                                                                 | Year   |                      | Month           | Day       | Hour(24) |              |
|                 | Default Value                                                                                                                                                                             | 20     | 24                   | 06              | 30        | 15       |              |

### 8.2.25 DID 0xF194: System Supplier ECU Software Part Number

It is up to the supplier to decide whether to use the DID, as well as to determine its format and content.

### 8.2.26 DID 0xF19E: ECU Diagnostic Database Version

| Data Identifier | DID Name                                                                                                                                                                                                                    | R/W            | Sessions         | Security Levels | Data Type | Len    | Appl or Boot |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------|-----------------|-----------|--------|--------------|
| 0xF19E          | ECU Diagnostic Database Version                                                                                                                                                                                             | Read           | Default Extended | None            | ASCII     | 16Byte | App          |
|                 |                                                                                                                                                                                                                             |                |                  |                 |           |        |              |
| Description     | Byte0~Byte7: platform number, example ‘V23’ (0x56 0x32 0x33 0x20 0x20 0x20 0x20 0x20) . If the platform number is less than 8, padding the byte which is useless by 0x20.<br>Byte8~Byte15: ECU Diagnostic Database Version. |                |                  |                 |           |        |              |
|                 | <b>Note:</b> This DID shall be supported by all ECUs.                                                                                                                                                                       |                |                  |                 |           |        |              |
|                 | Byte Position                                                                                                                                                                                                               | Byte1 – Byte10 |                  |                 |           |        |              |
|                 | Parameter                                                                                                                                                                                                                   | V23 V1.02.03   |                  |                 |           |        |              |
|                 | Default value                                                                                                                                                                                                               | All 0s         |                  |                 |           |        |              |

### 8.2.27 DID 0xF1A2: ECU Matrix Version of the ECU product

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                   | R/W            | Sessions            | Security Levels | Data Type | Len    | Appl or Boot |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|-----------------|-----------|--------|--------------|
| 0xF1A2          | Matrix Version of the ECU product                                                                                                                                                                                                                                                          | Read           | Default<br>Extended | None            | ASCII     | 16Byte | App          |
| Description     | Byte0~Byte7: platform number, example ‘V23’ (0x56 0x32 0x33 0x20 0x20 0x20 0x20 0x20) . If the platform number is less than 8, padding the byte which is useless by 0x20.<br>Byte8~Byte15: Matrix Version of the ECU product.<br><br><b>Note:</b> This DID shall be supported by all ECUs. |                |                     |                 |           |        |              |
|                 | Byte Position                                                                                                                                                                                                                                                                              | Byte1 – Byte10 |                     |                 |           |        |              |
|                 | Parameter                                                                                                                                                                                                                                                                                  | V23 V1.02.03   |                     |                 |           |        |              |
|                 | Default value                                                                                                                                                                                                                                                                              | All 0s         |                     |                 |           |        |              |

### 8.2.28 DID 0xF1B0: ECU Address

| Data Identifier | DID Name                                                                                                                                                                 | R/W   | Sessions                           | Security Levels | Data Type | Len    | Appl or Boot |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------|-----------------|-----------|--------|--------------|
| 0xF1B0          | ECU Address                                                                                                                                                              | Read  | Default<br>Programming<br>Extended | None            | HEX       | 1 Byte | App&Boot     |
| Description     | This DID description the name of ECU. Supplier can get the name from <QSMG-EEA-4001-UDS 诊断服务规范> Appendix A.<br><br><b>Note:</b> This DID shall be supported by all ECUs. |       |                                    |                 |           |        |              |
|                 | Byte Position                                                                                                                                                            | Byte0 |                                    |                 |           |        |              |
|                 | Parameter(Example)                                                                                                                                                       | VCCM  |                                    |                 |           |        |              |
|                 | value                                                                                                                                                                    | 0x10  |                                    |                 |           |        |              |

### 8.2.29 DID 0xF1B1: ECU PRID

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                                                                                                                            | R/W              | Sessions                     | Security Levels | Data Type | Len         | Appl or Boot |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------|-----------------|-----------|-------------|--------------|
| 0xF1B1          | ECU PRID                                                                                                                                                                                                                                                                                                                                                                                                            | Read             | Default Programming Extended | None            | HEX       | 3 – 3n Byte | App&Boot     |
| Description     | PRID is comes from the Headerinfo of a programming file, details please refer to “APP SW 包文件 Basic Header 格式信息” in <QSMG-EEA-6002-Utility 需求规范><br>Byte 0~2:<br>Byte0:File Type, Refer to “文件类型表” in <QSMG-EEA-6002-Utility 需求规范>.<br>Byte1-2: PRID of current software.<br>Byte 3n-3 ~ 3n-1:<br>n is the number of Program File.<br>Note: This DID shall be supported only by ECUs which has A/B system partition. |                  |                              |                 |           |             |              |
|                 | Byte Position                                                                                                                                                                                                                                                                                                                                                                                                       | Byte0-Byte2      |                              |                 |           |             |              |
|                 | Parameter(Example)                                                                                                                                                                                                                                                                                                                                                                                                  |                  |                              |                 |           |             |              |
|                 | value                                                                                                                                                                                                                                                                                                                                                                                                               | 0x10, 0x00, 0x01 |                              |                 |           |             |              |

### 8.2.30 DID 0xF1B2: Log File Information

| Data Identifier | DID Name                                                                                                                                                                                                                                                                                                                                                                                                                      | R/W  | Sessions | Security Levels | Data Type | Len | Appl or Boot |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|-----------------|-----------|-----|--------------|
| 0xF1B2          | Log File Information                                                                                                                                                                                                                                                                                                                                                                                                          | Read | 01/02/03 | None            |           | 不固定 | App&Boot     |
| Description     | The "Log File" here refers specifically to the ECU upgrade log file, and the detailed definition is shown in "QSMG-EEA-4012-文件系统刷写规范".<br><br>Byte 0 (Unsigned) : Log File Number<br>Byte 1-2 (Unsigned) : Log File 1 FilePathAndNameLength<br>Byte 3-n (ASCII) : Log File 1 FilePathAndName<br>Byte (n+1)-(n+2) (Unsigned) : Log File 2 FilePathAndNameLength<br>Byte (n+3)-m (ASCII) : Log File 2 FilePathAndName<br>... .. |      |          |                 |           |     |              |
|                 | Byte Position                                                                                                                                                                                                                                                                                                                                                                                                                 |      |          |                 |           |     |              |
|                 | Parameter(Example)                                                                                                                                                                                                                                                                                                                                                                                                            |      |          |                 |           |     |              |
|                 | value                                                                                                                                                                                                                                                                                                                                                                                                                         |      |          |                 |           |     |              |

### 8.2.31 0xF1E5: Link Status

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

### 8.2.32 0xF1E6: 100BASE-T1 SQI threshold

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

|                                                                                   |                                                          |         |                 |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|-----------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |                 |
|                                                                                   | V23                                                      | Rev 3.3 | Page 121 of 146 |

#### 8.2.33 **0xF1E7: 1000BASE-T1 SQI threshold**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.34 **0xF1E8: SQI**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.35 **0xF1E9: Polarity detection**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.36 **0xF1EB: Switch counter**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.37 **0xF1EC: MAC address**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.38 **0xF1D3: Signal lines short/open circuits Number**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.39 **0xF1D5: Signal lines current status**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.40 **0xF1D7: MASTER/SLAVE status**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.41 **0xF1D8: Test mode status**

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.42 0xF1D9: Switch ARL table

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.43 0xF1DA: Ethernet Package Error Rate

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.44 0xF1DB: IP Address

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

#### 8.2.45 0xF100: Vehicle Configuration

Refer to Enterprise-level feature database

#### 8.2.46 0xF101: ECU Configuration

It is design by ECU supplier.

### 8.3 Routine Identification

Below table shows the overview of the SmartMi Routine DID implementation. Cvt definition as below:

M: Mandatory

O: Optional

NA: Not Applicable

C4: Ethernet supported ECU

C6: Ethernet and switch supported ECU

C7: only used for the ECU with gateway function

C8: only used for the ECU with file system

| RID   | Description                                       | Application SW | Bootloader SW | Comment |
|-------|---------------------------------------------------|----------------|---------------|---------|
| 02 02 | Check Memory Routine                              | NA             | M             |         |
| 02 03 | Check Programming Precondition                    | M              | M             |         |
| FF 00 | Erase Memory                                      | NA             | M             |         |
| FF 01 | Check Programming Dependence                      | NA             | M             |         |
| 02 04 | Disable Programming Prohibition Rollback function | NA             | M             |         |
| 02 05 | Upgrade Package SHA256 Check                      | C8             | C8            |         |
| 02 06 | Check Programming Integrity-SHA256                | C8             | C8            |         |
| 02 07 | Upgrade Package Installation                      | C8             | C8            |         |
| 02 08 | Program Rollback                                  | C8             | C8            |         |
| 04 00 | SecurityKey Generation                            | M              | NA            |         |
| A1 A1 | Test Mode Active                                  | C4             | NA            |         |
| A1 A2 | Cable Diagnostics                                 | C4             | NA            |         |

|       |                         |    |    |  |
|-------|-------------------------|----|----|--|
| A1 A3 | Reset Switch Counter    | C6 | NA |  |
| A1 AF | Ethernet Message Mirror | C7 | NA |  |

### 8.3.1 RID 0x0202: Check Memory Routine

|                                      |                                                                                                                                                                                                                                                                                                                                                                               |                   |                              |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Routine Identifier (RID)             | 0x0202                                                                                                                                                                                                                                                                                                                                                                        |                   |                              |
| RID name                             | Check Memory Routine                                                                                                                                                                                                                                                                                                                                                          |                   |                              |
| Description                          | The DID is used to verify that the programming file is correct.<br>The DID shall not be skipped.<br>If the signature fails, or the RID is not sent; then the flashing fails, and the ECU shall clear the written data.<br>Note: This RID shall be supported by all programming ECUs.                                                                                          |                   |                              |
| Enable Conditions                    | Programming session, must be in boot                                                                                                                                                                                                                                                                                                                                          |                   |                              |
| Security Level                       | 05,09                                                                                                                                                                                                                                                                                                                                                                         |                   |                              |
| Availability                         | Bootloader(Session 01,02,03)                                                                                                                                                                                                                                                                                                                                                  |                   |                              |
| Request Message Definition           |                                                                                                                                                                                                                                                                                                                                                                               |                   |                              |
| Sub-Functions Required               | 0x01 Start Routine                                                                                                                                                                                                                                                                                                                                                            | 0x02 Stop Routine | 0x03 Request Routine Results |
| Routine Control Option Record        | 1 Byte = File Type, Refer to “文件类型表” in <QSMG-EEA-6002-Utility 需求规范><br>Each logical partition has a programming file.<br><br>For example File Type number (in hex) have the following meaning.<br>0x10 : MCU 处理器 1 的应用层软件<br>0x11 : MCU 处理器 1 的标定软件 1<br>0x12 : MCU 处理器 1 的标定软件 2<br>....<br>0x20 : MCU 处理器 2 的应用层软件<br>0x21 : MCU 处理器 2 的标定软件 1<br>0x22 : MCU 处理器 2 的标定软件 2 | Not Supported     | Not Supported                |
| Positive Response Message Definition |                                                                                                                                                                                                                                                                                                                                                                               |                   |                              |
| Routine Info                         | Not Supported                                                                                                                                                                                                                                                                                                                                                                 | Not Supported     | Not Supported                |
| Routine Status Record                | The positive response to the RID request contains 1 byte as defined below:<br>0x00: signature is ok<br>0x01: signature is not ok                                                                                                                                                                                                                                              | Not Supported     | Not Supported                |
| Supported Negative Response(s)       |                                                                                                                                                                                                                                                                                                                                                                               |                   |                              |
|                                      |                                                                                                                                                                                                                                                                                                                                                                               |                   |                              |

|                                                                                   |                                                          |         |                 |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|-----------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |                 |
|                                                                                   | V23                                                      | Rev 3.3 | Page 124 of 146 |

### 8.3.2 RID 0x0203: Check Programming Precondition

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |                              |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Routine Identifier (RID)             | 0x0203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                              |
| RID name                             | Check Programming Precondition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                              |
| Description                          | <p>For the routine of precondition checking, no optional parameters are defined. The exact pre-conditions should be decided by each ECU specifically, suggest to check the below information.</p> <p>1). Ignition switch must be located at IgnitionOn, but not Ready.</p> <p>2). SystemBatteryVoltage must be in the valid range : 9~16 Voltage</p> <p>3). Vehicle speed must be 0kph</p> <p>4). Engine rotate speed must be 0 rpm</p> <p>5). ECU must ensure all of the inputs and outputs stay in the safe and secure inactive status.</p> <p>Note: This RID shall be supported by all programming ECUs.</p> |                   |                              |
| Enable Conditions                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |                              |
| Security Level                       | 01,03,05,09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   |                              |
| Availability                         | APP(Session 03); Bootloader(Session 01,02,03)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                              |
| Request Message Definition           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |                              |
| Sub-Functions Required               | 0x01 Start Routine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0x02 Stop Routine | 0x03 Request Routine Results |
| Routine Control Option Record        | Not Supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Not Supported     | Not Supported                |
| Positive Response Message Definition |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |                              |
| Routine Info                         | Not Supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Not Supported     | Not Supported                |
| Routine Status Record                | The positive response to the RID request contains 1 byte as defined below:<br>0x00: check successful<br>0x01: check failed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Not Supported     | Not Supported                |
| Supported Negative Response(s)       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |                              |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |                              |

### 8.3.3 RID 0xFF00: Erase Memory

|                                 |                                                                                                                                                                                                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Routine Identifier (RID)</b> | <b>0xFF00</b>                                                                                                                                                                                                                                                      |
| <b>RID name</b>                 | <b>Erase Memory</b>                                                                                                                                                                                                                                                |
| Description                     | <p>The purpose of the Erase Memory routine control is to erase a logical partition (e.g.,APP file partition, Cal file partition) in flash memory with one of the supported File Type number.</p> <p>Note: This RID shall be supported by all programming ECUs.</p> |
| Enable Conditions               | Programming session, must be in boot                                                                                                                                                                                                                               |
| Security Level                  | 05,09                                                                                                                                                                                                                                                              |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                              |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Availability                         | Bootloader(Session 01,02,03)                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                              |
| Request Message Definition           |                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                              |
| Sub-Functions Required               | 0x01 Start Routine                                                                                                                                                                                                                                                                                                                                                                                                                | 0x02 Stop Routine | 0x03 Request Routine Results |
| Routine Control Option Record        | <p>1 Byte = File Type number, Refer to “文件类型表” in &lt;QSMG-EEA-6002-Utility 需求规范&gt;</p> <p>Each logical partition has a programming file.</p> <p>For example File Type number (in hex) have the following meaning.</p> <p>0x10 : MCU 处理器 1 的应用层软件</p> <p>0x11 : MCU 处理器 1 的标定软件 1</p> <p>0x12 : MCU 处理器 1 的标定软件 2</p> <p>....</p> <p>0x20 : MCU 处理器 2 的应用层软件</p> <p>0x21 : MCU 处理器 2 的标定软件 1</p> <p>0x22 : MCU 处理器 2 的标定软件 2</p> | Not Supported     | Not Supported                |
| Positive Response Message Definition |                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                              |
| Routine Info                         | Not Supported                                                                                                                                                                                                                                                                                                                                                                                                                     | Not Supported     | Not Supported                |
| Routine Status Record                | <p>The positive response to the RID request contains 1 byte as defined below:</p> <p>0x00: Erase successful</p> <p>0x01: Erase failure</p>                                                                                                                                                                                                                                                                                        | Not Supported     | Not Supported                |
| Supported Negative Response(s)       |                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                              |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |                              |

### 8.3.4 RID 0xFF01: Check Programming Dependencies

|                            |                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routine Identifier (RID)   | 0xFF01                                                                                                                                                                                                                                                                                                                                                                                                    |
| RID name                   | Check Programming Dependencies                                                                                                                                                                                                                                                                                                                                                                            |
| Description                | <p>This DID has be used to check the ECU inter dependencies (between boot SW and APP SW. Cal SW and APP SW.). Dependencies between ECUs should be checked by diagnostic tools (OTA master or off-board tester). The function of the DID can be achieved through the Compatibility ID (refer to &lt;QSMG-EEA-6002-Utility 需求规范&gt;).</p> <p>Note: This RID shall be supported by all programming ECUs.</p> |
| Enable Conditions          | Programming session, must be in boot                                                                                                                                                                                                                                                                                                                                                                      |
| Security Level             | 05,09                                                                                                                                                                                                                                                                                                                                                                                                     |
| Availability               | Bootloader(Session 01,02,03)                                                                                                                                                                                                                                                                                                                                                                              |
| Request Message Definition |                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                                                   |                                                          |         |                 |
|-----------------------------------------------------------------------------------|----------------------------------------------------------|---------|-----------------|
|  | <b>SmartMi Global Electrical/Electronic Architecture</b> |         |                 |
|                                                                                   | V23                                                      | Rev 3.3 | Page 126 of 146 |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Sub-Functions Required               | 0x01 Start Routine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0x02 Stop Routine | 0x03 Request Routine Results |
| Routine Control Option Record        | Not Supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Not Supported     | Not Supported                |
| Positive Response Message Definition |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                              |
| Routine Info                         | Not Supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Not Supported     | Not Supported                |
| Routine Status Record                | <p>The positive response to the RID request contains 1 byte as defined below:</p> <p>0x00: fully programmed</p> <p>0x01: application software missing</p> <p>0x02: calibration data missing</p> <p>0x03: no start or default calibrations present (needs to be programmed)</p> <p>0x04: Boot SW is incompatible with APP SW</p> <p>0x05: Boot SW is incompatible with Cal SW</p> <p>0x06: APP SW is incompatible with Cal SW (0x05 has a higher priority than 0x06)</p> <p>0x50: internal memory error detected (e.g.RAM, Flash, EEPROM, etc.)</p> <p>All other values are reserved.</p> | Not Supported     | Not Supported                |
| Supported Negative Response(s)       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                              |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                              |

### 8.3.5 RID 0x0204: Disable Programming Prohibition Rollback function

|                                 |                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Routine Identifier (RID)</b> | <b>0x0204</b>                                                                                                                                                                                                                                                                                                                                       |
| <b>RID name</b>                 | <b>Disable Programming Prohibition Rollback function</b>                                                                                                                                                                                                                                                                                            |
| Description                     | <p>This DID has be used to disable the ECU programming Prohibition Rollback function.</p> <p>When the ECU reset, the Programming Prohibition Rollback function will be automatic enable. Reset triggers include, but are not limited to Diagnostic Commands, Power reset.....</p> <p>Note: This RID shall be supported by all programming ECUs.</p> |
| Enable Conditions               | Programming session, must be in boot                                                                                                                                                                                                                                                                                                                |
| Security Level                  | 05,09                                                                                                                                                                                                                                                                                                                                               |
| Availability                    | Bootloader(Session 01,02,03)                                                                                                                                                                                                                                                                                                                        |
| Request Message Definition      |                                                                                                                                                                                                                                                                                                                                                     |

|                                      |                                                                                                                                                                                                                                   |                   |                              |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Sub-Functions Required               | 0x01 Start Routine                                                                                                                                                                                                                | 0x02 Stop Routine | 0x03 Request Routine Results |
| Routine Control Option Record        | Not Supported                                                                                                                                                                                                                     | Not Supported     | Not Supported                |
| Positive Response Message Definition |                                                                                                                                                                                                                                   |                   |                              |
| Routine Info                         | Not Supported                                                                                                                                                                                                                     | Not Supported     | Not Supported                |
| Routine Status Record                | The positive response to the RID request contains 1 byte as defined below:<br>0x00: Enable Programming Prohibition Rollback function<br>0x01: Disable Programming Prohibition Rollback function<br>All other values are reserved. | Not Supported     | Not Supported                |
| Supported Negative Response(s)       |                                                                                                                                                                                                                                   |                   |                              |
|                                      |                                                                                                                                                                                                                                   |                   |                              |

### 8.3.6 **RID 0x0205: Upgrade Package SHA256 Check**

Refer to < QSMG-EEA-4012-文件系统刷写规范> for more detail.

### 8.3.7 **RID 0x0206: Check Programming Integrity-SHA256**

Refer to < QSMG-EEA-4012-文件系统刷写规范> for more detail.

### 8.3.8 **RID 0x0207: Upgrade Package Installation**

Refer to < QSMG-EEA-4012-文件系统刷写规范> for more detail.

### 8.3.9 **RID 0x0208: Program Rollback**

Refer to < QSMG-EEA-4012-文件系统刷写规范> for more detail.

### 8.3.10 **RID 0x0400: SecurityKey Generation**

|                                 |                               |
|---------------------------------|-------------------------------|
| <b>Routine Identifier (RID)</b> | <b>0x0400</b>                 |
| <b>RID name</b>                 | <b>SecurityKey Generation</b> |

|                                      |                                                                                                                                                                                                                                                                                                                        |                   |                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| Description                          | This routine is used to generate a set of SecurityKey for a ECU, which is used in 27 SecurityAcces service. For detail, please refer “QSMG-EEA-4022-SecurityAccess 需求规范”.<br><br>Note: This RID must be supported by all ECUs. Once ECU get the correct SecurityKey, this routine service will no longer be available. |                   |                              |
| Enable Conditions                    | ECUs which only have a initial SecurityKey                                                                                                                                                                                                                                                                             |                   |                              |
| Security Level                       | 01,03                                                                                                                                                                                                                                                                                                                  |                   |                              |
| Availability                         | APP(Session 03)                                                                                                                                                                                                                                                                                                        |                   |                              |
| Request Message Definition           |                                                                                                                                                                                                                                                                                                                        |                   |                              |
| Sub-Functions Required               | 0x01 Start Routine                                                                                                                                                                                                                                                                                                     | 0x02 Stop Routine | 0x03 Request Routine Results |
| Routine Control Option Record        | root_key: 16 byte Hex                                                                                                                                                                                                                                                                                                  | Not Supported     | Not Supported                |
| Positive Response Message Definition |                                                                                                                                                                                                                                                                                                                        |                   |                              |
| Routine Info                         | Not Supported                                                                                                                                                                                                                                                                                                          | Not Supported     | Not Supported                |
| Routine Status Record                | The positive response to the RID request contains 1 byte as defined below:<br>0x00: successful<br>0x01: failed                                                                                                                                                                                                         | Not Supported     | Not Supported                |
| Supported Negative Response(s)       |                                                                                                                                                                                                                                                                                                                        |                   |                              |
|                                      |                                                                                                                                                                                                                                                                                                                        |                   |                              |

### 8.3.11 0xA1A1: Test Mode Active

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

### 8.3.12 0xA1A2: Cable Diagnostic Active

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

### 8.3.13 0xA1A3: Reset Switch Counter

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

### 8.3.14 0xA1AF: Ethernet Message Mirror

Refer to < QSMG-EEA-3001-Ethernet需求规范> for more detail.

## 9 Appendix C-Failure Type

The DTC Failure Type consists of 16 different failure categories, where each category is associated with 16 Sub Type Failures (also known as symptoms). The Sub Type Failures are logically grouped in a DTC Failure Type Category. This was done to simplify the selection of the appropriate Sub Type Failure (Symptom) for a DTC.

The DTC Failure Category is decoded in the High Nibble of the DTC Failure Type Byte and the Failure Sub Type is decoded in the Low Nibble of the DTC Failure Type Byte.

Table : DTC Failure Type Byte Definition

| DTC Failure Type Byte Definition |                          |
|----------------------------------|--------------------------|
| High Nibble (\$0x - \$Fx)        | Low Nibble (\$x0 - \$xF) |
| DTC Failure Category             | DTC Failure Sub Type     |

Table : DTC Failure Category Definition

| DTC Failure Type Byte Category Definitions |            |                                                                   |
|--------------------------------------------|------------|-------------------------------------------------------------------|
| High Nibble (0000b - 1111b)                | Category # | Category Description                                              |
| 0000                                       | 0          | General Failure Information                                       |
| 0001                                       | 1          | General Electrical Failures                                       |
| 0010                                       | 2          | General Signal Failures                                           |
| 0011                                       | 3          | FM (Frequency Modulation) / PWM (Pulse Width Modulation) Failures |
| 0100                                       | 4          | System Internal Failures                                          |
| 0101                                       | 5          | System Programming Failures                                       |
| 0110                                       | 6          | Algorithm Based Failures                                          |
| 0111                                       | 7          | Mechanical Failures                                               |
| 1000                                       | 8          | Bus Signal Failures                                               |
| 1001                                       | 9          | Component Failures                                                |
| 1010 – 1110                                | A - E      | Reserved By Document                                              |
| 1111                                       | F          | Vehicle Manufacturer / System Supplier Specific                   |

### 9.1 Definition and Appropriate Uses Of DTC Failure Type Value \$00

The DTC Failure Type value zero (\$00) is used to indicate that no additional information is available beyond that provided by the description for the DTC two (2) byte base code. The value of zero (\$00) can also be used for other types of DTC's as long as no other fault type value exists that would provide additional information. In cases where legislative requirements do not allow the reporting of DTC's with a fault type byte (e.g., emissions related DTC's), the ECU may choose to report its DTC's via enhanced diagnostic services with the fault type byte coded to zero (\$00). This approach allows the same DTC tables to be used for reporting DTC's with both legislated and enhanced diagnostic services.

## 9.2 DTC Failure Type Definition by Category

The DTC Failure Type Byte Definitions in the tables below shall be used to provide additional fault information for all DTC's which don't meet one of the exceptions outlined above.

### 9.2.1 DTC Failure Sub Type (Symptom) Definition of General Failure Information

| Failure Type | Sub Type        | General Failure Information                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                        |
|--------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | This range includes all other categories and is used when the fault within that failure category is unique (not amenable to standardization through assignment of a new Sub Type) or when the detected fault is best described by two or more Sub Types within that Failure Category. |                                                                                                                                                                                                                                                                        |
| Byte (hex)   | Nibble (binary) | Sub Type                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                        |
|              |                 | Description                                                                                                                                                                                                                                                                           | Definition                                                                                                                                                                                                                                                             |
| 00           | 0000            | No Sub Type Information                                                                                                                                                                                                                                                               | This sub type is used for failures where the base DTC text string provides the complete description of the failure itself (no Category and no Sub Type information used, e.g. emissions-related DTC (012700 hex): P0127 Intake Air Temperature Too High).              |
| 01           | 0001            | General Electrical Failure                                                                                                                                                                                                                                                            | This sub type is used for General Electrical Failures that cannot be assigned to a specific sub type (Category information and no Sub Type information, e.g. DTC (803901): B0039-01 Second Row Right Frontal Stage 1 Deployment Control - General Electrical Failure). |
| 02           | 0010            | General Signal Failure                                                                                                                                                                                                                                                                | This sub type is used for General Signal Failures that cannot be assigned to a specific sub type (Category information and no Sub Type information, e.g. DTC (403002): C0030 Left Front Tone Wheel - General Signal Failure).                                          |
| 03           | 0011            | FM (Frequency Modulated) / PWM (Pulse Width Modulated) Failure                                                                                                                                                                                                                        | This sub type is used for FM / PWM Failures that cannot be assigned to a specific sub type.                                                                                                                                                                            |
| 04           | 0100            | System Internal Failure                                                                                                                                                                                                                                                               | This sub type is used for control module Internal Failures that cannot be assigned to a specific sub type.                                                                                                                                                             |
| 05           | 0101            | System Programming Failure                                                                                                                                                                                                                                                            | This sub type is used for System Programming Failures that cannot be assigned to a specific sub type.                                                                                                                                                                  |
| 06           | 0110            | Algorithm Based Failure                                                                                                                                                                                                                                                               | This sub type is used for Algorithm Based Failures that cannot be assigned to a specific sub type.                                                                                                                                                                     |
| 07           | 0111            | Mechanical Failure                                                                                                                                                                                                                                                                    | This sub type is used for Mechanical Failures that cannot be assigned to a specific sub type.                                                                                                                                                                          |
| 08           | 1000            | Bus Signal / Message Failure                                                                                                                                                                                                                                                          | This sub type is used for Bus Signal / Message Failures that cannot be assigned to a specific sub type.                                                                                                                                                                |
| 09           | 1001            | Component Failure                                                                                                                                                                                                                                                                     | This sub type is used for Component Failures that cannot be assigned to a specific sub type.                                                                                                                                                                           |
| 0A           | 1010            | General Electrical Failures - 2                                                                                                                                                                                                                                                       | This range specifies the standard wiring failure modes (i.e., shorts and opens), and direct current (DC) quantities related by Ohm's Law.                                                                                                                              |
| 0B           | 1011            | ISO/SAE Reserved                                                                                                                                                                                                                                                                      | This value is reserved by the document for future expansion.                                                                                                                                                                                                           |
| 0C           | 1100            | ISO/SAE Reserved                                                                                                                                                                                                                                                                      | This value is reserved by the document for future expansion.                                                                                                                                                                                                           |
| 0D           | 1101            | ISO/SAE Reserved                                                                                                                                                                                                                                                                      | This value is reserved by the document for future expansion.                                                                                                                                                                                                           |
| 0E           | 1110            | ISO/SAE Reserved                                                                                                                                                                                                                                                                      | This value is reserved by the document for future expansion.                                                                                                                                                                                                           |
| 0F           | 1111            | ISO/SAE Reserved                                                                                                                                                                                                                                                                      | This value is reserved by the document for future expansion.                                                                                                                                                                                                           |

## 9.2.2 DTC Failure Sub Type (Symptom) Definition of General Electrical Failures

| Failure Type | Sub Type        | General Electrical Failures                                                                                                               |                                                                                                                                                                                                                                                  |
|--------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | This range specifies the standard wiring failure modes (i.e., shorts and opens), and direct current (DC) quantities related by Ohm's Law. |                                                                                                                                                                                                                                                  |
| Byte (hex)   | Nibble (binary) | Description                                                                                                                               | Definition                                                                                                                                                                                                                                       |
| 10           | 0000            | ISO/SAE Reserved                                                                                                                          | This value is reserved by the document for future expansion.                                                                                                                                                                                     |
| 11           | 0001            | Circuit Short To Ground                                                                                                                   | This sub type is used for failures where the control module measures ground (battery negative) potential for greater than a specified time period or when some other value less than the low voltage limit DTC Sub Type is measured.             |
| 12           | 0010            | Circuit Short To Battery                                                                                                                  | This sub type is used for failures, where the control module measures vehicle system (battery positive) potential for greater than a specified time period or when some other value higher than the high voltage limit DTC Sub Type is measured. |
| 13           | 0011            | Circuit Open                                                                                                                              | This sub type is used for failures, where the control module determines an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output, etc.                                            |
| 14           | 0100            | Circuit Short To Ground or Open                                                                                                           | This sub type is used for failures, where the condition detected by the control module is the same for either indicated failure mode.                                                                                                            |
| 15           | 0101            | Circuit Short To Battery or Open                                                                                                          | This sub type is used for failures, where the condition detected by the control module is the same for either indicated failure mode.                                                                                                            |
| 16           | 0110            | Circuit Voltage Below Threshold                                                                                                           | This sub type is used for failures, where the control module measures a voltage below a specified range.                                                                                                                                         |
| 17           | 0111            | Circuit Voltage Above Threshold                                                                                                           | This sub type is used for failures where, the control module measures a voltage above a specified range.                                                                                                                                         |
| 18           | 1000            | Circuit Current Below Threshold                                                                                                           | This sub type is used for failures, where the control module measures current flow below a specified range.                                                                                                                                      |
| 19           | 1001            | Circuit Current Above Threshold                                                                                                           | This sub type is used for failures, where the control module measures current flow above a specified range.                                                                                                                                      |
| 1A           | 1010            | Circuit Resistance Below Threshold                                                                                                        | This sub type is used for failures, where the control module infers a circuit resistance below a specified range.                                                                                                                                |
| 1B           | 1011            | Circuit Resistance Above Threshold                                                                                                        | This sub type is used for failures, where the control module infers a circuit resistance above a specified range.                                                                                                                                |
| 1C           | 1100            | Circuit Voltage Out of Range                                                                                                              | This sub type is used for failures, where the control module measures a voltage outside the expected range but not identified as too high or too low.                                                                                            |
| 1D           | 1101            | Circuit Current Out of Range                                                                                                              | This sub type is used for failures, where the control module measures a current outside the expected range but not identified as too high or too low.                                                                                            |
| 1E           | 1110            | Circuit Resistance Out of Range                                                                                                           | This sub type is used for failures, where the control module measures a resistance outside the expected range but not identified as too high or too low.                                                                                         |
| 1F           | 1111            | Circuit Intermittent                                                                                                                      | This sub type is used for failures, where the control module momentarily detects one of the conditions defined above, but not long enough to set a specific sub type.                                                                            |

### 9.2.3 DTC Failure Sub Type (Symptom) Definition of General Signal Failures

| Failure Type Byte (hex) | Sub Type Nibble (binary) | General Signal Failures                                                                            |                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|--------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                          | This range specifies quantities related to amplitude, frequency or rate of change, and wave shape. |                                                                                                                                                                                                                                                                                                                                          |
|                         |                          | Description                                                                                        | Definition                                                                                                                                                                                                                                                                                                                               |
| 20                      | 0000                     | ISO/SAE Reserved                                                                                   | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                             |
| 21                      | 0001                     | Signal Amplitude < Minimum                                                                         | This sub type is used for failures where the control module measures a signal voltage amplitude below a specified range (e.g., low gain).                                                                                                                                                                                                |
| 22                      | 0010                     | Signal Amplitude > Maximum                                                                         | This sub type is used for failures where the control module measures a signal voltage amplitude above a specified range (e.g., gain too high).                                                                                                                                                                                           |
| 23                      | 0011                     | Signal Stuck Low                                                                                   | This sub type is used for failures where the control module measures a signal that remains low when transitions are expected.                                                                                                                                                                                                            |
| 24                      | 0100                     | Signal Stuck High                                                                                  | This sub type is used for failures where the control module measures a signal that remains high when transitions are expected.                                                                                                                                                                                                           |
| 25                      | 0101                     | Signal Shape / Waveform Failure                                                                    | This sub type is used for failures where the shape of the signal (plot of the amplitude with respect to time) is not correct, e.g., improper circuit impedance.                                                                                                                                                                          |
| 26                      | 0110                     | Signal Rate of Change Below Threshold                                                              | This sub type is used for failures where the signal transitions more slowly than the specified limit.                                                                                                                                                                                                                                    |
| 27                      | 0111                     | Signal Rate of Change Above Threshold                                                              | This sub type is used for failures where the signal transitions more quickly than the specified limit.                                                                                                                                                                                                                                   |
| 28                      | 1000                     | Signal Bias Level Out of Range / Zero Adjustment Failure                                           | This sub type is used for failures where the control module applies a bias voltage or a zero signal level to a circuit upon which is superimposed a signal voltage (e.g., bias voltage to an Oxygen Sensor circuit, or a filtered digital m/sec <sup>2</sup> signal while vehicle stands still for a lateral accelerator sensor module.) |
| 29                      | 1001                     | Signal Invalid                                                                                     | This sub type is used for failures where the value of the signal is not plausible given the operating conditions.                                                                                                                                                                                                                        |
| 2A                      | 1010                     | Signal Stuck In Range                                                                              | This sub type is used for failures where the value of the signal is in the normal operating range, but not correct for current operating conditions.                                                                                                                                                                                     |
| 2B                      | 1011                     | Signal Cross Coupled                                                                               | This sub type is used when a signal is found to be incorrectly correlated to another signal that the control module is also monitoring, indicating that the signals are shorted together                                                                                                                                                 |
| 2C                      | 1100                     | ISO/SAE Reserved                                                                                   | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                             |
| 2D                      | 1101                     | ISO/SAE Reserved                                                                                   | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                             |
| 2E                      | 1110                     | ISO/SAE Reserved                                                                                   | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                             |
| 2F                      | 1111                     | Signal Erratic                                                                                     | This sub type is used for failures where the signal is momentarily implausible (not long enough for "signal invalid") or discontinuous.                                                                                                                                                                                                  |

## 9.2.4 DTC Failure Sub Type (Symptom) Definition of FM (Frequency Modulation) / PWM (Pulse Width Modulation) Failures

| Failure Type | Sub Type        | FM (Frequency Modulation) / PWM (Pulse Width Modulation) Failures                                                                                                                                                    |                                                                                                                                                                                                         |
|--------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | This range specifies faults related to Frequency Modulated (FM) and Pulse Width Modulated (PWM) inputs and outputs of the control module. This category also includes faults where position is determined by counts. |                                                                                                                                                                                                         |
| Byte (hex)   | Nibble (binary) | Sub Type                                                                                                                                                                                                             |                                                                                                                                                                                                         |
|              |                 | Description                                                                                                                                                                                                          | Definition                                                                                                                                                                                              |
| 30           | 0000            | ISO/SAE Reserved                                                                                                                                                                                                     | This value is reserved by the document for future expansion.                                                                                                                                            |
| 31           | 0001            | No Signal                                                                                                                                                                                                            | This sub type is used for failures where the control module does not detect a signal which ought to be present (e.g., wheel speed signals present for three of the four wheels and brakes not applied.) |
| 32           | 0010            | Signal Low Time < Minimum                                                                                                                                                                                            | This sub type is used for failures where the control module detects the low pulse is too narrow with respect to time.                                                                                   |
| 33           | 0011            | Signal Low Time > Maximum                                                                                                                                                                                            | This sub type is used for failures where the control module detects the low pulse is too wide with respect to time.                                                                                     |
| 34           | 0100            | Signal High Time < Minimum                                                                                                                                                                                           | This sub type is used for failures where the control module detects the high pulse is too narrow with respect to time.                                                                                  |
| 35           | 0101            | Signal High Time > Maximum                                                                                                                                                                                           | This sub type is used for failures where the control module detects the high pulse is too wide with respect to time.                                                                                    |
| 36           | 0110            | Signal Frequency Too Low                                                                                                                                                                                             | This sub type is used for failures where the control module detects excessive duration for one cycle of the output across a specified sample size.                                                      |
| 37           | 0111            | Signal Frequency Too High                                                                                                                                                                                            | This sub type is used for failures where the control module detects insufficient duration for one cycle of the output across a specified sample size.                                                   |
| 38           | 1000            | Signal Frequency Incorrect                                                                                                                                                                                           | This sub type is used for failures where the control module measures an incorrect number of cycles in a given time period.                                                                              |
| 39           | 1001            | Signal Has Too Few Pulses                                                                                                                                                                                            | This sub type is used for failures where the control module measures too few pulses (e.g., position is calibrated in counts from one extreme to the other).                                             |
| 3A           | 1010            | Signal Has Too Many Pulses                                                                                                                                                                                           | This sub type is used for failures where the control module measures too many pulses (e.g., position is calibrated in counts from one extreme to the other).                                            |
| 3B           | 1011            | ISO/SAE Reserved                                                                                                                                                                                                     | This value is reserved by the document for future expansion.                                                                                                                                            |
| 3C           | 1100            | ISO/SAE Reserved                                                                                                                                                                                                     | This value is reserved by the document for future expansion.                                                                                                                                            |
| 3D           | 1101            | ISO/SAE Reserved                                                                                                                                                                                                     | This value is reserved by the document for future expansion.                                                                                                                                            |
| 3E           | 1110            | ISO/SAE Reserved                                                                                                                                                                                                     | This value is reserved by the document for future expansion.                                                                                                                                            |
| 3F           | 1111            | ISO/SAE Reserved                                                                                                                                                                                                     | This value is reserved by the document for future expansion.                                                                                                                                            |

### 9.2.5 DTC Failure Sub Type (Symptom) Definition of System Internal Failures

| Failure Type | Sub Type        | System Internal Failures                                                                                                                                    |                                                                                                                                                                                                            |
|--------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | This range specifies faults related to memory, software, and internal electrical circuitry; requiring component (control module, sensor, etc.) replacement. |                                                                                                                                                                                                            |
| Byte (hex)   | Nibble (binary) | Sub Type                                                                                                                                                    |                                                                                                                                                                                                            |
|              |                 | Description                                                                                                                                                 | Definition                                                                                                                                                                                                 |
| 40           | 0000            | ISO/SAE Reserved                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                               |
| 41           | 0001            | General Checksum Failure                                                                                                                                    | This sub type is used by the control module to indicate an incorrect checksum calculation where memory type is not specified.                                                                              |
| 42           | 0010            | General Memory Failure                                                                                                                                      | This sub type is used by the control module to indicate a memory failure where memory type is not specified.                                                                                               |
| 43           | 0011            | Special Memory Failure                                                                                                                                      | This sub type is used by the control module to indicate a memory failure where the specific memory type is not defined in this category.                                                                   |
| 44           | 0100            | Data Memory Failure                                                                                                                                         | This sub type is used by the control module to indicate a data (or working) memory failure for embedded systems using FLASH memory. This is equivalent to RAM in RAM/ROM/EEPROM embedded systems.          |
| 45           | 0101            | Program Memory Failure                                                                                                                                      | This sub type is used by the control module to indicate a program memory failure for embedded systems using FLASH memory. This is equivalent to ROM in RAM/ROM/EEPROM embedded systems.                    |
| 46           | 0110            | Calibration / Parameter Memory Failure                                                                                                                      | This sub type is used by the control module to indicate a calibration / parameter memory failure for embedded systems using FLASH memory. This is equivalent to EEPROM in RAM/ROM/EEPROM embedded systems. |
| 47           | 0111            | Watchdog / Safety $\mu$ C Failure                                                                                                                           | This sub type is used by the control module to indicate a watchdog / safety $\mu$ C failure, or a failure in the execution of operational software.                                                        |
| 48           | 1000            | Supervision Software Failure                                                                                                                                | This sub type is used by the control module to indicate a supervision software failure.                                                                                                                    |
| 49           | 1001            | Internal Electronic Failure                                                                                                                                 | This sub type is used by the control module to indicate the detection of an internal circuit failure.                                                                                                      |
| 4A           | 1010            | Incorrect Component Installed                                                                                                                               | This sub type is used by the control module to indicate a mismatch between the hardware connected to the control module and the hardware expected by the control module.                                   |
| 4B           | 1011            | Over Temperature                                                                                                                                            | This sub type is used by the control module to indicate the detection of an internal temperature above the expected range.                                                                                 |
| 4C           | 1100            | ISO/SAE Reserved                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                               |
| 4D           | 1101            | ISO/SAE Reserved                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                               |
| 4E           | 1110            | ISO/SAE Reserved                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                               |
| 4F           | 1111            | ISO/SAE Reserved                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                               |

## 9.2.6 DTC Failure Sub Type (Symptom) Definition of System Programming Failures

| Failure Type | Sub Type        | System Programming Failures                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | This range specifies faults related to operational software, calibrations, and options; remedied by configuring/programming a part of the system (control module, sensor, etc.) |                                                                                                                                                                                                                                                                                                                                                                                 |
| Byte (hex)   | Nibble (binary) | Sub Type                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                 |
|              |                 | Description                                                                                                                                                                     | Definition                                                                                                                                                                                                                                                                                                                                                                      |
| 50           | 0000            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |
| 51           | 0001            | Not Programmed                                                                                                                                                                  | This sub type is used by the control module to indicate that programming is required.                                                                                                                                                                                                                                                                                           |
| 52           | 0010            | Not Activated                                                                                                                                                                   | This sub type is used by the control module to indicate that some portion of the program has not been enabled.                                                                                                                                                                                                                                                                  |
| 53           | 0011            | Deactivated                                                                                                                                                                     | This sub type is used by the control module to indicate that that some portion of the program has been disabled.                                                                                                                                                                                                                                                                |
| 54           | 0100            | Missing Calibration                                                                                                                                                             | This sub type is used by the control module to indicate that an operational range, etc., for a sensor or actuator must be taught to the control module, e.g. by programming or learning.                                                                                                                                                                                        |
| 55           | 0101            | Not Configured                                                                                                                                                                  | This sub type is used by the control module to indicate the need to enter (program) the sub system option content or the vehicle option content.                                                                                                                                                                                                                                |
| 56           | 0110            | Invalid / Incompatible Configuration                                                                                                                                            | This sub type indicates a control module or system configuration that cannot be valid, e.g. to have mutually exclusive options set on at the same time, or a set up that is not supported by the currently installed hardware/software.                                                                                                                                         |
| 57           | 0111            | Invalid / Incompatible Software Component                                                                                                                                       | This sub type is used by the control module to indicate that a software component (calibration or program) has been identified as invalid for the control module or incompatible with other hardware or software identified by the control module, e.g. a downloaded calibration software component is incompatible with a permanent or downloaded strategy software component. |
| 58           | 1000            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |
| 59           | 1001            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |
| 5A           | 1010            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |
| 5B           | 1011            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |
| 5C           | 1100            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |
| 5D           | 1101            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |
| 5E           | 1110            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |
| 5F           | 1111            | ISO/SAE Reserved                                                                                                                                                                | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                                                                                    |

### 9.2.7 DTC Failure Sub Type (Symptom) Definition of Algorithm Based Failures

| Failure Type | Sub Type        | Algorithm Based Failures                                                                                                                                                                                                    |                                                                                                                                                                                                                                                        |
|--------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | This range specifies faults based on comparing two or more input parameters for plausibility, comparing a single parameter to itself with respect to time, or inhibits operation due to a reported failure of that circuit. |                                                                                                                                                                                                                                                        |
| Byte (hex)   | Nibble (binary) | Description                                                                                                                                                                                                                 | Sub Type Definition                                                                                                                                                                                                                                    |
| 60           | 0000            | ISO/SAE Reserved                                                                                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                           |
| 61           | 0001            | Signal Calculation Failure                                                                                                                                                                                                  | This sub type is used for algorithm based calculation failures where the calculated value is outside the expected range.                                                                                                                               |
| 62           | 0010            | Signal Compare Failure                                                                                                                                                                                                      | This sub type is used for failures where the control module compares two or more input parameters for plausibility but cannot specifically identify the component that is faulted.                                                                     |
| 63           | 0011            | Circuit / Component Protection Time-Out                                                                                                                                                                                     | This sub type is used for failures where the control module detects a function is active for greater than a specified time period.                                                                                                                     |
| 64           | 0100            | Signal Plausibility Failure                                                                                                                                                                                                 | This sub type is used for failures where the control module detects a single input parameter is operating outside the plausible range.                                                                                                                 |
| 65           | 0101            | Signal Has Too Few Transitions / Events                                                                                                                                                                                     | This sub type is used for failures where the control module monitors a parameter over time within specified limits and detects fewer than the expected number of transitions.                                                                          |
| 66           | 0110            | Signal Has Too Many Transitions / Events                                                                                                                                                                                    | This sub type is used for failures where the control module monitors a parameter over time within specified limits and detects more than the expected number of transitions.                                                                           |
| 67           | 0111            | Signal Incorrect After Event                                                                                                                                                                                                | This sub type is used for failures where the control module does not see the correct change of a parameter or group of parameters in response to a particular event.                                                                                   |
| 68           | 1000            | Event Information                                                                                                                                                                                                           | This sub type is used by the control module to indicate the detection of a system event that was not caused by the control module itself but forces the control module to store a DTC (e.g. missing functionality from another system/control module). |
| 69           | 1001            | ISO/SAE Reserved                                                                                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                           |
| 6A           | 1010            | ISO/SAE Reserved                                                                                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                           |
| 6B           | 1011            | ISO/SAE Reserved                                                                                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                           |
| 6C           | 1100            | ISO/SAE Reserved                                                                                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                           |
| 6D           | 1101            | ISO/SAE Reserved                                                                                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                           |
| 6E           | 1110            | ISO/SAE Reserved                                                                                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                           |
| 6F           | 1111            | ISO/SAE Reserved                                                                                                                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                           |

### 9.2.8 DTC Failure Sub Type (Symptom) Definition of Mechanical Failures

| Failure Type | Sub Type        | Mechanical Failures                                                                                                         |                                                                                                                                                                                                                                                |
|--------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | This range specifies faults detected by inappropriate motion in response to control module related input/controlled output. |                                                                                                                                                                                                                                                |
| Byte (hex)   | Nibble (binary) | Description                                                                                                                 | Definition                                                                                                                                                                                                                                     |
| 70           | 0000            | ISO/SAE Reserved                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                   |
| 71           | 0001            | Actuator Stuck                                                                                                              | This sub type is used for failures where the control module does not detect any mechanical motion in response to energizing a motor, solenoid, relay, etc.                                                                                     |
| 72           | 0010            | Actuator Stuck Open                                                                                                         | This sub type is used for failures where the control module does not detect any mechanical motion upon commanding the operation of a motor, solenoid, relay, etc., to close some piece of equipment.                                           |
| 73           | 0011            | Actuator Stuck Closed                                                                                                       | This sub type is used for failures where the control module does not detect any mechanical motion upon commanding the operation of a motor, solenoid, relay, etc., to open some piece of equipment.                                            |
| 74           | 0100            | Actuator Slipping                                                                                                           | This sub type is used for failures where the control module detects excessive duration to command a motor, solenoid, relay, etc., to move a piece of equipment to a desired position.                                                          |
| 75           | 0101            | Emergency Position Not Reachable                                                                                            | This sub type is used for failures where the control module is unable to command a motor, solenoid, relay, etc., to move a piece of equipment to the emergency position.                                                                       |
| 76           | 0110            | Wrong Mounting Position                                                                                                     | This sub type is used for failures where the control module detects incorrectly mounted components, e.g., acceleration sensor showing a position error of 90°.                                                                                 |
| 77           | 0111            | Commanded Position Not Reachable                                                                                            | This sub type is used for failures where the control module is unable to command a motor, solenoid, relay, etc., to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment. |
| 78           | 1000            | Alignment or Adjustment Incorrect                                                                                           | This sub type is used for failures where the control module detects incorrectly adjusted or aligned components.                                                                                                                                |
| 79           | 1001            | Mechanical Linkage Failure                                                                                                  | This sub type is used for failures where the control module detects that the actuator is operational but the driven device is not operating, e.g., drive cable for power sliding door broken.                                                  |
| 7A           | 1010            | Fluid Leak or Seal Failure                                                                                                  | This sub type is used for failures where the control module detects that a mechanical component has an unexpected gas or liquid flow in, out, or through the component.                                                                        |
| 7B           | 1011            | Low Fluid Level                                                                                                             | This sub type is used for failures where the control module detects that a fluid level is too low for proper operation of the system.                                                                                                          |
| 7C           | 1100            | Slow Response                                                                                                               | This sub type is used where control module has detected a response that is longer than expected or a degraded rate of change in the monitored component or system.                                                                             |
| 7D           | 1101            | ISO/SAE Reserved                                                                                                            | This value is reserved by the document for future expansion.                                                                                                                                                                                   |
| 7E           | 1110            | Actuator Stuck On                                                                                                           | This sub type is used for failures where the control module detects the actuator mechanical function is fixed in the ON / applied state                                                                                                        |
| 7F           | 1111            | Actuator Stuck Off                                                                                                          | This sub type is used for failures where the control module detects the actuator mechanical function is fixed in the OFF / Released state                                                                                                      |

### 9.2.9 DTC Failure Sub Type (Symptom) Definition of Bus Signal Failures

| Failure Type | Sub Type        | Bus Signal Failures                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                              |
|--------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | This range specifies faults related to bus hardware and signal integrity. This category is also used when the physical input for a signal is located in one control module and another control module diagnoses the circuit. |                                                                                                                                                                                                                                                                                                              |
| Byte (hex)   | Nibble (binary) | Description                                                                                                                                                                                                                  | Definition                                                                                                                                                                                                                                                                                                   |
| 80           | 0000            | ISO/SAE Reserved                                                                                                                                                                                                             | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                 |
| 81           | 0001            | Invalid Serial Data Received                                                                                                                                                                                                 | This sub type is used by the control module to indicate a signal was received with the corresponding validity bit equal to "invalid" or post processing of the signal determines it is invalid.                                                                                                              |
| 82           | 0010            | Alive / Sequence Counter Incorrect / Not Updated                                                                                                                                                                             | This sub type is used by the control module to indicate that a signal was received without the corresponding rolling count value being properly updated.                                                                                                                                                     |
| 83           | 0011            | Value of Signal Protection Calculation Incorrect                                                                                                                                                                             | This sub type is used by the control module to indicate, that a message was processed with an incorrect protection (checksum) calculation.                                                                                                                                                                   |
| 84           | 0100            | Signal Below Allowable Range                                                                                                                                                                                                 | This sub type is used for failures where some circuit quantity, reported via serial data, is below a specified range.                                                                                                                                                                                        |
| 85           | 0101            | Signal Above Allowable Range                                                                                                                                                                                                 | This sub type is used for failures where some circuit quantity, reported via serial data, is above a specified range.                                                                                                                                                                                        |
| 86           | 0110            | Signal Invalid                                                                                                                                                                                                               | This sub type is used for failures where some circuit quantity, reported via serial data, is not plausible given the operating conditions.                                                                                                                                                                   |
| 87           | 0111            | Missing Message                                                                                                                                                                                                              | This sub type is used for failures where one (or more) expected message(s) is not received, e.g., periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control unit communicating with ABS). |
| 88           | 1000            | Bus off                                                                                                                                                                                                                      | This sub type is used for failures where a data bus is not available.                                                                                                                                                                                                                                        |
| 89           | 1001            | ISO/SAE Reserved                                                                                                                                                                                                             | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                 |
| 8A           | 1010            | ISO/SAE Reserved                                                                                                                                                                                                             | This value is reserved by the document for future expansion.                                                                                                                                                                                                                                                 |

## 10 Appendix D-Corporate Standard DTC

### 10.1 ECU Communication CAN Bus Off

Each ECU, connected to a CAN bus shall support the Bus off DTC.

|                                      |                                     |
|--------------------------------------|-------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b> | 0xC07X<br>(X is the CAN bus number) |
| <b>FTB</b>                           | 0x00: No Sub Type Information       |
| <b>Failure Lamp Flag</b>             | Depends on the ECU.                 |

|                                            |                                     |                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Failure Level</b>                       |                                     | 0x80                                                                                                                                                                                                                                                                                                                                             |
| <b>DTC name&amp;Description</b>            |                                     | XX ECU CAN channel X bus off                                                                                                                                                                                                                                                                                                                     |
| <b>Enable Criteria</b>                     | <b>Operation Cycle</b>              | DEM_OPCYC_IGNITION                                                                                                                                                                                                                                                                                                                               |
|                                            | <b>Vehicle Power Mode(s)</b>        | On for 5 seconds.                                                                                                                                                                                                                                                                                                                                |
|                                            | <b>Vehicle Supply Voltage</b>       | 9.0 V ≤ Voltage ≤16.0 V                                                                                                                                                                                                                                                                                                                          |
|                                            | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                                                                                                                                                                                                                  |
|                                            | <b>ECU Operating Conditions</b>     | ECU communication active state = True.                                                                                                                                                                                                                                                                                                           |
|                                            | <b>Others</b>                       | None                                                                                                                                                                                                                                                                                                                                             |
| <b>Exceptions</b>                          |                                     | 1. C07X_00_ENABLE = disabled<br>2. Within the first 5 seconds of a recovery from an under or over voltage condition.                                                                                                                                                                                                                             |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | Refer to “<QSMG-EEA-1001-CAN(FD)需求规范> Bus OFF Handling”.                                                                                                                                                                                                                                                                                         |
| <b>DTC Failure criteria</b>                |                                     | Refer to “<QSMG-EEA-1001-CAN(FD)需求规范> Bus OFF Handling”.                                                                                                                                                                                                                                                                                         |
| <b>Test Pass criteria</b>                  |                                     | Refer to “<QSMG-EEA-1001-CAN(FD)需求规范> Bus OFF Handling”.                                                                                                                                                                                                                                                                                         |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                              |
| <b>Fail soft Actions</b>                   |                                     | The ECU suspends all message transmissions on the network where the fault was detected. The ECU will automatically attempt to recover from the bus off condition. The setting of all other network supervision DTCs and bus performance DTCs on the indicated network are inhibited by U007X 00.                                                 |
| <b>Diagnostic investigation method</b>     |                                     | Check the ECU and the wiring.                                                                                                                                                                                                                                                                                                                    |
| <b>After-sales maintenance Suggestions</b> |                                     | This condition is not necessarily caused by the ECU that has reported the DTC. Possible causes for this DTC include wiring faults (e.g., a short between the two communication lines), improper impedance of the network (e.g., terminating resistor disconnected, too many ECUs on the network), or improper operation of the ECU CAN hardware. |

## 10.2 Lost Communication with CAN ECU

Each ECU, connected to a CAN bus shall support the lost communication DTC.

|                                         |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b>    |                                     | 0xC100 – C1FF<br>(C1XX; XX is the ECU address)                                                                                                                                                                                                                                                                                                                                                                               |
| <b>FTB</b>                              |                                     | 0x00: No Sub Type Information                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Failure Lamp Flag</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Failure Level</b>                    |                                     | 0x80                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>DTC name&amp;Description</b>         |                                     | Lost communication with XX ECU (e.g., C110 00 is lost communication with VCCM)                                                                                                                                                                                                                                                                                                                                               |
| <b>Enable Criteria</b>                  | <b>Operation Cycle</b>              | DEM_OPCYC_IGNITION                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                         | <b>Vehicle Power Mode(s)</b>        | On for 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                         | <b>Vehicle Supply Voltage</b>       | 9.0 V ≤ Voltage ≤16.0 V                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                         | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                                                                                                                                                                                                                                                                                              |
|                                         | <b>ECU Operating Conditions</b>     | ECU communication active state = True.                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | <b>Others</b>                       | None                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Exceptions</b>                       |                                     | 1. U0XXX_00_ENABLE = disabled; however, fail-soft actions shall still be taken if failure conditions are met.<br>2. When a bus off condition (C07X) is For DTCs, these Lost Communications DTCs shall not set but the fail-soft action shall occur if conditions to set the DTC are met.<br>3. Within the first 5 seconds of a recovery from an under or over voltage condition.<br>4. Vehicle is in logistic mode. if used. |
| <b>Test Frequency/ Monitoring Cycle</b> |                                     | Monitored continuously while frames are being transmitted and received. Upon notification of supervision timeout. Refer to “<QSMG-EEA-1001-CAN(FD)需求规范> Node Monitoring”.                                                                                                                                                                                                                                                    |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Failure criteria</b>                | Upon notification by the handler that associated supervised signal has failed supervision. A calibrate able debounce time is permitted at the time that notification of failed supervision has been received by the application to delay storing of DTC. Refer to “<QSMG-EEA-1001-CAN(FD)需求规范> Node Monitoring”.                                                                                                                          |
| <b>Test Pass criteria</b>                  | Reception of the previously failed supervised signal shall clear the associated signal supervision failure indication. A calibrate able debounce time may be used. Refer to “<QSMG-EEA-1001-CAN(FD)需求规范> Node Monitoring”.                                                                                                                                                                                                                |
| <b>Snapshot Record Data</b>                | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Fail soft Actions</b>                   | The ECU shall take fail-soft action to ensure safe operation of the system. Typically, this involves simple parameter substitution for failed signals though other methods may be used. All signals for all module devices associated with failed module shall be considered failed. No action need be taken until activation of a Network in which the module device associated with failed module participates.                         |
| <b>Diagnostic investigation method</b>     | Check the ECU and the wiring.                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>After-sales maintenance Suggestions</b> | This is not true when there is an intermittent connection of the data line(s) to the controller that reported the DTCs. When a single ECU reports multiple Lost Communication DTCs, but other ECUs connected to that network do not, diagnosis should be directed toward the reporting ECU. When multiple ECUs report a single Lost Communication DTC against the same ECU, diagnosis should be directed toward the ECU named by the DTC. |

### 10.3 Invalid Data Received From CAN ECU

Each ECU, that received the signal or message with “invalid flag” or “sequence counter” or “protection calculation” shall support this DTC.

|                                         |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b>    |                                     | 0xC400 – C4FF<br>(C4XX; XX is the ECU address)                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>FTB</b>                              |                                     | 0x00: No Sub Type Information<br>0x82: Alive/Sequence Counter Incorrect/Not Updated<br>0x83: Value of Signal Protection Calculation Incorrect                                                                                                                                                                                                                                                                                                                                        |
| <b>Failure Lamp Flag</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Failure Level</b>                    |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>DTC name&amp;Description</b>         |                                     | Invalid data received from XX ECU (e.g., C410 00 is Invalid data received from VCCM)                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Enable Criteria</b>                  | <b>Operation Cycle</b>              | DEM OPCYC IGNITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                         | <b>Vehicle Power Mode(s)</b>        | On for 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                         | <b>Vehicle Supply Voltage</b>       | 9.0 V ≤ Voltage ≤ 16.0 V                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                         | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                         | <b>ECU Operating Conditions</b>     | ECU communication active state = True.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                         | <b>Others</b>                       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                         | <b>Exceptions</b>                   | <ol style="list-style-type: none"> <li>U0XXX_00_ENABLE = disabled; however, fail-soft actions shall still be taken if failure conditions are met.</li> <li>When a bus off condition (C07X) is for DTCs, these Lost Communications DTCs shall not set but the fail-soft action shall occur if conditions to set the DTC are met.</li> <li>Within the first 5 seconds of a recovery from an under or over voltage condition.</li> <li>Vehicle is in logistic mode, if used.</li> </ol> |
| <b>Test Frequency/ Monitoring Cycle</b> |                                     | Monitored continuously while frames are being transmitted.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>DTC Failure criteria</b>             |                                     | Upon receipt of a monitored signal which is indicated as invalid on consecutive receptions or for a period of time (whichever is longer).                                                                                                                                                                                                                                                                                                                                            |

|                                            |                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Pass criteria</b>                  | Reception of the monitored signal with the corresponding rolling count value being properly updated.                                                                                       |
| <b>Snapshot Record Data</b>                | Depends on the ECU.                                                                                                                                                                        |
| <b>Fail soft Actions</b>                   | The module shall take fail soft action to ensure safe operation of the system. Typically, this involves simple parameter substitution for failed signals though other methods may be used. |
| <b>Diagnostic investigation method</b>     | Check the ECU.                                                                                                                                                                             |
| <b>After-sales maintenance Suggestions</b> | Check the ECU signal and the software version.                                                                                                                                             |

## 10.4 No Communication on LIN Bus

No Communication on LIN Bus DTCs represent a condition where no communication is possible on Lin Bus.

|                                            |                                     |                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b>       |                                     | 0xC080 – C08F<br>(C08X, X is the number of LIN bus;)                                                                                                                                                                                                                                    |
| <b>FTB</b>                                 |                                     | 0x00: No Sub Type Information                                                                                                                                                                                                                                                           |
| <b>Failure Lamp Flag</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                     |
| <b>Failure Level</b>                       |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                     |
| <b>DTC name&amp;Description</b>            |                                     | No Communication on LIN Bus X (e.g., C08100 is No Communication on LIN Bus 1)                                                                                                                                                                                                           |
| <b>Enable Criteria</b>                     | <b>Operation Cycle</b>              | DEM_OPCYC_IGNITION                                                                                                                                                                                                                                                                      |
|                                            | <b>Vehicle Power Mode(s)</b>        | On for 5 seconds.                                                                                                                                                                                                                                                                       |
|                                            | <b>Vehicle Supply Voltage</b>       | 9.0 V ≤ Voltage ≤ 16.0 V                                                                                                                                                                                                                                                                |
|                                            | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                                                                                                                                                         |
|                                            | <b>ECU Operating Conditions</b>     | The specified LIN bus is enabled and in the FULL_COMM state.                                                                                                                                                                                                                            |
|                                            | <b>Others</b>                       | None                                                                                                                                                                                                                                                                                    |
|                                            | <b>Exceptions</b>                   | <ol style="list-style-type: none"> <li>Test Results shall not be considered valid</li> <li>If the specified LIN bus is not enabled</li> <li>Within the first 5 seconds of a recovery from an under or over voltage condition.</li> <li>Vehicle is in logistic mode. if used.</li> </ol> |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | At least one LIN schedule table is active for the specified LIN bus.                                                                                                                                                                                                                    |
| <b>DTC Failure criteria</b>                |                                     | LIN Master has a Lost Communication DTC confirmed for every LIN slave on the specified LIN bus.                                                                                                                                                                                         |
| <b>Test Pass criteria</b>                  |                                     | Reception of the monitored signal with the corresponding rolling count value being properly updated.                                                                                                                                                                                    |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                     |
| <b>Fail soft Actions</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                     |
| <b>Diagnostic investigation method</b>     |                                     | Check the ECU.                                                                                                                                                                                                                                                                          |
| <b>After-sales maintenance Suggestions</b> |                                     | Check the Lin line and the power supply. Check the NM function.                                                                                                                                                                                                                         |

## 10.5 Lost Communication with LIN Slave ECU

The DTCs represent the loss of communication of the LIN master ECU with a LIN slave ECU.

|                                      |                              |                                                                                              |
|--------------------------------------|------------------------------|----------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b> |                              | C100 – C1FF<br>(C1XX, XX is the LIN slave ECU address)                                       |
| <b>FTB</b>                           |                              | 0x00: No Sub Type Information                                                                |
| <b>Failure Lamp Flag</b>             |                              | Depends on the ECU.                                                                          |
| <b>Failure Level</b>                 |                              | Depends on the ECU.                                                                          |
| <b>DTC name&amp;Description</b>      |                              | Lost Communication with LIN Slave X (e.g., C103 00 is Lost Communication with LIN Slave EBS) |
| <b>Enable Criteria</b>               | <b>Operation Cycle</b>       | DEM_OPCYC_IGNITION                                                                           |
|                                      | <b>Vehicle Power Mode(s)</b> | On for 5 seconds.                                                                            |

|                                            |                                     |                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | <b>Vehicle Supply Voltage</b>       | 9.0 V ≤ Voltage ≤16.0 V                                                                                                                                                                                                                                                               |
|                                            | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                                                                                                                                                       |
|                                            | <b>ECU Operating Conditions</b>     | The specified LIN bus is enabled and in the FULL COMM state.                                                                                                                                                                                                                          |
|                                            | <b>Others</b>                       | None                                                                                                                                                                                                                                                                                  |
|                                            | <b>Exceptions</b>                   | 1. Test Results shall not be considered valid, If the specified LIN bus is not enabled<br>2. Algorithm shall not run if Cxxx_00_ENABLE = disabled<br>3. Within the first 5 seconds of a recovery from an under or over voltage condition.<br>4. Vehicle is in logistic mode. if used. |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | Monitored continuously while LIN frames are being transmitted.                                                                                                                                                                                                                        |
| <b>DTC Failure criteria</b>                |                                     | No response from a slave device to a correctly transmitted header for 5 (calibratable) consecutive transmissions to that slave device.                                                                                                                                                |
| <b>Test Pass criteria</b>                  |                                     | Receipt of a response from that slave device to a transmitted header and the J2602 Status Byte in that response message does not report any errors in the defined Error Field.                                                                                                        |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                   |
| <b>Fail soft Actions</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                   |
| <b>Diagnostic investigation method</b>     |                                     | Check the ECU.                                                                                                                                                                                                                                                                        |
| <b>After-sales maintenance Suggestions</b> |                                     | Check the LIN Slave ECU.                                                                                                                                                                                                                                                              |

## 10.6 Loss of Link on Ethernet Port X

Loss of Link on Ethernet Port {x} DTCs represent the Lost Link with a network component on the Ethernet Bus. The loss of link on port (x) shall be determined by the switch and reported by the network component that houses the switch.

|                                      |                                         |                                                                                                                                                                                                                                                                                      |
|--------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b> |                                         | C090 – C09F;<br>(C09X, X is the switch port number)                                                                                                                                                                                                                                  |
| <b>FTB</b>                           |                                         | 0x00: No Sub Type Information                                                                                                                                                                                                                                                        |
| <b>Failure Lamp Flag</b>             |                                         | Depends on the ECU.                                                                                                                                                                                                                                                                  |
| <b>Failure Level</b>                 |                                         | Depends on the ECU.                                                                                                                                                                                                                                                                  |
| <b>DTC name&amp;Description</b>      |                                         | Lost of Link on Ethernet Port X (e.g., C090 00 is Lost of Link on Ethernet Port 0)                                                                                                                                                                                                   |
| <b>Enable Criteria</b>               | <b>Operation Cycle</b>                  | DEM OPCYC IGNITION                                                                                                                                                                                                                                                                   |
|                                      | <b>Vehicle Power Mode(s)</b>            | On for 5 seconds.                                                                                                                                                                                                                                                                    |
|                                      | <b>Vehicle Supply Voltage</b>           | 9.0 V ≤ Voltage ≤16.0 V                                                                                                                                                                                                                                                              |
|                                      | <b>Vehicle Operating Conditions</b>     | Network that the ECU participates in is active.                                                                                                                                                                                                                                      |
|                                      | <b>ECU Operating Conditions</b>         | Ethernet communication is active .                                                                                                                                                                                                                                                   |
|                                      | <b>Others</b>                           | None                                                                                                                                                                                                                                                                                 |
|                                      | <b>Exceptions</b>                       | 1. Test Results shall not be considered valid if the calibration “ Lost of Link on Ethernet Port X DTC” is disabled.<br>2. Within the first 5 seconds of a recovery from an under or over voltage condition.<br>3. Vehicle is in logistic mode. if used.                             |
|                                      | <b>Test Frequency/ Monitoring Cycle</b> | Report by register                                                                                                                                                                                                                                                                   |
|                                      | <b>DTC Failure criteria</b>             | Upon receipt of Link status =Down for a specific port for a calibratable amount of time (reported from the embedded switch) (please note that the calibration value is left to the discretion of the program team to determine whether it should be a calibration or configuration). |
| <b>Test Pass criteria</b>            |                                         | Upon receipt of Link status =Up for the specific port after a calibratable amount of time (reported from the embedded switch) (please note that the calibration value is left to the discretion of the program team to determine                                                     |

|                                            |                                                       |
|--------------------------------------------|-------------------------------------------------------|
|                                            | whether it should be a calibration or configuration). |
| <b>Snapshot Record Data</b>                | Depends on the ECU.                                   |
| <b>Fail soft Actions</b>                   | Depends on the ECU.                                   |
| <b>Diagnostic investigation method</b>     | Check the ECU.                                        |
| <b>After-sales maintenance Suggestions</b> | Check the ECU.                                        |

## 10.7 Invalid Data Received from Ethernet ECU (Checksum Failure)

This DTC will set when there are significant number of Cyclic Redundancy Check (CRC) errors received by the ECU. It is used to capture Ethernet CRC errors and indicates an issue with the source ECU.

|                                            |                                     |                                                                                                                                                                                                                                             |
|--------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b>       |                                     | C400 41 - C4FF 41                                                                                                                                                                                                                           |
| <b>FTB</b>                                 |                                     | 0x41: General Checksum Failure                                                                                                                                                                                                              |
| <b>Failure Lamp Flag</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                                         |
| <b>Failure Level</b>                       |                                     | Depends on the ECU.                                                                                                                                                                                                                         |
| <b>DTC name&amp;Description</b>            |                                     | Invalid Data Received from Ethernet(Checksum Failure)                                                                                                                                                                                       |
| <b>Enable Criteria</b>                     | <b>Operation Cycle</b>              | DEM OPCYC IGNITION                                                                                                                                                                                                                          |
|                                            | <b>Vehicle Power Mode(s)</b>        | On for 5 seconds.                                                                                                                                                                                                                           |
|                                            | <b>Vehicle Supply Voltage</b>       | 9.0 V ≤ Voltage ≤16.0 V                                                                                                                                                                                                                     |
|                                            | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                                                                                                             |
|                                            | <b>ECU Operating Conditions</b>     | Ethernet communication is active .                                                                                                                                                                                                          |
|                                            | <b>Others</b>                       | None                                                                                                                                                                                                                                        |
| <b>Exceptions</b>                          |                                     | 1. Test Results shall not be considered valid If CXXX41_ENABLE = disabled.<br>2. Loss of Link DTC = SET<br>3. Within the first 5 seconds of a recovery from an under or over voltage condition.<br>4. Vehicle is in logistic mode. if used. |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | Report by register                                                                                                                                                                                                                          |
| <b>DTC Failure criteria</b>                |                                     | Receiving ECU has greater than or equals to a calibratable number of Cyclic Redundancy Check (CRC) Errors for Ethernet frames.                                                                                                              |
| <b>Test Pass criteria</b>                  |                                     | Receiving ECU has less than a calibratable number of Cyclic Redundancy Check (CRC) Errors for Ethernet frames.                                                                                                                              |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                         |
| <b>Fail soft Actions</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                                         |
| <b>Diagnostic investigation method</b>     |                                     | Check the ECU.                                                                                                                                                                                                                              |
| <b>After-sales maintenance Suggestions</b> |                                     | Check the ECU.                                                                                                                                                                                                                              |

## 10.8 Lost Communication with Ethernet ECU

The purpose of this DTC is to detect message communication failures due to microcontroller/software issues.

The diagnostic shall run when both the sending ECU and receiving ECU for a particular message are awake and Ethernet communication is active in both ECUs.

|                                      |                        |                                                       |
|--------------------------------------|------------------------|-------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b> |                        | C300 – C3FF<br>(C3XX, XX is the Ethernet ECU address) |
| <b>FTB</b>                           |                        | 0x00: No Sub Type Information                         |
| <b>Failure Lamp Flag</b>             |                        | Depends on the ECU.                                   |
| <b>Failure Level</b>                 |                        | Depends on the ECU.                                   |
| <b>DTC name&amp;Description</b>      |                        | Lost Communication with Ethernet ECU                  |
| <b>Enable</b>                        | <b>Operation Cycle</b> | DEM OPCYC IGNITION                                    |

|                                            |                                     |                                                                                                                                                                                                                                                           |
|--------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Criteria</b>                            | <b>Vehicle Power Mode(s)</b>        | On for 5 seconds.                                                                                                                                                                                                                                         |
|                                            | <b>Vehicle Supply Voltage</b>       | 9.0 V ≤ Voltage ≤16.0 V                                                                                                                                                                                                                                   |
|                                            | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                                                                                                                           |
|                                            | <b>ECU Operating Conditions</b>     | Ethernet communication is active.                                                                                                                                                                                                                         |
|                                            | <b>Others</b>                       | None                                                                                                                                                                                                                                                      |
|                                            | <b>Exceptions</b>                   | 1. Test Results shall not be considered valid If CXXX00_ENABLE = disabled.<br>2. Loss of Link DTC = SET<br>3. Within the first 5 seconds of a recovery from an under or over voltage condition.<br>4. Vehicle is in logistic mode, if used.               |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | Monitored continuously while frames are being transmitted and received. Upon notification of supervision timeout.                                                                                                                                         |
| <b>DTC Failure criteria</b>                |                                     | Upon notification by the handler that associated supervised signal has failed supervision. A calibrate able de-bounce time is permitted at the time that notification of failed supervision has been received by the application to delay storing of DTC. |
| <b>Test Pass criteria</b>                  |                                     | Reception of the previously failed supervised signal shall clear the associated signal supervision failure indication.                                                                                                                                    |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                                       |
| <b>Fail soft Actions</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                                                       |
| <b>Diagnostic investigation method</b>     |                                     | Check the ECU.                                                                                                                                                                                                                                            |
| <b>After-sales maintenance Suggestions</b> |                                     | Check the ECU.                                                                                                                                                                                                                                            |

## 10.9 ECU Power Under Voltage DTC

|                                            |                                     |                                                                                                                                                        |
|--------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b>       |                                     | F003                                                                                                                                                   |
| <b>FTB</b>                                 |                                     | 0x16:Circuit Voltage Below Threshold                                                                                                                   |
| <b>Failure Lamp Flag</b>                   |                                     | Depends on the ECU.                                                                                                                                    |
| <b>Failure Level</b>                       |                                     | Depends on the ECU.                                                                                                                                    |
| <b>DTC name&amp;Description</b>            |                                     | ECU Power Under Voltage                                                                                                                                |
| <b>Enable Criteria</b>                     | <b>Operation Cycle</b>              | DEM_OPCYC_IGNITION                                                                                                                                     |
|                                            | <b>Vehicle Power Mode(s)</b>        | On for 5 seconds.                                                                                                                                      |
|                                            | <b>Vehicle Supply Voltage</b>       | N/A                                                                                                                                                    |
|                                            | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                        |
|                                            | <b>ECU Operating Conditions</b>     | The control module is executing operational(APP) software.                                                                                             |
|                                            | <b>Others</b>                       | None                                                                                                                                                   |
|                                            | <b>Exceptions</b>                   | 1. Test Results shall not be considered valid If F00316_ENABLE = disabled.                                                                             |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | The control module power circuit shall be monitored continuously while the control module is executing operational software at least once every 500ms. |
| <b>DTC Failure criteria</b>                |                                     | ECU supply voltage Vsup < 9.0 +/-0.5 volts for 3000 ms (calibratable).                                                                                 |
| <b>Test Pass criteria</b>                  |                                     | 9.0 +/-0.5 volts < ECU supply voltage Vsup < 16.0 +/-0.5 volts for 3000ms (calibratable).                                                              |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                    |
| <b>Fail soft Actions</b>                   |                                     | Depends on the ECU.                                                                                                                                    |
| <b>Diagnostic investigation method</b>     |                                     | Check the ECU.                                                                                                                                         |
| <b>After-sales maintenance Suggestions</b> |                                     | Check the Power supply.                                                                                                                                |

## 10.10 ECU Power Over Voltage DTC

|                                      |  |                                       |
|--------------------------------------|--|---------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b> |  | F003                                  |
| <b>FTB</b>                           |  | 0x17: Circuit Voltage Above Threshold |

|                                            |                                     |                                                                                                                                                        |
|--------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Failure Lamp Flag</b>                   |                                     | Depends on the ECU.                                                                                                                                    |
| <b>Failure Level</b>                       |                                     | Depends on the ECU.                                                                                                                                    |
| <b>DTC name&amp;Description</b>            |                                     | ECU Power Over Voltage                                                                                                                                 |
| <b>Enable Criteria</b>                     | <b>Operation Cycle</b>              | DEM OPCYC IGNITION                                                                                                                                     |
|                                            | <b>Vehicle Power Mode(s)</b>        | On for 5 seconds.                                                                                                                                      |
|                                            | <b>Vehicle Supply Voltage</b>       | N/A                                                                                                                                                    |
|                                            | <b>Vehicle Operating Conditions</b> | Network that the ECU participates in is active.                                                                                                        |
|                                            | <b>ECU Operating Conditions</b>     | The control module is executing operational (APP) software.                                                                                            |
|                                            | <b>Others</b>                       | None                                                                                                                                                   |
| <b>Exceptions</b>                          |                                     | 1. Test Results shall not be considered valid If F00317_ENABLE = disabled.                                                                             |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | The control module power circuit shall be monitored continuously while the control module is executing operational software at least once every 500ms. |
| <b>DTC Failure criteria</b>                |                                     | ECU supply voltage $V_{sup} > 16.0 \pm 0.5$ volts for 3000ms (calibratable).                                                                           |
| <b>Test Pass criteria</b>                  |                                     | $9.0 \pm 0.5$ volts < ECU supply voltage $V_{sup} < 16.0 \pm 0.5$ volts for 3000ms (calibratable).                                                     |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                    |
| <b>Fail soft Actions</b>                   |                                     | Depends on the ECU.                                                                                                                                    |
| <b>Diagnostic investigation method</b>     |                                     | Check the ECU.                                                                                                                                         |
| <b>After-sales maintenance Suggestions</b> |                                     | Check the Power supply.                                                                                                                                |

## 10.11 Some ECU Hardware DTCs

|                                            |                                     |                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b>       |                                     | F000                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>FTB</b>                                 |                                     | 0x41: General Checksum Failure<br>0x42: General Memory Failure<br>0x43: Special Memory Failure<br>0x44: Data Memory Failure<br>0x45: Program Memory Failure<br>0x46: Calibration / Parameter Memory Failure<br>0x47: Watchdog / Safety $\mu$ C Failure<br>0x48: Supervision Software Failure<br>0x49: Internal Electronic Failure<br>0x4A: Incorrect Component Installed<br>0x4B: Over Temperature |
| <b>Failure Lamp Flag</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Failure Level</b>                       |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>DTC name&amp;Description</b>            |                                     |                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Enable Criteria</b>                     | <b>Operation Cycle</b>              | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
|                                            | <b>Vehicle Power Mode(s)</b>        | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
|                                            | <b>Vehicle Supply Voltage</b>       | $9.0 \text{ V} \leq \text{Voltage} \leq 16.0 \text{ V}$                                                                                                                                                                                                                                                                                                                                            |
|                                            | <b>Vehicle Operating Conditions</b> | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
|                                            | <b>ECU Operating Conditions</b>     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
|                                            | <b>Others</b>                       | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Exceptions</b>                          |                                     | 1. Algorithm shall not run if F000 xx_ENABLE = disabled.                                                                                                                                                                                                                                                                                                                                           |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>DTC Failure criteria</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Test Pass criteria</b>                  |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Fail soft Actions</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Diagnostic investigation method</b>     |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>After-sales maintenance Suggestions</b> |                                     | Depends on the ECU.                                                                                                                                                                                                                                                                                                                                                                                |

## 10.12 Some ECU Software DTCs

|                                            |                                     |                                                                                                                                                                                                                        |
|--------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTC Number/High&amp;Mid (Hex)</b>       |                                     | F000                                                                                                                                                                                                                   |
| <b>FTB</b>                                 |                                     | 0x51: Not Programmed<br>0x52: Not Activated<br>0x53: Deactivated<br>0x54: Missing Calibration<br>0x55: Not Configured<br>0x56: Invalid / Incompatible Configuration<br>0x57: Invalid / Incompatible Software Component |
| <b>Failure Lamp Flag</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                    |
| <b>Failure Level</b>                       |                                     | Depends on the ECU.                                                                                                                                                                                                    |
| <b>DTC name&amp;Description</b>            |                                     |                                                                                                                                                                                                                        |
| <b>Enable Criteria</b>                     | <b>Operation Cycle</b>              | Depends on the ECU.                                                                                                                                                                                                    |
|                                            | <b>Vehicle Power Mode(s)</b>        | Depends on the ECU.                                                                                                                                                                                                    |
|                                            | <b>Vehicle Supply Voltage</b>       | 9.0 V ≤ Voltage ≤ 16.0 V                                                                                                                                                                                               |
|                                            | <b>Vehicle Operating Conditions</b> | Depends on the ECU.                                                                                                                                                                                                    |
|                                            | <b>ECU Operating Conditions</b>     | Depends on the ECU.                                                                                                                                                                                                    |
|                                            | <b>Others</b>                       | Depends on the ECU.                                                                                                                                                                                                    |
|                                            | <b>Exceptions</b>                   | 1. Algorithm shall not run if F000 xx_ENABLE = disabled.                                                                                                                                                               |
| <b>Test Frequency/ Monitoring Cycle</b>    |                                     | Depends on the ECU.                                                                                                                                                                                                    |
| <b>DTC Failure criteria</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                    |
| <b>Test Pass criteria</b>                  |                                     | Depends on the ECU.                                                                                                                                                                                                    |
| <b>Snapshot Record Data</b>                |                                     | Depends on the ECU.                                                                                                                                                                                                    |
| <b>Fail soft Actions</b>                   |                                     | Depends on the ECU.                                                                                                                                                                                                    |
| <b>Diagnostic investigation method</b>     |                                     | Depends on the ECU.                                                                                                                                                                                                    |
| <b>After-sales maintenance Suggestions</b> |                                     | Depends on the ECU.                                                                                                                                                                                                    |